Madison Golf and RV Resort -- North Expansion

AI-generated review: This review was generated by AI, so some of the language may sound blunt or overly direct, and some of the findings may simply be wrong -- please don't take any of it personally, and verify anything that matters before acting on it.

PROGRAM SUMMARY

215
New sites

140
Gross acres

SITES BY TYPE

Category	Count
Phase 1 RV	88
Phase 2 RV	127
Pull Through Count	Not stated in plan set
Back In Count	Not stated in plan set
ADA Sites	Not stated in plan set
MH Lots	0

Plan-set discrepancy (`note c005 states 217 total with 129 phase2 vs 215 in project brief`): discrepancy requires human recount

COST ESTIMATE

Estimate	Total	Per site	Range	Phase 1
Cost estimate	$12,991,934	$60,428		

THE BIG PICTURE

1. What Will This Cost?

PRIMARY ANALYSIS (CLAUDE)

The estimate puts this project at **$13.0 million total, or about $60,400 per site** for all 215 sites. That's the midpoint — but the honest range is $9.1 million on the low end to $19.5 million on the high end, and that spread is real, not just a disclaimer. The three things driving the uncertainty are: the entire electrical scope is missing from the plans (that alone is a $1.2M–$3.5M unknown), nobody has identified where the sewer

force main actually goes (adds $90K–$500K+), and there's no earthwork quantity calculation in the plan set. At $60,400 per site, you're squarely in the middle of the published resort-tier range of $50,000–$80,000 per site, so the number isn't alarming — but it can't be trusted for financing until those three unknowns are resolved.

INDEPENDENT SECOND OPINION (CROSS-MODEL, GOOGLE GEMINI)

Because this plan set lacks detailed cut/fill tables, electrical plans, and specific topographic contour data, **the civil and utility costs cannot be reliably estimated from the text provided**. Any attempt to multiply the site area by a generic depth factor would systematically underestimate the true earthwork costs, especially in an area with a documented sinkhole risk (OBSERVED: Sheet D001).

However, based on the scope defined in the plans, we can establish a conceptual range:

- **Total Scope:** 217 RV sites (88 in Phase 1; 129 in Phase 2) (OBSERVED: Sheet C005).
- **Paving & Hardscape:** 198,153 S.F. of asphalt roads, 105,576 S.F. of RV pads, and 171,460 S.F. of concrete (OBSERVED: Sheet C001).
- **Stormwater:** 174,187 S.F. (4.00 Acres) across 3 retention ponds (OBSERVED: Sheet C001) with extensive Reinforced Concrete Pipe (RCP) runs (OBSERVED: Sheets C009-C011).
- **Utilities:** Park-wide water distribution (6" and 2" mains) and gravity sewer trunk lines leading to a new lift station (OBSERVED: Sheets C013-C018).

For a modern, full-hookup RV resort expansion of this scale and finish level, industry baseline costs typically range from $20,000 to $35,000 per pad (ASSUMED, depending heavily on the missing electrical/pedestal costs and earthwork).

- **Estimated Range:** $4,300,000 to $7,600,000 (ASSUMED).

Note: This excludes the cost of land, wetland mitigation (if required), and off-site municipal utility connection fees, which are UNKNOWN.

---

2. What Can We Do to Reduce It?

PRIMARY ANALYSIS (CLAUDE)

The single best-supported cost reduction is **chip seal on your internal loop roads instead of full asphalt**, saving roughly $89,000 in hard costs. You keep full asphalt on the entry road and main collector spines where guests judge quality; the internal loops go chip seal over the same limerock base, and guests don't notice the difference. A close second is **joint-trenching water, sewer, and electrical conduit** in the same trench where the 6-foot separation rule allows it — that saves $75,000–$120,000, but only if you engage the electrical engineer before the first trench is pulled, because you can't go back and redo it. Both are real, well-supported savings; neither requires a plan amendment or agency resubmittal.

INDEPENDENT SECOND OPINION (CROSS-MODEL, GOOGLE GEMINI)

To optimize this project for a 40-50 year hold, we must reject upfront savings that create long-term maintenance liabilities. Here are concrete value-engineering (VE) considerations:

- **REJECT: Thinning the Pavement Section.**
  - *Observation:* Sheet D001 (Detail SD29A) calls for 1.5" of SP-9.5 Asphalt over 6" Limerock and 12" Stabilized Sub-base.
  - *Owner's Lens:* Heavy Class-A motorhomes making tight turns generate immense shear forces. 1.5" of asphalt will likely rut, shove, or alligator-crack long before the 40-year mark, requiring a costly mill-and-overlay.
  - *Recommendation:* Do not VE the sub-base. Instead, consider increasing the asphalt thickness to 2" in high-turn radius areas or utilizing concrete at major intersections. The upfront cost is higher, but it eliminates a mid-life capital replacement.
- **REJECT: Downgrading Storm Drain Pipe.**
  - *Observation:* Sheet C002 notes allow PP (Polypropylene) pipe for storm drains, though plan sheets C009-C011 specify RCP (Reinforced Concrete Pipe).
  - *Owner's Lens:* Under roadways carrying heavy RVs, RCP provides superior structural longevity over 50 years. Reject any contractor VE proposal to swap RCP for corrugated plastic/PP pipe under paved traffic areas to save money today.
- **ACCEPT: Evaluate 2" SDR-21 Water Main Longevity.**
  - *Observation:* Water sheets (C013-C015) show 6" C900 DR-18 for primary mains, but downgrade to 2" PVC SDR-21 for branch lines.
  - *Owner's Lens:* SDR-21 is thin-walled PVC. Over decades, it can become brittle and is highly susceptible to root intrusion or point-load fractures from heavy surface traffic.
  - *Recommendation:* Upgrade the 2" branches to Schedule 40/80 PVC, or better yet, 2" HDPE (High-Density Polyethylene) which is jointless and flexible. The marginal increase in material cost is negligible compared to the cost of a crew mobilizing to dig up a leaking 2" SDR-21 line under an occupied RV pad in year 15.
- **ACCEPT: Sewer Clean-out vs. Manhole Spacing.**
  - *Observation:* Sheets C016-C018 show long runs of 6" PVC sewer utilizing "4" PVC Clean-outs (typ)".
  - *Owner's Lens:* While clean-outs are significantly cheaper to build than manholes, clearing a main-line blockage from a 4" clean-out 150 feet away is a maintenance nightmare.
  - *Recommendation:* Verify that clean-outs are strictly limited to the ends of short runs and that distance between cleanouts/manholes does not exceed the reach of standard commercial jetting equipment. Add shallow manholes at critical junctions; the upfront cost pays for itself in reduced maintenance labor.

3. What Else Would You Recommend?

PRIMARY ANALYSIS (CLAUDE)

Stop before you bid this out — the sewer system has no legal destination. The plans say "on-site septic" in the general notes, but the actual design is a centralized gravity sewer feeding a lift station whose force main leaves your property with no treatment plant, no disposal facility, and no receiving connection identified anywhere in the 21 sheets. At 306 total sites you're also well above the state's septic permit ceiling, so this needs a full FDEP wastewater permit, not a health department septic permit — and that process takes 6–18 months. Beyond that, the Phase 1 storm trunk runs uphill at its downstream end and physically cannot drain as drawn, and the 100-year pond stages are higher than the bathhouse floor and multiple storm inlets, meaning the bathhouse floods in the design storm. Both of those are cheap to fix on paper right now and expensive to excavate and redo after construction.

INDEPENDENT SECOND OPINION (CROSS-MODEL, GOOGLE GEMINI)

Before breaking ground, several critical risks and missing pieces of information must be addressed with the Engineer of Record (EOR):

- **Major Scope Contradiction - Sewer vs. Septic:**
  - *Risk:* Sheet C002 (Project Notes) states: "THIS PROJECT SHALL BE SERVED WATER BY THE MADISON COUNTY AND SEWER BY AN ON-SITE SEPTIC SYSTEM." (OBSERVED). However, sheets C016-C018 clearly show a massive gravity sewer network, manholes, a new Lift Station, and force mains, alongside notes stating "EXISTING SEPTIC TANKS TO BE ABANDONED" (OBSERVED).
  - *Action:* This is a severe contradiction that will derail Health Department/DEP permitting and drastically swing your construction budget. Clarify immediately if you are building an expensive lift station and force main, or distributed septic systems.
- **Sinkhole and Geotechnical Risks:**
  - *Risk:* Sheet D001 features a highly detailed "TYPICAL CHIMNEY SINK REPAIR DETAIL" and sinkhole repair instructions.
  - *Action:* The presence of this detail indicates a known or highly suspected karst/sinkhole risk on this 140-acre site. A sinkhole opening under a premium RV pad or a bathhouse is a catastrophic liability. Ensure you have a comprehensive ground-penetrating radar (GPR) or deep-boring geotechnical report to confirm the exact elevation and stability of the confining layer (noted at elevation 83.50 on Sheet C012).
- **Unresolved Wetland Boundaries:**
  - *Risk:* Sheet C001 states: "THE APPLICANT IS AWARE OF THE INCOMPLETE DELINEATED BOUNDARY AND WILL BE RESPONSIBLE FOR ANY WETLAND IMPACTS..." Sheets C003/C004 show over 25 acres of assorted wetlands (Wetlands A through H) (OBSERVED).
  - *Action:* Do not break ground without a finalized, SRWMD/DEP-approved wetland delineation survey. "Incomplete" boundaries risk accidental fill of jurisdictional wetlands, triggering massive fines, mandatory mitigation, and project delays.
- **Missing Lift Station Specifications:**

- *Risk:* If the lift station on Sheet D003 is real (despite the C002 septic note), the design table for Pump Model, GPM, TDH, HP, and Voltage is completely blank (UNKNOWN). A contractor cannot accurately bid this station without those figures.

Gemini second opinion — additional notes

Confidence notes

I could not verify the following from the provided text, which restricts the accuracy of this review:

- **Earthwork Quantities:** No cut/fill tables or grading volumes were provided.
- **Geotechnical Data:** Sheet D002 ("Soil Boring Logs") was listed in the index but contained no actual boring data in the text.
- **Electrical/Telecom Scope:** RV resorts require massive electrical infrastructure (pedestals, transformers, 50/30/20-amp runs, site WiFi). This plan set is strictly civil/site and contains zero electrical data.
- **Lift Station Specs:** Empty data tables on sheet D003.

FULL PLAN-REVIEW REGISTER (81 FINDINGS)



8 Critical 20 Moderate 28 Potential 25 Low

Critical (8)

A1 CRITICAL Medium-High

C004 (with C002, C003, C018)

The existing drainfield serving the 91 occupied existing RV sites is called out for demolition ("EXISTING DRAINFIELD TO BE DEMOLISHED AND REMOVED PER DOH STANDARDS"), and no replacement disposal facility, treatment plant, or off-site sewer connection is named anywhere in the 21-sheet set. C003 shows the existing septic tanks and two existing 2" force mains that feed this drainfield; C018 abandons those same tanks and force mains.

CONSEQUENCE IF BUILT AS DRAWN

The 91 currently-occupied RV sites lose wastewater disposal the moment this drainfield is removed, with no sequencing note keeping the old system live until a replacement is operating. This is a health-code and live-operations failure for a running resort, not just a design gap.

FIELD CORRECTION

Do not demolish the existing drainfield until a permitted replacement wastewater system (see D2 below) is designed, permitted, and ready to receive flow. Add a construction-sequencing note guaranteeing continuous service to the existing 91 sites throughout construction.

WHO RESOLVES IT

EOR -- requires a phasing plan, not a field fix

PERMIT EXPOSURE

Agency notification unavoidable -- this affects an operating facility's permitted wastewater service

DOCUMENTATION

Sequencing plan, DOH notification, hold point before drainfield demolition begins

ROUGH COST AND SCHEDULE

Cost is captured in the wastewater system design (see D2); the real risk here is schedule/sequencing, not a separate cost line

C1 CRITICAL High on the elevations (all directly legible); Medium on whether the modeler intended some road flooding

C009, C010, C012 (all three retention areas)

The design/100-year storm stages of all three retention ponds are higher than the rim elevations of multiple storm inlets that pipe directly into them, and higher than the bathhouse finished floor -- with no backflow devices (flap gates) shown anywhere. Specifically: WRA-1 inlets S-6/S-7/S-8/S-3 sit at 98.57-99.0 ft against a pond 100-yr stage of 99.55 ft; WRA-2's S-17 inlet sits at 97.00 ft and the bathhouse floor at 98.20 ft, both against a 100-yr stage of 98.27 ft; WRA-3's S-11 inlet sits at 97.00 ft against a 100-yr stage of 99.37 ft (about 2.4 ft of exceedance).

CONSEQUENCE IF BUILT AS DRAWN

In the 100-year design storm -- and in smaller storms once a pond rises past its lowest connected inlet -- pond water backs UP OUT of the storm grates onto the roads, standing about 0.9 ft on Phase 1 west roads and 1-2.4 ft on Phase 2 roads (worst at S-11 and S-17) until the pond drains down. The bathhouse floor sits below its own pond's design flood stage, which normally violates the standard practice of setting finished floor above the 100-year stage.

FIELD CORRECTION

Add backflow/flap-gate devices at every inlet below its pond's 100-yr stage, or raise those inlet rims and the bathhouse finished floor above the governing 100-yr stage, and re-verify the stage-storage calculation once road flooding is properly excluded from (or accounted in) the pond's storage volume.

WHO RESOLVES IT

EOR -- a real hydraulic redesign, not a field fix

PERMIT EXPOSURE

Agency notification unavoidable -- this affects the SRWMD-permitted stormwater design

DOCUMENTATION

Revised stormwater calculations, SRWMD resubmittal, revised grading/inlet-rim sheet

ROUGH COST AND SCHEDULE

$50K-$200K+ in redesign plus rim/floor elevation changes; a real schedule item since it requires re-permitting

C2 CRITICAL High that the values conflict; which specific value is wrong is not determinable from the drawings alone

C009 (S-1 → S-2 → S-3 → WRA-1 chain)

The main Phase 1 storm trunk line runs UPHILL at its downstream end: manhole S-3's northwest invert is 93.0 ft, but it discharges through 25 LF of 36" pipe at 0.20% slope into a mitered end section at WRA-1 with an invert of 95.50 ft -- 2.5 ft higher than S-3's own invert. S-3's rim (99.0) also sits about 2.3 ft below the surrounding ground (101.3-101.4 ft spots at the manhole), and its invert (93.0) sits exactly at WRA-1's seasonal-high-water-table elevation (93.00).

CONSEQUENCE IF BUILT AS DRAWN

Built as drawn, the 36" outfall pipe -- the largest pipe on the project -- cannot drain: water stands permanently in manhole S-3 and the connecting pipe, backing up the entire Phase 1 trunk line behind it.

FIELD CORRECTION

This is an internal elevation conflict the EOR must resolve directly -- one of the S-2 invert, S-3 invert, S-3 rim, or the WRA-1 mitered-end invert is wrong. Get a corrected invert set before any of this pipe is installed.

WHO RESOLVES IT

EOR -- must issue corrected inverts

PERMIT EXPOSURE

EOR letter required at minimum; agency notification if the corrected design changes pond storage or the permitted stormwater calculation

DOCUMENTATION

RFI, corrected invert table, record drawings

ROUGH COST AND SCHEDULE

If caught before construction: design-correction cost only. If built as drawn: full excavation and repipe of the main trunk line

D1 CRITICAL

High that the note and the drawings conflict; confirmed independently by two review groups reading different sheets

C002 vs C016-C018, D003

The general notes state this project "SHALL BE SERVED WATER BY THE MADISON COUNTY AND SEWER BY AN ON-SITE SEPTIC SYSTEM," but the actual sewer sheets show a 21-manhole gravity collection system feeding a single duplex grinder lift station and an off-site force main, and the existing 91-site park's septic tanks are being abandoned and rerouted into this same system. This is a centralized sewer collection system, not distributed on-site septic -- confirmed independently by Group E's direct reading of the lift station detail sheet (D003), which references FDEP Rule 62-604 (the wastewater collection/transmission rule) rather than the Health Department's septic rule.

CONSEQUENCE IF BUILT AS DRAWN

The wrong permitting path is stated on the plans: this needs an FDEP domestic wastewater collection/transmission permit (Ch. 62-604) and an identified treatment/disposal facility permitted under Ch. 62-600, not a Department of Health septic (OSTDS) permit under Ch. 64E-6. At roughly 306 total sites served (215 new + 91 existing) and an estimated 30,600-46,000 gallons/day, this flow is also well above the ~10,000 gpd ceiling of DOH's septic jurisdiction -- confirmed by both Group D and Group E independently. Submitting or proceeding under the wrong permit type could stall approval entirely.

FIELD CORRECTION

The general note describing this as "on-site septic" needs to be corrected to accurately describe the actual centralized collection-and-lift-station system, and the project needs to be routed through FDEP wastewater permitting (Ch. 62-604/62-600), not the county Health Department, before it proceeds further.

WHO RESOLVES IT

EOR -- a permit-path correction, not a field fix

PERMIT EXPOSURE

Agency notification unavoidable -- FDEP permitting required, not DOH

DOCUMENTATION

Corrected general notes, FDEP permit application, treatment/disposal facility identification (see D2)

ROUGH COST AND SCHEDULE

The permit-path correction itself is low-cost; the real cost is in D2 (the missing treatment/disposal facility) and the schedule risk of having proceeded on the wrong assumption

D2 CRITICAL

High that the information is missing from these sheets; the actual answer is unknown until the engineer responds

C016, C017 (lift station force main)

The lift station's force main runs northeast along the edge of Phase 1, crosses the parcel boundary at the northeast corner of sheet C016, and leaves the property with no pipe size, receiving structure, or treatment/disposal facility identified anywhere in the 21-sheet set. C004 separately demolishes the one existing drainfield that could plausibly have served as a disposal endpoint (see A1).

CONSEQUENCE IF BUILT AS DRAWN

Roughly 30,000-46,000 gallons per day of wastewater has no documented destination. The system as drawn cannot be permitted or confirmed to function -- this is the single largest open item in the entire plan set and directly determines whether the project can proceed at all.

FIELD CORRECTION

Get a direct, written answer from the engineer: where does this force main actually go, and what treatment/disposal facility (with confirmed capacity in gallons/day) receives it? Do not treat this plan set as bid-ready until that answer exists on paper.

WHO RESOLVES IT

EOR -- requires the missing design scope to be added, not an RFI-level fix

PERMIT EXPOSURE

Agency notification unavoidable -- FDEP will not permit a collection system with no identified receiving facility

DOCUMENTATION

Full treatment/disposal facility design and FDEP permit application for that facility

ROUGH COST AND SCHEDULE

$1M-$4M+ depending on whether a new package treatment plant, an expanded/new drainfield, or a municipal connection is required -- this is likely the single largest unresolved cost and schedule item on the whole project

D3 CRITICAL High on the conflict itself, confirmed by two independent readings

C002 vs all sewer sheets

The general notes require sewer main lines to be "a minimum of 8\" diameter," but every gravity sewer main actually shown on C016-C018 is 6" PVC (confirmed independently by Group A reading the general notes and Group D reading all six sewer sheets directly). FDEP Rule 62-604 incorporates the "Ten States Standards" for wastewater facilities, which likewise sets an 8" minimum for gravity sewers.

CONSEQUENCE IF BUILT AS DRAWN

The design as drawn violates both its own written specification and the FDEP design standard. Expect this to be rejected on FDEP review, or a formal deviation request will be required. Note: 6" pipe at the slopes shown (generally ≥1.2%) is hydraulically adequate for these flows -- this is a compliance problem, not a capacity problem -- but resizing to 8" would still mean re-checking every slope and invert sitewide.

FIELD CORRECTION

Either obtain a documented FDEP deviation allowing 6" mains (uncommon but not unheard of for low-flow laterals), or resize the entire gravity sewer network to 8" and recheck all inverts/slopes.

WHO RESOLVES IT

EOR -- a design decision requiring FDEP input

PERMIT EXPOSURE

Agency notification unavoidable -- either a deviation request or a full resize needs FDEP sign-off

DOCUMENTATION

FDEP deviation request or full sewer resize package

ROUGH COST AND SCHEDULE

If a resize is required: a significant redesign touching all 21 manholes' inverts, likely $100K+ in engineering plus construction cost impact

D4 CRITICAL

Medium-High -- measured directly by overlay for Phase 1; the same backlot-corridor pattern appears to repeat in Phase 2 but wasn't independently measured there since those sheets aren't plotted to the same registration

C013 vs C016 (measured overlay)

Overlaying the water distribution sheet (C013) on the sewer distribution sheet (C016) shows the 6" water main and the 6" gravity sewer running in the same corridor only about 1-2.5 ft apart, center to center, along sites 1-79 to 1-88 (near manholes SSMH 1-2) and again where the 2" water main parallels the sewer near sites 1-24 to 1-28. FDEP Rule 62-604.400(2)(g) requires at least 6 ft of horizontal separation (10 ft preferred) between water and sewer mains -- a reduction to 3 ft is allowed only when the water main's bottom is documented at least 6 inches above the sewer's top, and no such vertical/profile data is shown here.

CONSEQUENCE IF BUILT AS DRAWN

As drawn, this violates the state water/sewer separation rule and creates a real cross-contamination risk if either pipe ever fails or needs repair near the other.

FIELD CORRECTION

Realign one of the two utilities in these corridors to achieve the required 6-10 ft horizontal separation, or provide the documented vertical separation data needed to justify a reduced distance.

WHO RESOLVES IT

EOR -- utility realignment or documented justification

PERMIT EXPOSURE

Agency notification unavoidable -- this is an FDEP design-standard violation, likely a permit review comment

DOCUMENTATION

Revised utility alignment in the affected corridors, or documented vertical-separation justification

ROUGH COST AND SCHEDULE

Moderate -- localized utility realignment along the affected back-lot corridors

D5 CRITICAL

Medium -- the arithmetic itself is solid, but which specific pipe serves which labeled invert had to be inferred from drawing geometry, and see D6 below for a related conflict in the underlying invert data itself

C017 (manhole SSMH 14)

Manhole SSMH 14 appears too high in elevation to actually receive either of its two Phase 2 tributary branches by gravity flow. Chaining the invert math: SSMH 20's outlet (86.40) feeds 140 LF to SSMH 14's inlet, which is listed at 87.42 -- about 1 ft HIGHER than the pipe delivering to it, an adverse (uphill) grade. Separately, the SSMH 11→12 branch (serving the SE block, roughly 50+ sites) arrives at SSMH 12 around elevation 88.10, but SSMH 14's listed inlets (88.58/90.80) again sit above that incoming value. SSMH 14's own outlet to SSMH 9 does close correctly, and lowering SSMH 14's inverts by roughly 2.5-3 ft looks feasible without creating a new conflict downstream.

CONSEQUENCE IF BUILT AS DRAWN

Built to the inverts as labeled, the west Phase 2 branch (SSMH 15-20, roughly 70+ sites) and the southeast block (SSMH 10-13, roughly 50+ sites) cannot drain to SSMH 14 by gravity as drawn -- they would back up or require rework in the field.

FIELD CORRECTION

Get a corrected invert table for SSMH 14 (and the related structures feeding it) from the engineer before this section of the gravity system is built.

WHO RESOLVES IT

EOR -- must issue corrected inverts

PERMIT EXPOSURE

EOR letter required at minimum

DOCUMENTATION

RFI, corrected invert table, record drawings

ROUGH COST AND SCHEDULE

Low if caught before construction (a design correction); a real excavation/repipe cost if built as labeled

Moderate (20)

A2 MODERATE High that the conflict exists

C002 vs C017/C018

The general notes require sewer main lines to be "a minimum of 8\" diameter," but the actual sewer distribution sheets specify 6" PVC mains throughout (confirmed independently by Group D's direct sheet reading).

CONSEQUENCE IF BUILT AS DRAWN

The mains contradict the project's own written specification. A reviewing agency (FDEP or the County) may reject the design as submitted, or a contractor could bid the wrong pipe size, creating rework risk.

FIELD CORRECTION

Either revise the C002 note to match the as-designed 6" mains (if FDEP/County accept a documented deviation), or upsize the sewer network to 8" throughout -- ask the EOR which direction is intended before bidding.

WHO RESOLVES IT

EOR clarification (RFI)

PERMIT EXPOSURE

EOR letter required -- a documented deviation from a stated design standard

DOCUMENTATION

RFI, EOR response letter, record drawing note

ROUGH COST AND SCHEDULE

If 8" is actually required: full sewer network resize, six-figure redesign and reconstruction cost, weeks of schedule

A4 MODERATE Medium

C004

No silt fence is shown on the downslope (east) side of the drainfield demolition and Phase 1 grading area, which is the side facing the currently-occupied RV sites. Contours fall from about 111 ft (west) to 103-104 ft (southeast) -- directly toward the existing park.

CONSEQUENCE IF BUILT AS DRAWN

Sediment-laden runoff from drainfield demolition and site grading flows onto occupied RV sites and existing gravel drives during construction, creating NPDES compliance exposure and guest complaint/cleanup cost.

FIELD CORRECTION

Add silt fence along the east edge of the Phase 1/drainfield work area, protecting the existing occupied sites.

WHO RESOLVES IT

GC field decision, or EOR erosion-control revision

PERMIT EXPOSURE

Within approved plans -- a standard erosion-control addition

DOCUMENTATION

Field change order or revised erosion-control sheet, inspection hold point

ROUGH COST AND SCHEDULE

Low -- additional silt fence is a minor line item

A5 MODERATE Medium (aerial photo date is unknown)

C004

The site aerial shows at least 6-8 RVs, trailers, sheds, or awnings sitting inside the Phase 2 clearing limits, under the proposed new layout, with no demolition, relocation, or "to remain" callout for them. Existing septic tank lids are also shown right next to the proposed gravel-driveway work band.

CONSEQUENCE IF BUILT AS DRAWN

The contractor cannot price or schedule relocation of units the drawings don't acknowledge. If these are occupied, long-term guest sites, the owner may not yet have a plan to vacate them, and existing septic tanks risk being crushed under new road construction.

FIELD CORRECTION

Field-verify current occupancy of the Phase 2 footprint before mobilization; add relocation/removal callouts for any unit still in place, and protect or properly abandon the septic tanks shown in the work band.

WHO RESOLVES IT

GC/Owner coordination before mobilization

PERMIT EXPOSURE

Within approved plans, once field-verified

DOCUMENTATION

Site walk documentation, relocation notices if occupied, photo log

ROUGH COST AND SCHEDULE

Depends on occupancy -- could range from zero (already vacated) to real relocation/compensation cost

A6 MODERATE Medium

C003

The main existing asphalt driveway linking the existing park to the pro shop and recreation area is shown removed ("existing pavement to be removed"), along with the existing 3" PVC water main it runs beside (which also branches to serve the recreation area's pavilion and fire pit). No temporary access or water-service maintenance note is shown for this period.

CONSEQUENCE IF BUILT AS DRAWN

Guests and resort operations may lose vehicle access and/or water service to the recreation area during construction, with no documented interim plan.

FIELD CORRECTION

Add a construction-phasing note describing temporary access and a temporary or early-relocated water service to the recreation area before the existing 3" main is abandoned.

WHO RESOLVES IT

GC field decision with EOR sequencing note

PERMIT EXPOSURE

Within approved plans -- a sequencing/phasing addition

DOCUMENTATION

Phasing plan addendum, temporary service work order

ROUGH COST AND SCHEDULE

Modest -- temporary water service and traffic control, a few thousand dollars

A8 MODERATE High

C002, C003, C004

The general notes prohibit construction activity "beyond the CONSTRUCTION BARRIER FENCE," but no such fence is drawn on either erosion-control sheet or shown in the legend. C004 instead uses vague language ("existing trees in this area to be removed as needed") in place of a defined clearing limit.

CONSEQUENCE IF BUILT AS DRAWN

There is no enforceable limit of disturbance next to the wetlands and the occupied park, and tree removal is unquantified for any county tree-ordinance mitigation requirement.

FIELD CORRECTION

Add a drawn, dimensioned construction barrier fence / limit-of-disturbance line matching the note's requirement.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- drafting completeness

DOCUMENTATION

Revised erosion-control sheet

ROUGH COST AND SCHEDULE

Low

A9 MODERATE Medium that the pro shop is the same building as the clubhouse addition (same location, only building there)

C003 vs C013/C016, and the separate clubhouse permits

The existing pro shop/clubhouse building (labeled "One story building (Pro Shop) F.F.E. = 117.54" on C003) carries no demolition callout on this set, so this project does not double-book the separate clubhouse demolition permit (26-0243). However, C013/C016 run new water and sewer service stubs to that same "existing clubhouse building," and the overall layout sheet (C005) shows a "PROPOSED ADDITION" at the same location -- yet neither this civil set nor the separate architectural clubhouse addition set references the other project.

CONSEQUENCE IF BUILT AS DRAWN

The new utility service stubs are designed to a building that is being demolished and rebuilt under a separate permit; the connection points and demand may change once the clubhouse's own design is accounted for. There is also no erosion control shown here for clubhouse demolition if the two projects' schedules overlap.

FIELD CORRECTION

Have the civil engineer (Menadier) and the clubhouse architect (Hammond Design Group) coordinate directly -- confirm the utility stub locations and sizing against the actual clubhouse addition design before either is built.

WHO RESOLVES IT

EOR-to-EOR coordination (both firms)

PERMIT EXPOSURE

Within approved plans once coordinated; could require a plan revision if the stub locations conflict with the actual clubhouse addition footprint

DOCUMENTATION

Coordination memo between the two design teams, RFI if a conflict is found

ROUGH COST AND SCHEDULE

Low if caught now; a field conflict discovered during construction would be a real change-order cost

B5 MODERATE Medium-High

C006, C007, C008

Several road-junction inside turning radii are tight for RV traffic: R8' and R10' at the C006 north entrance, R10' at many C007/C008 junctions, and some acute (~60°) junctions near the bathhouse and the 2-72 to 2-127 block. No turning-movement exhibit (e.g. AutoTURN for a 45-ft Class A motorhome or a 40-ft fifth-wheel-plus-truck) is provided.

CONSEQUENCE IF BUILT AS DRAWN

Large rigs will track over curb returns, islands, and lot corners, causing edge-of-pavement breakup, rutting into lots, and damaged utility pedestals -- an ongoing repair cost.

FIELD CORRECTION

Run an AutoTURN (or equivalent) turning analysis for the largest expected RV/tow-vehicle combination at every tight junction and widen radii where the analysis shows overtracking.

WHO RESOLVES IT

EOR design verification

PERMIT EXPOSURE

Within approved plans if radii are increased before construction; EOR letter if adjusted after

DOCUMENTATION

Turning-movement study, revised layout at flagged junctions

ROUGH COST AND SCHEDULE

Moderate -- localized paving/curb radius increases at specific junctions

B6 MODERATE Medium

C008 (near sites 2-17, 2-21), C005

Existing 2-inch force mains run through the new Phase 2 south block, under or near the island and lot/pad corners of sites 2-17 and 2-21, with no relocation or protection note on the layout sheets.

CONSEQUENCE IF BUILT AS DRAWN

New pads and road subgrade may be constructed directly over pressurized existing sewer lines, creating a real risk of damage during construction and access problems for any future repair.

FIELD CORRECTION

Confirm with the sewer distribution sheets (C016-C018) whether these lines are abandoned or rerouted before this area is graded; if they remain active, add a protection or relocation note to the layout sheet.

WHO RESOLVES IT

EOR cross-sheet coordination

PERMIT EXPOSURE

Within approved plans once coordinated

DOCUMENTATION

Cross-sheet coordination note, utility protection detail if lines remain active

ROUGH COST AND SCHEDULE

Low if already addressed elsewhere; real repair/access cost if not

C5 MODERATE High on the geometry; Medium on whether it is actually permitted

C009, C012

Water Retention Area 1 is designed with 2:1 side slopes (versus 4:1 for WRA-2 and WRA-3), and it is also the deepest of the three ponds (5.1 ft), with a perimeter fence at the top of bank.

CONSEQUENCE IF BUILT AS DRAWN

2:1 sodded slopes cannot be safely mowed with ordinary equipment, erode more readily, and are harder to inspect -- adding real cost to both routine O&M and the state-mandated 5-year professional pond inspection (Rule 62-330.311).

FIELD CORRECTION

Confirm the 2:1 slope is actually permitted under the SRWMD ERP and county stormwater criteria for this size/depth pond; consider flattening to match the other two ponds if maintenance cost is a concern.

WHO RESOLVES IT

EOR/SRWMD confirmation

PERMIT EXPOSURE

Within approved plans if already permitted at 2:1; agency notification if a permit deviation is needed to flatten it

DOCUMENTATION

SRWMD confirmation, possible pond regrading if flattened

ROUGH COST AND SCHEDULE

Low if confirmed acceptable as-is; moderate regrading cost if it must be flattened

C9 MODERATE High

C010

The bathhouse finished floor (98.20 ft) sits at or below the adjacent road edges (98.20-98.36 ft) and only 1.2 ft above the nearest storm inlet (S-17, rim 97.00 ft), with essentially no freeboard above surrounding grade separate from the pond-backup issue in C1.

CONSEQUENCE IF BUILT AS DRAWN

Water can run directly into the building in a heavy rainfall event. Standard practice sets finished floor at least 0.5-1 ft above adjacent grade or road crown.

FIELD CORRECTION

Raise the bathhouse finished floor elevation, or regrade immediately around it to provide positive drainage away from the building.

WHO RESOLVES IT

EOR design correction

PERMIT EXPOSURE

Within approved plans if corrected before construction; EOR letter if corrected after

DOCUMENTATION

Revised architectural/grading coordination, corrected floor elevation

ROUGH COST AND SCHEDULE

Low if caught now (a plan revision); real water-intrusion repair cost if built as-is and it floods

D10 MODERATE

High on the depths (directly legible); groundwater level itself is not directly confirmed in these structures' immediate vicinity

C016, C017

Several sewer structures are unusually deep, near wetlands: SSMH 9 is about 15.9 ft deep, SSMH 8 about 14 ft, SSMH 14 about 11.7 ft, SSMH 20 about 11.6 ft, and the lift station's inlet sits about 14.9 ft below its rim, near Wetland A and the retention areas.

CONSEQUENCE IF BUILT AS DRAWN

Deep excavation this close to wetlands and (per Group E's soil borings) a shallow water table means high dewatering and excavation cost, plus a real buoyancy/flotation concern for the deep structures (see the lift station's own buoyancy question, item E3).

FIELD CORRECTION

Confirm dewatering permit requirements (SRWMD) and buoyancy design for these deep structures before construction, and budget for the dewatering cost this depth implies.

WHO RESOLVES IT

EOR/geotechnical confirmation

PERMIT EXPOSURE

Agency notification -- SRWMD dewatering permit required per the general notes

DOCUMENTATION

Dewatering permit, buoyancy calculations for deep structures

ROUGH COST AND SCHEDULE

Real cost item -- deep excavation with dewatering in this soil/water-table condition is meaningfully more expensive than shallow trenching

D16 MODERATE Medium-High

C013-C018

Water and sewer mains (2" water, 6" sewer) are routed directly under the concrete RV pads at multiple locations (e.g., sites 1-24 to 1-28, 1-79 to 1-88) rather than in the roadway or an open utility strip.

CONSEQUENCE IF BUILT AS DRAWN

Any future repair to these lines means breaking out the concrete pad and displacing the guest on that site -- an ongoing operating cost and guest-relations issue for the life of the resort, not a one-time construction cost.

FIELD CORRECTION

Where feasible, relocate main runs to roadways or dedicated utility strips rather than under pads, particularly for lines likely to need future maintenance access.

WHO RESOLVES IT

EOR design revision (where feasible)

PERMIT EXPOSURE

Within approved plans -- a design choice, not a compliance requirement

DOCUMENTATION

Revised utility alignment where pad relocation is feasible

ROUGH COST AND SCHEDULE

Depends on how much re-routing is feasible at this design stage; likely modest if addressed now, real ongoing cost over the resort's life if not

D6 MODERATE High that the sheets conflict; confirmed directly from the tables on each sheet

C016 vs C017 vs C018 (multiple structures)

Several manholes show different invert elevations on different sheets for the same structure. SSMH 9: C016 lists NE 82.84/NW 85.80; C017 lists NE 82.84/S 86.16/NW 82.80 (if 85.80 were the true outlet, it would sit 3 ft above the inlet -- C017's 82.80 value is the one that correctly closes to the lift station). SSMH 12: C016 lists SW 93.00/SE 90.90/NW 92.90; C017 lists SW 88.10/SE 90.90/NW 90.80 for the same structure. SSMH 20: C017 lists W 86.50/NE 86.40; C018 lists W 84.00/NE 85.90. The lift station itself: C017 lists W 83.76/SE 82.65; C018 lists W 83.76/E 85.66 for what should be the same two inlets.

CONSEQUENCE IF BUILT AS DRAWN

A contractor could build to the wrong sheet's values for any of these structures. Several of the conflicting values would produce a reverse (uphill) slope if used, generating RFIs and real rework risk during construction.

FIELD CORRECTION

Have the engineer issue one reconciled, correct invert table for SSMH 9, SSMH 12, SSMH 20, and the lift station, superseding whichever values are wrong on each sheet.

WHO RESOLVES IT

EOR -- must issue a reconciled invert table

PERMIT EXPOSURE

EOR letter required

DOCUMENTATION

RFI, corrected invert table applied to all affected sheets, record drawings

ROUGH COST AND SCHEDULE

Low if caught in design review; real excavation rework if discovered in the field

D7 MODERATE High

C016-C018

Several manhole connections show an inlet more than 2 ft above the outlet with no "DROP" callout, unlike SSMH 21 which is properly labeled as a drop manhole: SSMH 9 (inlet 86.16 vs. outlet 82.80, a 3.4 ft drop), SSMH 5 (96.00 vs. 92.50, 3.5 ft), SSMH 14 (90.80 vs. 87.30, 3.5 ft), and SSMH 2 (98.00 vs. 95.60, 2.4 ft). The Ten States Standards (incorporated by FDEP 62-604) call for a drop connection detail at any inlet-outlet difference of 24 inches or more.

CONSEQUENCE IF BUILT AS DRAWN

Without the drop structure, turbulence and hydrogen-sulfide gas release accelerate corrosion of the manhole walls and benches at these locations -- a real maintenance burden and eventual structural repair cost.

FIELD CORRECTION

Add drop-manhole callouts and details at each of these locations, matching the convention already used at SSMH 21.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- a detailing addition, not a design change

DOCUMENTATION

Revised manhole callouts, drop-connection detail added to D001

ROUGH COST AND SCHEDULE

Modest -- a standard detail applied at 4 additional locations

D8 MODERATE High

C017, C018 vs C002

Sewer lateral slopes are called out below the general notes' own stated minimums: "4\" PVC SEWER LATERAL @ 1.00% MIN" (C017, C018) versus the note's required 4"@2.00%; "6\" PVC SEWER LATERAL @ 1.00% MIN" (C017) versus the note's required 6"@1.04%. One 245 LF run of 6" main is also shown at 1.00%.

CONSEQUENCE IF BUILT AS DRAWN

Laterals installed at these minimums will run flatter than specified, increasing the risk of solids deposition, site-level backups, and maintenance calls -- and they conflict with the project's own written specification.

FIELD CORRECTION

Correct the lateral slope callouts to match the general notes' minimums, or document why a lower slope is acceptable for this pipe diameter and get it confirmed against FDEP standards.

WHO RESOLVES IT

EOR drafting/design correction

PERMIT EXPOSURE

EOR letter required for the deviation from the stated minimum

DOCUMENTATION

RFI, corrected lateral slope callouts

ROUGH COST AND SCHEDULE

Low if corrected before construction

D9 MODERATE High

C016

The 6" sewer main between manholes SSMH 3 and SSMH 4 changes direction at a 4" cleanout (near site 1-55) instead of a manhole -- a 407 LF run (185+222 LF) between manholes with a directional change in the middle. Other 6" mains similarly terminate upstream in 4" cleanouts rather than manholes (e.g., 45 LF of 6" main ending at a 4" cleanout near site 1-68). The Ten States Standards call for a manhole at every change of direction and a maximum manhole spacing of 400 ft.

CONSEQUENCE IF BUILT AS DRAWN

A 6" main cannot be jetted or TV-inspected through a 4" cleanout, creating an ongoing maintenance access problem, and the 407 ft manhole spacing already exceeds the 400 ft standard.

FIELD CORRECTION

Replace the mid-run cleanout with a proper manhole at the direction change, and verify manhole spacing throughout the system against the 400 ft standard.

WHO RESOLVES IT

EOR design correction

PERMIT EXPOSURE

EOR letter required, likely also a permit review comment

DOCUMENTATION

Revised sewer layout adding manholes where needed

ROUGH COST AND SCHEDULE

Moderate -- additional manhole structures at specific locations

E2 MODERATE

High that no such detail exists on D001; not checked whether it might exist on a sheet outside this review group

D001

The concrete overflow weir / pond control structure for all three retention ponds, and the aluminum skimmers called out on C012, are each labeled "(SEE DETAIL)" on the grading and profile sheets, but no such detail actually appears on D001 -- an independent confirmation of Group C's same finding (C10) from the details-sheet side.

CONSEQUENCE IF BUILT AS DRAWN

The control structures cannot be built or inspected as intended without a dimensioned detail, and the Water Management District permit reviewer will typically require one.

FIELD CORRECTION

Add the missing weir/skimmer detail to D001.

WHO RESOLVES IT

EOR -- a real missing-detail gap

PERMIT EXPOSURE

Agency notification likely -- SRWMD typically requires this detail as part of permit review

DOCUMENTATION

Added detail content, SRWMD resubmittal if already submitted without it

ROUGH COST AND SCHEDULE

Low direct cost; a real permitting-timeline dependency

E4 MODERATE Medium

D001

General structural/site details are largely present (typical pavement section, concrete sidewalk, sinkhole-repair instructions, mitered end section, silt fence, stabilized construction entrance, concrete washout, inlet protection, tree protection, chain-link fence), but the typical pavement section calls out no prime or tack coat between the limerock base and the asphalt surface course.

CONSEQUENCE IF BUILT AS DRAWN

Missing a prime/tack coat callout can lead to poor bond between pavement layers and premature delamination/potholing, especially given the loose sandy soils documented in the nearby soil borings.

FIELD CORRECTION

Add a prime coat (base) and tack coat (between asphalt lifts, if more than one) callout to the typical pavement section detail.

WHO RESOLVES IT

EOR drafting/design addition

PERMIT EXPOSURE

Within approved plans -- a standard detail completeness item

DOCUMENTATION

Revised pavement section detail

ROUGH COST AND SCHEDULE

Negligible added cost if specified from the start

E8 MODERATE High

D003

The lift station's emergency contact sign detail names "FGUA" (Florida Governmental Utility Authority) as the emergency contact, with the phone number still shown as an unfilled placeholder ("(XXX) XXX-XXXX"). This appears to be leftover vendor-template content from Barney's Pumps, since FGUA is a public utility authority and this is a private system.

CONSEQUENCE IF BUILT AS DRAWN

A real emergency (an overflow event) would route a caller to the wrong organization entirely, and the phone number on the sign doesn't even work as printed.

FIELD CORRECTION

Replace the emergency sign's contact information with the actual responsible party (Jennings Realty / the resort's own operations contact) and a real, working phone number.

WHO RESOLVES IT

GC/Owner at final inspection, or EOR drafting correction now

PERMIT EXPOSURE

Within approved plans -- a labeling correction

DOCUMENTATION

Corrected emergency sign, verified before final inspection/occupancy

ROUGH COST AND SCHEDULE

Negligible

E9 MODERATE Medium

D003

Standby power for the lift station is limited to a generator receptacle with an interlock -- there is no permanent on-site generator, no automatic transfer switch, and no stated emergency operating plan beyond a bypass-pumping connection. FDEP Rule 62-604 requires documented emergency operation provisions for a system of this kind.

CONSEQUENCE IF BUILT AS DRAWN

A power outage at this single lift station (which will serve roughly 306 sites once the existing park is folded in) has no automatic backup -- someone has to physically bring in and connect a portable generator, or arrange pump-and-haul, before an overflow occurs.

FIELD CORRECTION

Develop and document a real emergency response plan for this lift station -- either a permanent standby generator with automatic transfer switch, or a written, resourced pump-and-haul/portable-generator protocol with response-time commitments.

WHO RESOLVES IT

Owner/Operations, with EOR input on FDEP compliance

PERMIT EXPOSURE

Agency notification -- FDEP 62-604 requires this to be documented as part of the permit

DOCUMENTATION

Emergency operations plan document, possible generator/ATS installation

ROUGH COST AND SCHEDULE

A permanent generator + ATS installation is a real, meaningful cost; a documented pump-and-haul plan is comparatively low-cost but less resilient

Potential (28)

A10 POTENTIAL High that the three numbers genuinely differ

C001 vs C005 vs C008

Three different total site counts appear across the plan set: the cover sheet's traffic statement uses 223 "occupied campsites," the overall layout sheet's own count box says 217 (88 Phase 1 + 129 Phase 2), and the site labels actually drawn total 215 (88 + 127) -- matching the confirmed program. (See B's independent confirmation, item C-1.)

CONSEQUENCE IF BUILT AS DRAWN

Traffic generation, impervious-area totals, and wastewater flow calculations may be based on the wrong count, and the count also affects which septic/sewer permitting threshold applies (see D1).

FIELD CORRECTION

Confirm the correct total with the engineer and correct the cover-sheet traffic statement and the C005 count box to match the as-drawn 215 sites, or explain where the additional 2-8 sites are.

WHO RESOLVES IT

EOR clarification

PERMIT EXPOSURE

EOR letter required if any agency-facing calculation used the wrong count

DOCUMENTATION

RFI, corrected cover sheet and count box

ROUGH COST AND SCHEDULE

Low -- a correction, not a redesign, unless it reveals missing sites that need to be added

A3 POTENTIAL Medium (scaled measurement, ±5 ft)

C003, C004 (Wetland B/C/D south and west edges)

Clearing limits and silt fence scale to roughly 5-20 ft from the Wetland B/C/D flag line on its south and west sides, inside the 30-ft buffer the plan itself labels only on the northeast side. Several trees marked for removal sit on or just outside that fence line.

CONSEQUENCE IF BUILT AS DRAWN

If the 30-ft buffer is an SRWMD permit condition, clearing inside it is a permit violation -- this would escalate to Critical if confirmed. At minimum it is a self-contradiction on the drawings.

FIELD CORRECTION

Confirm the buffer requirement with SRWMD and either pull the silt fence/clearing limit back to a consistent 30 ft on all sides, or document why the buffer varies by side.

WHO RESOLVES IT

EOR / SRWMD coordination

PERMIT EXPOSURE

Agency notification unavoidable if the buffer is a permit condition

DOCUMENTATION

RFI to EOR, SRWMD confirmation, revised erosion-control sheet if needed

ROUGH COST AND SCHEDULE

Minor if just a drafting fix; a real wetland-impact violation could mean a stop-work and mitigation cost

A7 POTENTIAL Medium

C003

A construction entrance ties into the existing asphalt driveway right where it crosses the railroad (Florida Gulf & Atlantic / Pinsly-line) right-of-way at grade. No railroad crossing agreement, flagging plan, or permit reference is shown for construction traffic use of this at-grade crossing.

CONSEQUENCE IF BUILT AS DRAWN

Heavy construction traffic on a private at-grade rail crossing may require a formal crossing agreement or flagging protocol from the railroad; using it without one is a real liability and schedule risk.

FIELD CORRECTION

Confirm the crossing's legal status and obtain a written crossing/flagging agreement with the railroad (confirm the actual current owner -- see A9 -- before contacting them) before using it for construction traffic.

WHO RESOLVES IT

Owner/GC coordination with the railroad

PERMIT EXPOSURE

Agency (railroad) notification likely required

DOCUMENTATION

Crossing agreement, flagging plan, insurance/indemnification as required by the railroad

ROUGH COST AND SCHEDULE

Low direct cost, but a real schedule dependency if the railroad requires lead time to approve

B1 POTENTIAL Medium (distances are scaled, ±2-3 ft)

C006 (pads 1-68, 1-69, 1-70), C005

Phase 1 pads 1-68 through 1-70 sit close to the parcel line / railroad south right-of-way, with no rail setback or perimeter buffer drawn. Scaled nearest-corner distances: 1-68 about 7-14 ft, 1-69 about 6-12 ft, 1-70 about 22-25 ft from the parcel/R.O.W. line. The cover sheet itself lists building setbacks of 25 ft front / 20 ft corner / 20 ft rear / 10 ft side.

CONSEQUENCE IF BUILT AS DRAWN

If Madison County applies the stated rear (20 ft) or side (10 ft) setback -- or any separate RV-park perimeter buffer -- to these pads, 1-68 and possibly 1-69/1-70 would be non-compliant and need to be moved or dropped, costing 2-3 sites.

FIELD CORRECTION

Confirm with Madison County Planning whether the stated setbacks (or a separate rail setback) apply to RV pads at this location; relocate or remove non-compliant pads if so.

WHO RESOLVES IT

County P&Z confirmation, then EOR redesign of the affected pads if needed

PERMIT EXPOSURE

Agency notification unavoidable if setback non-compliance is confirmed

DOCUMENTATION

County written confirmation, revised layout sheet if pads must move

ROUGH COST AND SCHEDULE

Loss of 2-3 revenue-generating sites if pads must be dropped; modest redesign cost if they can be relocated

B2 **POTENTIAL** Medium (survey text partly illegible; ownership history is outside knowledge, not confirmed on the drawings)

C005

The layout sheet labels the railroad "CSX RAILROAD," but the survey text at the at-grade crossing appears to read approximately "...SLEY RR COMPANY (F... GULF ATLANTIC RAILROAD)" -- consistent with CSX having sold this Jacksonville-Pensacola line segment to Florida Gulf & Atlantic Railroad (a Pinsly Railroad Company line) around 2019.

CONSEQUENCE IF BUILT AS DRAWN

If crossing, encroachment, or setback permits are routed to CSX instead of the actual current owner, they would go to the wrong party, delaying approval.

FIELD CORRECTION

Confirm the current owner of this rail line directly with Madison County or the FRA before submitting any crossing/encroachment permit application.

WHO RESOLVES IT

Owner/EOR verification

PERMIT EXPOSURE

Agency notification -- confirm correct railroad entity before applying

DOCUMENTATION

Written confirmation of current rail line ownership

ROUGH COST AND SCHEDULE

Low direct cost, but a real schedule risk if permits are misrouted

B3 **POTENTIAL** Medium (the crossing is clearly drawn; its permitted status is not shown)

C005 (off-site force main ~Sta. 2+00), C006 (north entrance)

The off-site sewer force main is shown crossing the railroad at approximately station 2+00, with no casing pipe, crossing detail, or railroad permit reference on these sheets. The Phase 1 north entrance also ties into an existing asphalt driveway that appears to cross the tracks at grade (a gated crossing, "METAL GATE POST (TYP. FOR 3)").

CONSEQUENCE IF BUILT AS DRAWN

A utility crossing under an active rail line typically requires a railroad license and usually a casing pipe; without them, the utility work cannot proceed. The at-grade driveway crossing would also need its permitted status reviewed once it carries RV-park traffic (39-44 peak-hour trips per the cover sheet).

FIELD CORRECTION

Obtain the railroad utility-crossing permit (with casing pipe design if required) before this crossing is built, and confirm the driveway crossing's permitted status for the new traffic volume.

WHO RESOLVES IT

EOR design (casing) plus Owner/railroad permitting

PERMIT EXPOSURE

Agency notification unavoidable

DOCUMENTATION

Railroad crossing permit application, casing pipe design if required

ROUGH COST AND SCHEDULE

Real cost and schedule item -- railroad permitting timelines can run months

B4 **POTENTIAL** Medium (widths are clearly dimensioned; whether the 20-ft rule applies here is not confirmed)

C005-C008 (all internal roads)

Most internal roads are drawn as 16-ft one-way lanes. No fire lane callout, fire apparatus access route, apparatus turning template, or fire department turnaround is shown on any layout sheet. Florida Fire Prevention Code / NFPA 1 §18.2.3.4.1.1 generally requires 20 ft of clear fire-apparatus access unless the fire marshal approves a lesser width. The 22-ft two-way segments do meet that width; the 16-ft one-way lanes do not.

CONSEQUENCE IF BUILT AS DRAWN

If the Madison County Fire Marshal designates these roads as required fire access (plausible given the bathhouse building and the density of RV sites), the 16-ft lanes would need widening -- a large paving and grading redesign affecting site layout, lot counts, and the already-designed grading/drainage plan.

FIELD CORRECTION

Get the Fire Marshal's written determination on required fire-apparatus access width for this layout before finalizing paving widths; if 20 ft is required, redesign the affected road segments now rather than after permitting.

WHO RESOLVES IT

Fire Marshal determination, then EOR redesign if required

PERMIT EXPOSURE

Agency notification unavoidable

DOCUMENTATION

Fire Marshal written determination, revised layout/grading if roads must widen

ROUGH COST AND SCHEDULE

Potentially large -- a road-width increase cascades into grading, drainage, and site-layout redesign

B7 **POTENTIAL** Low-Medium (label is faint, could be a leftover survey artifact)

C007

A ghosted wetland label ("WETLAND B/C/D, SEE DETAIL ON SHEET 9") and linework appear to fall inside the footprint of Water Detention/Retention Area 3.

CONSEQUENCE IF BUILT AS DRAWN

If this represents a real, remaining wetland polygon being excavated for the pond, it would be an unpermitted wetland impact unless specifically authorized by SRWMD.

FIELD CORRECTION

Confirm with the engineer/surveyor whether this label represents an actual wetland boundary conflicting with the pond footprint, or a leftover survey reference.

WHO RESOLVES IT

EOR/surveyor clarification

PERMIT EXPOSURE

Agency notification unavoidable if a real wetland impact is confirmed

DOCUMENTATION

RFI, SRWMD confirmation if needed

ROUGH COST AND SCHEDULE

Could be significant if a real, unpermitted wetland impact is confirmed

B9 POTENTIAL High that the schedule is missing; the area-reconciliation math itself is approximate

C005-C008

No schedule anywhere in the set states which sites are pull-through vs. back-in configuration -- the type can only be determined by visually inspecting each pad. The cover sheet's concrete total (171,460 SF) implies a mix of the two pad types across 215 sites, but the mix is never stated, and the asphalt "RV PADS" total (105,576 SF, about 491 SF/site) does not reconcile cleanly with the 20'x20' (400 SF) apron shown in the typical-lot detail.

CONSEQUENCE IF BUILT AS DRAWN

Without a stated pull-through/back-in count, cost estimating and rate-setting (pull-throughs typically rent at a premium) cannot be done exactly, and the site-count cross-check (A10/C-1) cannot be independently verified this way either.

FIELD CORRECTION

Have the engineer provide a pull-through/back-in schedule by site number.

WHO RESOLVES IT

EOR documentation addition

PERMIT EXPOSURE

Within approved plans -- documentation only

DOCUMENTATION

Site-type schedule

ROUGH COST AND SCHEDULE

None -- a documentation request, not a redesign

C10 POTENTIAL Medium -- a detail could exist as pure graphics the OCR text-check missed; not fully ruled out

C009, C010, C012

All three pond overflow weirs (and their aluminum skimmers) are called out only as "SEE DETAIL" on the grading and profile sheets, but no weir or skimmer detail actually exists anywhere in the 21-sheet set (the only drainage detail present, on D001, is for a mitered end section, a different structure entirely). Where each weir's overflow actually discharges downstream is also not shown: WRA-1 appears to overflow north toward the existing road/force-main corridor, and WRA-2/WRA-3 appear to overflow toward the wetlands.

CONSEQUENCE IF BUILT AS DRAWN

The contractor cannot build the pond control structures without a dimensioned detail, and the permitting reviewer cannot verify the weir's hydraulic rating or confirm the downstream discharge path is acceptable.

FIELD CORRECTION

Add a dimensioned weir/skimmer detail to the details sheet (D001), and show or describe each pond's downstream discharge path.

WHO RESOLVES IT

EOR -- a real missing detail, not just a documentation gap

PERMIT EXPOSURE

Agency notification likely -- SRWMD typically requires the control-structure detail as part of permit review

DOCUMENTATION

New detail sheet content, SRWMD resubmittal if already submitted without it

ROUGH COST AND SCHEDULE

Low direct cost (an engineering deliverable), but a real permitting-timeline dependency

C11 POTENTIAL Medium

C012

WRA-1 uses a 100-year/24-hour design storm while WRA-2 and WRA-3 use a 100-year/4-hour design storm for their critical stage calculations, with no stage-storage table, pre/post peak-flow comparison, or other supporting drainage calculation shown anywhere in the 21 sheets to justify the different storm durations.

CONSEQUENCE IF BUILT AS DRAWN

These may be legitimately different critical durations for different drainage basins, but nothing on the drawings themselves can confirm or refute that.

FIELD CORRECTION

Request the engineer's full stormwater report / SRWMD permit calculations to confirm the storm-duration basis for each pond.

WHO RESOLVES IT

EOR documentation request

PERMIT EXPOSURE

Within approved plans, pending confirmation the calculations support this

DOCUMENTATION

Stormwater report / SRWMD calculation package

ROUGH COST AND SCHEDULE

None -- a documentation request

C15 POTENTIAL Medium-High that fill is needed; the actual quantity is genuinely unknown from these sheets

C009, C010, C011

No earthwork (cut/fill) quantities or a site-wide balance are shown anywhere in the grading sheets. The pond soil-profile columns on C012 show all three ponds built up with new fill (3.6 ft at WRA-1, 2.5 ft at WRA-2, 2.4 ft at WRA-3, since WRA-2/3 bottoms sit essentially at existing grade and yield almost no excavated spoil), and several Phase 2 pads are shown raised above existing ground. The general notes leave "off-site borrow" quantity estimation entirely to the contractor.

CONSEQUENCE IF BUILT AS DRAWN

The site almost certainly requires net imported fill in an unknown quantity -- a real, currently unbudgeted cost item, and one that will vary meaningfully between contractor bids without a stated earthwork balance to bid against.

FIELD CORRECTION

Have the engineer calculate and state a site-wide cut/fill balance and an estimated import-fill quantity before this goes out to bid.

WHO RESOLVES IT

EOR -- earthwork calculation

PERMIT EXPOSURE

Within approved plans -- a design deliverable, not an agency approval item

DOCUMENTATION

Earthwork balance calculation added to the plan set or bid documents

ROUGH COST AND SCHEDULE

Potentially a significant line item once quantified -- import fill for a site this size could run well into six figures

C3 POTENTIAL

High on the elevations; Medium on the real-world consequence, which depends on the confining layer's true properties

C010, C012, D002 (soil borings)

Retention pond sumps and several feeder pipe inverts sit below the seasonal high groundwater table AND through the "confining layer" the pond design itself relies on. WRA-2: sumps at 92.50-93.00 ft are below the 94.50 ft water table and the 93.50 ft confining layer. WRA-3: a sump at 92.00 ft is 3.5 ft below the 95.50 ft water table and 3 ft below the 95.00 ft confining layer. Feeder pipes S-19, S-26, S-13, and S-14 are also below the water table. The plan set separately includes a sinkhole-repair detail specifically for "the water retention basins," confirming the design team already recognizes karst risk here.

CONSEQUENCE IF BUILT AS DRAWN

These sumps will hold standing groundwater year-round (a permanent wet spot, mosquito breeding, and an eyesore in what's designed as a dry retention pond), pipe joints below the water table need watertight/filter-wrapped joints, and digging through a confining clay layer under a stormwater pond in known karst terrain raises real sinkhole and groundwater-contamination risk that SRWMD may challenge.

FIELD CORRECTION

Have the geotechnical engineer confirm the confining layer's actual integrity at these specific depths, and either redesign the sumps to stay above it or add a documented, SRWMD-accepted mitigation (liner, deeper confining verification) before construction.

WHO RESOLVES IT

Geotechnical engineer + EOR

PERMIT EXPOSURE

Agency notification likely -- SRWMD may require additional geotechnical justification

DOCUMENTATION

Geotechnical addendum, possible SRWMD resubmittal

ROUGH COST AND SCHEDULE

Could range from minor (geotechnical confirmation only) to significant (pond redesign) depending on findings

C4 POTENTIAL Medium

C012, D002 (soil borings)

The soil-infiltration (soak-away) rates assumed for WRA-2 and WRA-3 in the pond design (Kh 11.50 ft/day and 4.80 ft/day respectively) look optimistic against the actual soil borings in those pond footprints: boring B-4 (inside WRA-2) is clayey sand from the surface to 13.5 ft then clay below; boring B-6 (WRA-3) hits clayey sand starting at 1.5 ft. WRA-3 also has only 0.5 ft of separation between its design water table (95.50 ft) and its confining layer (95.00 ft). No borings encountered actual groundwater (drilled in the dry season, April), so the water-table elevations are estimates, and no pond-recovery calculation is shown anywhere in the set.

CONSEQUENCE IF BUILT AS DRAWN

If the ponds don't recover (drain down) within the time window SRWMD requires between storms, repeat storms stack up on top of each other -- compounding the road-flooding issue in C1 into longer-duration standing water and an ongoing maintenance problem.

FIELD CORRECTION

Obtain a real infiltration/recovery calculation from the geotechnical engineer using the actual measured soil types in B-4 and B-6, not the assumed Kh values, before relying on this pond design.

WHO RESOLVES IT

Geotechnical engineer

PERMIT EXPOSURE

Agency notification likely if the recovery calculation doesn't meet SRWMD criteria

DOCUMENTATION

Geotechnical recovery calculation, possible pond redesign or SRWMD resubmittal

ROUGH COST AND SCHEDULE

Moderate -- a geotechnical study plus possible pond capacity increase

C7 POTENTIAL Medium

C011

The northwest Phase 2 lot cluster (sites 2-1 to 2-16, road grades 110.50 down to 104.40 near the dog park) has no storm inlets, pipes, or swales draining to any of the three retention areas anywhere on these sheets. Runoff instead flows to a "match existing pavement" tie-in point and over the rear lot line toward existing, currently-occupied RV sites and the existing dry retention area. A "DRAINFIELD AREA" label also sits directly in this flow path across lots 2-12/2-13.

CONSEQUENCE IF BUILT AS DRAWN

New impervious runoff from this cluster may reach existing developed sites without being captured or counted in the WRA-1/2/3 stormwater calculations at all, risking nuisance flooding of existing sites and a real gap in the permitted stormwater volume calculation.

FIELD CORRECTION

Add a drainage collection system (inlets, pipe, or swale) for this cluster tying into one of the three retention areas, and revise the stormwater calculations to include its contributing area.

WHO RESOLVES IT

EOR -- a real design gap to close

PERMIT EXPOSURE

Agency notification likely -- this area appears to be missing from the permitted stormwater volume

DOCUMENTATION

Revised grading/drainage sheet for this area, updated SRWMD stormwater calculation

ROUGH COST AND SCHEDULE

Moderate -- new pipe/inlet infrastructure plus a stormwater recalculation

C8 POTENTIAL Medium

C009

The rear-yard swale behind lots 1-79 through 1-81 (east side of Phase 1) has a trapped low point at elevation 104.8 ft with no storm inlet, before the grade rises again to the north and south. The adjacent existing asphalt driveway sits at 104-105 ft, above existing ground that drops to about 99-100 ft at the Wetland E line.

CONSEQUENCE IF BUILT AS DRAWN

Water either ponds permanently in the rear yards of lots 1-79 to 1-81, or spills over the existing driveway directly into Wetland E without ever passing through a permitted retention area.

FIELD CORRECTION

Add a storm inlet at the low point (elev. ~104.8) tying into the nearest retention system, or regrade the swale to eliminate the trapped low point.

WHO RESOLVES IT

EOR design addition

PERMIT EXPOSURE

Agency notification likely if this represents an unpermitted direct discharge to Wetland E

DOCUMENTATION

Revised grading sheet, possible SRWMD coordination

ROUGH COST AND SCHEDULE

Modest -- one additional inlet and pipe run, unless a direct wetland discharge requires broader mitigation

D11 POTENTIAL Medium

C016-C018

This single lift station now serves approximately 306 total sites (215 new plus the 91 existing sites routed in via SSMH 21), making it a single point of failure for wastewater service to the entire resort. The discharge force main size is not labeled on these sheets, and the design flow, peaking factor, pump data, emergency storage, and backup power (reviewed separately on D003, see Group E's findings) could not be independently verified from the water/sewer distribution sheets alone.

CONSEQUENCE IF BUILT AS DRAWN

Undersizing or a lack of redundancy at this single station means a pump failure or power outage could cause a sewage overflow affecting the entire resort, not just the new expansion.

FIELD CORRECTION

Verify the D003 pump station design (see Group E) against the full expected average and peak flow for all ~306 sites, and confirm an adequate emergency response plan (backup power, storage, or pump-and-haul readiness) is in place.

WHO RESOLVES IT

EOR verification against D003

PERMIT EXPOSURE

Agency notification -- FDEP 62-604 requires documented emergency provisions for a collection system this size

DOCUMENTATION

Design verification memo, emergency operations plan

ROUGH COST AND SCHEDULE

Could range from none (if D003's design is confirmed adequate) to significant (generator/redundancy addition) once verified

D12 POTENTIAL Medium

C018

The conversion of the existing 91-site park onto the new system is underspecified: existing septic tanks are abandoned and the existing 2" force mains are rerouted through wye connections into new 2" force mains feeding a 10-ft drop at manhole SSMH 21, but no source pumps (grinder or STEP units) are shown for the existing sites once their own septic tanks are removed, and no pressure/velocity calculation is given for 2" force mains feeding that drop connection.

CONSEQUENCE IF BUILT AS DRAWN

Existing, currently-occupied sites could lose sewer service during or immediately after this cutover if the actual pumping mechanism and hydraulics for their converted service aren't properly designed -- and the cost/scope of that conversion isn't currently captured anywhere in the plan set.

FIELD CORRECTION

Get the engineer to specify the actual pumping mechanism for the converted existing sites and confirm the 2" force main hydraulics at the SSMH 21 drop connection, with a construction sequencing plan that avoids a service gap for existing guests.

WHO RESOLVES IT

EOR -- a real design gap for the existing-site conversion

PERMIT EXPOSURE

Agency notification likely, since this affects an operating facility's permitted service

DOCUMENTATION

Design addendum for the existing-site conversion, sequencing plan

ROUGH COST AND SCHEDULE

Real, currently unbudgeted scope -- the existing-park conversion is its own meaningful cost item

D13 POTENTIAL Medium

C013-C015

Fire hydrant coverage looks thin for the site: only about 5 hydrants serve roughly 215 new sites across 20+ acres, with none along the Phase 1 east arc (sites 1-68 to 1-88) or the center block (sites 1-37 to 1-67). By scaled distance, several sites sit roughly 600-900 ft from the nearest hydrant. No fire-flow test or hydraulic calculation is shown for the 6" network, which is fed through a single 6" meter and double-detector-check assembly.

CONSEQUENCE IF BUILT AS DRAWN

This coverage and lack of a fire-flow calculation could fail the Fire Marshal's review, especially combined with the fire-lane width concern already flagged (B4).

FIELD CORRECTION

Obtain a fire-flow test/hydraulic analysis for the proposed 6" network and add hydrants to close the coverage gaps, particularly along the Phase 1 east arc and center block.

WHO RESOLVES IT

EOR fire-flow analysis, Fire Marshal review

PERMIT EXPOSURE

Agency notification unavoidable

DOCUMENTATION

Fire-flow calculation, revised hydrant layout, Fire Marshal sign-off

ROUGH COST AND SCHEDULE

Moderate -- additional hydrants and possibly a larger main if the fire-flow calculation shows a deficiency

D14 POTENTIAL Medium

C013-C015

Several 2" SDR-21 water main branches are long dead-ends (e.g., about 16 sites plus a hose bibb on the northwest-finger 2" main) terminating in a plain 90° elbow, with only a temporary blow-off called for per the general notes -- no permanent flushing provision is shown at these dead ends.

CONSEQUENCE IF BUILT AS DRAWN

Water age and chlorine-residual issues develop at dead-end line ends over time, along with pressure drop at peak seasonal occupancy and an ongoing maintenance-flushing burden with no permanent fixture to do it through.

FIELD CORRECTION

Add permanent flushing hydrants or blow-off assemblies at the long dead-end runs identified.

WHO RESOLVES IT

EOR design addition

PERMIT EXPOSURE

Within approved plans -- a design enhancement, not a compliance issue

DOCUMENTATION

Revised water distribution sheet adding permanent flushing points

ROUGH COST AND SCHEDULE

Low -- a handful of additional flushing assemblies

D15 POTENTIAL Low-Medium

C013, C014, C015

The new 6" water network is fed through a new meter and double-detector-check assembly at the northeast corner, but it is also tied into the existing 3" main (via a 6"x3"x6" tee) and existing 2" mains elsewhere on C015 -- if either of those existing mains is separately metered or connects to the public side without its own backflow protection, it could bypass the new DDCV assembly entirely.

CONSEQUENCE IF BUILT AS DRAWN

A cross-connection bypassing the intended backflow protection is a real water-quality risk, and the utility provider may reject the connection configuration on review.

FIELD CORRECTION

Confirm with Madison County Utilities that the existing main tie-ins don't create a backflow-protection bypass around the new DDCV assembly.

WHO RESOLVES IT

EOR/utility coordination

PERMIT EXPOSURE

Agency notification likely -- utility provider review of the connection configuration

DOCUMENTATION

Utility provider written confirmation

ROUGH COST AND SCHEDULE

Low if confirmed fine; a redesign of the tie-in points if not

D19 POTENTIAL Medium

C016, C017

The Phase 1 gravity sewer system drains to manhole SSMH 9 and then the lift station, but the lift station's own rim/invert data is only labeled on the Phase 2 sheets (C017/C018), and no phasing note anywhere states the lift station must be constructed as part of Phase 1.

CONSEQUENCE IF BUILT AS DRAWN

If the lift station were deferred to a Phase 2 construction sequence, Phase 1 (built and potentially occupied first) would have no working sewer discharge point at all.

FIELD CORRECTION

Add an explicit phasing note requiring the lift station to be constructed and operational concurrent with or before Phase 1 occupancy.

WHO RESOLVES IT

EOR documentation addition

PERMIT EXPOSURE

Within approved plans -- a sequencing clarification

DOCUMENTATION

Phasing note added to the sewer sheets or a project phasing schedule

ROUGH COST AND SCHEDULE

None if simply clarified; a real problem if actually built out of sequence

E1 POTENTIAL High

D001

No detail exists anywhere in the 21-sheet set for a sanitary manhole, a drop manhole, a cleanout, or pipe trench/bedding/backfill -- despite the sewer sheets showing 21 sanitary manholes (several with drop connections around 10 ft, per D7), deep mains below the water table, and traffic-bearing cleanout lids required by the general notes. The general notes only defer generically to "Madison County standard specifications" rather than including an actual detail.

CONSEQUENCE IF BUILT AS DRAWN

Without a drawn detail, the contractor and inspector are left to interpret county standard details from memory or a separate document not included here -- a real source of RFIs, bid variance, and rework risk, especially for the drop-manhole and deep-trench conditions already flagged elsewhere in this review (D7, D10).

FIELD CORRECTION

Add the missing standard details (sanitary manhole, drop manhole, cleanout, pipe trench/bedding/backfill) to sheet D001, referencing the applicable Madison County standard.

WHO RESOLVES IT

EOR -- a real missing-detail gap

PERMIT EXPOSURE

Within approved plans -- a documentation completeness item, not an agency approval issue by itself

DOCUMENTATION

Added detail sheets or an explicit reference to the county's own published standard details

ROUGH COST AND SCHEDULE

Low direct cost (an engineering deliverable); real rework/RFI risk if left unaddressed through construction

E10 POTENTIAL Medium

D003 vs C018

The lift station's pump design point is 80 GPM at 50 ft of total dynamic head, but the only force mains shown anywhere in the sewer sheets are 2" PVC (confirmed by Group D). Pushing 80 GPM through a 2" pipe would run at roughly 7.6 ft/second with high friction loss, calling into question whether the stated 50-ft-TDH design point is actually achievable through a 2" line; the pump station's own discharge is 4", implying a 4"-to-2" transition somewhere that isn't shown or explained.

CONSEQUENCE IF BUILT AS DRAWN

If the real discharge path is genuinely 2" pipe for any meaningful distance, the pumps may not achieve their design flow and head, reducing actual pumping capacity below what the system needs -- and increasing pump wear.

FIELD CORRECTION

Confirm the actual force main size and routing from the lift station discharge to wherever it ultimately connects (see D2), and re-verify the pump selection against the real pipe size and length.

WHO RESOLVES IT

EOR -- hydraulic verification

PERMIT EXPOSURE

Within approved plans once verified; EOR letter if the pump selection needs to change

DOCUMENTATION

Hydraulic calculation for the actual force main path, pump curve re-verification

ROUGH COST AND SCHEDULE

Low if the pumps check out; a pump re-selection cost if not

E3 POTENTIAL

Medium -- the depth/elevation numbers are directly legible; the buoyancy shortfall is a rough independent check, not the engineer's own stated calculation, which does not appear on this sheet

D003 (lift station wet well)

The lift station's fiberglass wet well is 72 inches in diameter and 22.5 ft deep (top elevation 97.50, bottom 75.00) -- deeper than every soil boring in the set (all drilled to only 15 ft), with no boring taken at the lift station location itself. Seasonal high groundwater is estimated as shallow as 1.5-4 ft across the site, and the plan set separately includes a sinkhole-repair detail confirming the design team treats this as karst terrain. No buoyancy (flotation) calculation is shown for this structure; a rough check suggests the 11.5 cubic yards of anchor concrete specified (roughly 27,000 lb submerged weight) may not be enough to counteract the uplift force on a 72" basin submerged some 19-21 ft in groundwater (very roughly 40,000+ lb, before accounting for soil weight over the anti-flotation ring).

CONSEQUENCE IF BUILT AS DRAWN

This is the deepest excavation on the entire project, in unverified soil conditions, in karst terrain with a shallow water table -- real risk of the tank floating (flotation) if groundwater rises during or after construction, and real risk of triggering a sinkhole during the required dewatering, neither of which is addressed by any calculation in this plan set.

FIELD CORRECTION

Obtain a soil boring specifically at the lift station location (all existing borings stop well short of its depth), and require the engineer to provide and verify an actual buoyancy calculation for the wet well before construction.

WHO RESOLVES IT

Geotechnical engineer + EOR -- a real missing design step, not a field fix

PERMIT EXPOSURE

Agency notification likely if the buoyancy calculation reveals a design change is needed

DOCUMENTATION

Site-specific geotechnical boring, buoyancy calculation, possible anchor-concrete redesign

ROUGH COST AND SCHEDULE

Could be significant if the buoyancy check fails and the anchor system needs to be upsized, plus the cost of the missing site-specific boring and any karst mitigation it reveals is needed

E5 POTENTIAL Medium

D002 (soil borings) vs C012 (pond design)

Of the six soil borings, B5 (in the WRA-2/WRA-3 vicinity) estimates seasonal high groundwater at just 1.5 ft below grade -- shallower than any water-table value actually used in the C012 pond design (which uses 2.5 ft separation for both WRA-2 and WRA-3). Borings B4 and B6, both showing clayey soils at or near the surface, give no groundwater estimate at all. No lab permeability or fines-content data backs up the infiltration rates (Kh/Kv) used in the pond design (an independent confirmation of Group C's same concern, item C4, from the geotechnical-sheet side).

CONSEQUENCE IF BUILT AS DRAWN

If the true seasonal high water table in this area is actually closer to B5's 1.5 ft than the 2.5 ft the pond design assumes, the ponds may not meet SRWMD's required separation criteria and could hold standing water more of the year than designed.

FIELD CORRECTION

Have the geotechnical engineer resolve the discrepancy between B5's shallower water-table estimate and the pond design's assumed value before relying on the current pond design.

WHO RESOLVES IT

Geotechnical engineer

PERMIT EXPOSURE

Agency notification likely if the pond design needs to change as a result

DOCUMENTATION

Geotechnical clarification memo, possible pond redesign

ROUGH COST AND SCHEDULE

Could range from none (if reconciled favorably) to a real pond-capacity increase

E6 POTENTIAL Medium

D002, D001

Water Retention Area 1 is designed with 2:1 side slopes (an independent confirmation of Group C's finding C5) sitting in soil the borings describe as very loose fine sand (SPT blow counts of only 2-5 through the top 10 ft in borings B1/B2, taken in the WRA-1 area). No geotechnical recommendations for slope stabilization, proof-rolling, or removal of unsuitable material are given anywhere in the set, even though the general notes require fill to be compacted "in accordance with the Geotechnical Investigation Report" -- a report that is not actually included in this plan set.

CONSEQUENCE IF BUILT AS DRAWN

A 2:1 slope in loose, low-blow-count sand is a real slope-stability risk (sloughing, raveling) and could cause settlement under nearby pads/roads; as written, the contractor bears this risk since the notes assign unsuitable-material quantification to them without giving them the underlying geotechnical report to quantify it from.

FIELD CORRECTION

Provide the actual geotechnical investigation report referenced in the general notes, including specific slope-stability and compaction recommendations for the WRA-1 area's loose sand.

WHO RESOLVES IT

Geotechnical engineer

PERMIT EXPOSURE

Within approved plans once the report is provided and its recommendations incorporated

DOCUMENTATION

Geotechnical report as an attached/referenced document, revised slope design if recommended

ROUGH COST AND SCHEDULE

Could range from none (if 2:1 is confirmed acceptable) to a real pond-slope redesign

E7 POTENTIAL Low-Medium

D001

The 6" limerock base pavement section (LBR 100) is a material sensitive to saturation, and the soil borings show seasonal high groundwater as shallow as 1.5 ft in places -- the plan set doesn't confirm the base of the pavement section maintains the typically recommended 12-18 inches of separation above groundwater in the lower-elevation roadway areas.

CONSEQUENCE IF BUILT AS DRAWN

Pavement built with an inadequately separated limerock base in a saturated condition is prone to premature failure in the lower-lying road segments.

FIELD CORRECTION

Verify pavement-base-to-groundwater separation in the lowest roadway areas against the actual boring data, and add underdrain or additional fill if needed in those specific locations.

WHO RESOLVES IT

EOR/geotechnical verification

PERMIT EXPOSURE

Within approved plans once verified

DOCUMENTATION

Verification memo, localized pavement-section revision if needed

ROUGH COST AND SCHEDULE

Low if verified fine; localized underdrain/fill cost if not

Low (25)

A11 LOW High

C002

Stormwater O&M section contains boilerplate carried over from another project: "The previously permitted retention basin was sized to meet both local and state requirements" (this project proposes three NEW retention areas, not one previously-permitted basin), plus other leftover references to "abutting city streets," "city storm water system," "grease traps," curbing/ADA ramps, and "architectural plans"/"landscape plan" not part of this civil set.

CONSEQUENCE IF BUILT AS DRAWN

The O&M plan is a permit-record document and currently misdescribes the system; the field crew is unaffected.

FIELD CORRECTION

Clean up the general notes to remove carried-over language from a prior project template.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- documentation only

DOCUMENTATION

Revised general notes sheet

ROUGH COST AND SCHEDULE

Negligible

A12 LOW High

C002

Water main material specifications contradict themselves: ASTM D2241 (an IPS-OD SDR pipe standard) is cited alongside AWWA C900 DR-18 (a different, CI-OD pipe standard) for the same pipe; "ASTM 1784" should read D1784; ASTM D1869 (for asbestos-cement pipe rubber rings) is misapplied where D3139/F477 would be correct; and "ASTM 3034" should read D3034.

CONSEQUENCE IF BUILT AS DRAWN

Submittal confusion, and a possible pipe outside-diameter/fitting mismatch if a contractor orders to the literal (incorrect) D2241 spec instead of the intended C900 DR-18.

FIELD CORRECTION

Correct the material callouts in the general notes to the intended standards.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- documentation only

DOCUMENTATION

Revised general notes sheet

ROUGH COST AND SCHEDULE

Negligible if caught before ordering; a real submittal delay if not

A13 LOW High

C003, C004, C002

The legend omits several symbols actually used on the drawings (the "pavement to be removed" cross-hatch, the dense hatch along the C004 gravel driveway, the wetland flag line, the buffer line, washout/staging symbols, the barrier fence). The notes refer to "the gravel construction entrance" and "the internal construction entrance" (singular) while two entrances are actually drawn.

CONSEQUENCE IF BUILT AS DRAWN

Minor field-crew confusion; will not change what gets built once clarified verbally.

FIELD CORRECTION

Update the legend to match everything actually drawn; correct singular/plural language in the notes.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans

DOCUMENTATION

Revised legend

ROUGH COST AND SCHEDULE

Negligible

A14 LOW Medium

C003

The tree-removal table has roughly 80 consecutive entries (trees 4063-4146) all identically labeled "TREE 30\" PINE," which reads as a defaulted survey field rather than 80 separately measured diameters. One entry (tree 4153) has a blank description.

CONSEQUENCE IF BUILT AS DRAWN

Unreliable input if the county requires a tree-mitigation calculation based on actual measured diameters.

FIELD CORRECTION

Have the surveyor re-verify and re-populate the actual measured diameters for the tree-removal table.

WHO RESOLVES IT

Surveyor re-verification

PERMIT EXPOSURE

Within approved plans, unless the county tree ordinance requires certified measurements first

DOCUMENTATION

Corrected tree table

ROUGH COST AND SCHEDULE

Modest -- a re-survey of ~80 trees

A15 LOW Medium

C003/C004

Wetland "E" (the lake) is shown only by flag symbols on C003, without a text label; Wetlands "A" and "F/G" do not appear on either demolition/erosion-control sheet at all (they appear on other sheets in the set). Only Wetland "B/C/D" (2.10 ac) and "H" (1.03 ac) are labeled and match the confirmed acreages on these two sheets.

CONSEQUENCE IF BUILT AS DRAWN

Protection measures for Wetlands A and E cannot be independently verified from the erosion-control sheets alone.

FIELD CORRECTION

Add labels for all confirmed wetlands on the erosion-control sheets, or reference which sheet shows each one's protection measures.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans

DOCUMENTATION

Revised erosion-control sheet

ROUGH COST AND SCHEDULE

Negligible

A16 LOW High that the delineation is stated as incomplete

C001

The survey is credited to "Menadier Engineering" itself, but the benchmark note cites "Trav. Rogers LB 4074" as the surveying business -- no surveyor of record or survey date is separately named. The cover sheet itself states the wetland boundary delineation is "incomplete," with the applicant accepting responsibility for any impacts beyond what's shown.

CONSEQUENCE IF BUILT AS DRAWN

A documentation gap; more importantly, the incomplete wetland delineation is itself a real permitting risk that the owner has already accepted by note.

FIELD CORRECTION

Confirm the surveyor of record and survey date for the record; treat the incomplete wetland delineation as a live, unresolved permitting risk, not a formality.

WHO RESOLVES IT

EOR/surveyor clarification

PERMIT EXPOSURE

Agency notification likely -- an incomplete wetland delineation is a real SRWMD permitting exposure

DOCUMENTATION

Surveyor certification, SRWMD coordination on delineation completeness

ROUGH COST AND SCHEDULE

Could be significant if additional wetland area is found once delineation is completed

A17 LOW High that the dates differ

C001-C004

The CAD plot stamp on all four sheets reads "Tuesday, September 30, 2025," about one year before the 09/16/2026 title-block date. All revision blocks are blank.

CONSEQUENCE IF BUILT AS DRAWN

It is not clear from the drawings alone whether the current issued sheets reflect the most recent design revision.

FIELD CORRECTION

Confirm with the engineer that the reviewed set is the current, sealed revision.

WHO RESOLVES IT

EOR confirmation

PERMIT EXPOSURE

Within approved plans -- a documentation check, not a design issue

DOCUMENTATION

Written confirmation from EOR of current revision status

ROUGH COST AND SCHEDULE

None if confirmed current

A18 LOW Medium (zoning/flood item); High (notation item)

C001

The PM peak-hour trip equation is written as "0.71 LN(223) – 0.06 = 44," which as literally written evaluates to about 3.78, not 44 -- the correct ITE form is Ln(T) = 0.71·Ln(X) – 0.06, which does correctly produce T≈44. The stated answer (44) is right; only the written notation is wrong. Separately, zoning is listed as R1 (Residential) for a "RV Park Expansion" use, and the property includes Flood Zone A (no base flood elevation established) with no flood-zone limit line drawn on the erosion-control sheets.

CONSEQUENCE IF BUILT AS DRAWN

The trip-equation notation issue is cosmetic only, since the answer is correct. The zoning/flood-zone items are real verification items: if any new sites fall within Zone A, county floodplain rules for RV placement would apply, and this needs confirming against the actual zoning approval (e.g. a special exception or PUD) since R1 is not typically an RV park zoning classification.

FIELD CORRECTION

Correct the trip-equation notation for clarity. Separately, confirm the zoning approval mechanism (special exception/PUD) with Madison County P&Z, and map the Flood Zone A boundary onto the site plan to confirm no new sites fall within it without proper floodplain compliance.

WHO RESOLVES IT

EOR clarification; County P&Z confirmation on zoning

PERMIT EXPOSURE

Agency notification likely for the zoning/flood-zone question if unresolved

DOCUMENTATION

RFI, County P&Z written confirmation, floodplain compliance documentation if needed

ROUGH COST AND SCHEDULE

Low for the notation fix; potentially significant if sites must be removed from Zone A

B10 LOW High

C005, C007, C008

"Retention" and "detention" are used inconsistently for the same ponds across sheets (cover sheet and C006 say "Water Retention Area," C007 says "Water Detention Area" for the same features -- see also C12 below, an independent confirmation from Group C).

CONSEQUENCE IF BUILT AS DRAWN

The two terms carry different legal/hydraulic meaning in stormwater permitting; this could cause confusion in permit review and O&M documentation.

FIELD CORRECTION

Standardize on one term across all sheets, matching whichever is correct per the SRWMD permit application.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- documentation only

DOCUMENTATION

Revised sheet titles/labels

ROUGH COST AND SCHEDULE

Negligible

B11 LOW High

All four layout sheets

Revision blocks are empty on all four sheets. The CAD plot stamp reads "Tuesday, September 30, 2025," about a year before the 09/16/2026 title-block date.

CONSEQUENCE IF BUILT AS DRAWN

Uncertainty about whether the reviewed sheets are the current, sealed revision.

FIELD CORRECTION

Confirm current revision status with the engineer.

WHO RESOLVES IT

EOR confirmation

PERMIT EXPOSURE

Within approved plans -- a documentation check

DOCUMENTATION

Written confirmation from EOR

ROUGH COST AND SCHEDULE

None if confirmed current

B12 LOW High

C007/C008

Site 2-35 is labeled only on the overall layout sheet (C005, 1"=100' scale). On the enlarged C007 sheet its pad is cut off by the sheet border, and C008's border stops just short of it, with no match line drawn between the two enlarged sheets to cover it.

CONSEQUENCE IF BUILT AS DRAWN

The field crew has no dimensioned 1"=40' layout for site 2-35 or its adjacent road return.

FIELD CORRECTION

Add a match line and ensure site 2-35 is fully shown, dimensioned, on one of the two enlarged sheets.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- drafting completeness

DOCUMENTATION

Revised enlarged layout sheet

ROUGH COST AND SCHEDULE

Negligible

B13 LOW High

C005, C007, C008

Site labels 2-88 and roughly 2-92/2-93 are partly obscured by the magenta enlarged-sheet viewport outlines and callouts. Sewer structure callouts S-1 through S-8, S-17, and S-23 appear on the layout sheets with no legend explaining them (they are likely sewer manhole IDs from C016-C018). The legend's "WETLAND BOUNDARY" symbol (a dashed line) doesn't match the dash-dot-with-flags line actually used for wetlands on the drawings, and the heavy dashed "PHASE 2" limit line and the dashed parcel line (B8) aren't in the legend at all.

CONSEQUENCE IF BUILT AS DRAWN

Minor readability and cross-referencing confusion for the field crew; will not change construction once clarified.

FIELD CORRECTION

Clean up viewport overlaps, add the sewer-structure callouts to the legend, and correct the wetland-boundary legend symbol.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- documentation only

DOCUMENTATION

Revised legend and viewport placement

ROUGH COST AND SCHEDULE

Negligible

B8 LOW Low (line type is undefined on the drawings)

C005, C007, C008

A heavy dashed line (not shown in the legend) runs north-south along the west edge of the Phase 2 development, with the loop road and several sites (2-45, 2-55, 2-64) sitting directly on it. It appears to be the boundary between the two county parcels listed on the cover sheet.

CONSEQUENCE IF BUILT AS DRAWN

If this is in fact the inter-parcel boundary, building across it may require a unity-of-title filing or lot combination, and internal zoning setbacks could technically apply across it.

FIELD CORRECTION

Confirm the nature of this line with the surveyor/engineer and add it to the legend; file a unity of title if it is the parcel boundary.

WHO RESOLVES IT

EOR/surveyor clarification, unity-of-title filing if needed

PERMIT EXPOSURE

Agency notification if a unity of title is required

DOCUMENTATION

RFI, unity-of-title filing if applicable

ROUGH COST AND SCHEDULE

Low -- a filing fee and paperwork if needed

C12 LOW High

C010 vs C009/C012

The same three ponds are labeled "WATER DETENTION AREA" on C010 but "WATER RETENTION AREA" on C009 and C012 (an independent confirmation of the same inconsistency Group B found, item B10).

CONSEQUENCE IF BUILT AS DRAWN

Retention and detention carry different legal/hydraulic meaning; this should be made consistent across the permit and O&M documents.

FIELD CORRECTION

Standardize the terminology across all sheets.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- documentation only

DOCUMENTATION

Revised sheet labels

ROUGH COST AND SCHEDULE

Negligible

C13 LOW High

C012

A legend typo reads "KH = 40 FT/BY" where it should read "FT/DAY." The CAD plot stamp on every sheet reads "September 30, 2025" against the 09/16/2026 title-block date (an independent confirmation of the plot-date discrepancy already flagged in A17/B11).

CONSEQUENCE IF BUILT AS DRAWN

Documentation only.

FIELD CORRECTION

Correct the typo; confirm current revision status with the engineer.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans

DOCUMENTATION

Revised legend

ROUGH COST AND SCHEDULE

Negligible

C14 LOW High

C010

Storm structures S-9 through S-26 are shown on the plan as ovals, but their on-sheet callouts don't repeat the structure number the way the C009 structure table does.

CONSEQUENCE IF BUILT AS DRAWN

Matching a callout to the correct structure in the field is more error-prone than it needs to be.

FIELD CORRECTION

Add structure-number callouts consistent with C009's convention.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- documentation only

DOCUMENTATION

Revised structure callouts

ROUGH COST AND SCHEDULE

Negligible

C6 LOW High

C010

The mitered end section discharging into WRA-3 at invert 92.00 is labeled "15\" CONC. M.E.S.," but it is actually fed by 140 LF of 24" RCP pipe from structure S-14 (a real size mismatch confirmed by chaining the invert math: S-14's outlet at 92.70 minus the pipe's fall equals exactly 92.00).

CONSEQUENCE IF BUILT AS DRAWN

A contractor ordering to the label would order the wrong-size mitered end section -- caught in the field, but a real change order.

FIELD CORRECTION

Correct the M.E.S. callout to match the actual 24" pipe it receives.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- a labeling fix

DOCUMENTATION

Revised callout on the grading sheet

ROUGH COST AND SCHEDULE

Negligible if caught before ordering

D17 LOW High

C013 vs C002

The water meter callout on C013 reads "LIMITS OF CITY OF MADISON OWNERSHIP & MAINTENANCE," while the general notes (C002) state water service is by "MADISON COUNTY."

CONSEQUENCE IF BUILT AS DRAWN

Confusion over which utility's standards, fees, and permitting process actually apply to this connection.

FIELD CORRECTION

Confirm with the engineer which entity (City of Madison or Madison County) actually provides and maintains water service here, and correct whichever reference is wrong.

WHO RESOLVES IT

EOR clarification

PERMIT EXPOSURE

Within approved plans -- a documentation correction

DOCUMENTATION

RFI, corrected sheet reference

ROUGH COST AND SCHEDULE

Negligible

D18 LOW High

C017

A manhole appearing between SSMH 11 and SSMH 12 is labeled "SSMH-8," duplicating the Phase 1 structure already named SSMH 8, with no rim or invert data given for this second structure at all.

CONSEQUENCE IF BUILT AS DRAWN

Field confusion between two differently-located structures sharing the same name, and this structure's own elevation data is simply missing.

FIELD CORRECTION

Rename this structure to a unique ID and add its rim/invert data.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- documentation completeness

DOCUMENTATION

Corrected structure ID and invert table entry

ROUGH COST AND SCHEDULE

Negligible

E11 LOW High that the labels conflict; not confirmed which value (or whether both) is correct

D003 vs C017/C018

The lift station's second inlet invert is labeled "S.E. INV. 82.65" on both C017 and D003, but the same connection point is labeled "E. INV. 85.66" on C018 -- either a labeling inconsistency for the same inlet, or these are genuinely two separate connections and the second one is missing its own detail.

CONSEQUENCE IF BUILT AS DRAWN

Field confusion about which elevation actually governs at this connection point.

FIELD CORRECTION

Reconcile the two labels with the engineer and confirm whether this is one inlet or two.

WHO RESOLVES IT

EOR clarification

PERMIT EXPOSURE

Within approved plans -- a documentation reconciliation

DOCUMENTATION

RFI, corrected invert label

ROUGH COST AND SCHEDULE

Negligible if it's a simple labeling fix

E12 LOW Medium

D003

The lift station's high-water alarm is set at elevation 82.60, only about 0.05 ft (roughly half an inch) below the lowest connected inlet invert (S.E. 82.65).

CONSEQUENCE IF BUILT AS DRAWN

This leaves almost no warning time between the alarm triggering and that inlet line actually surcharging (backing up) -- an operator gets essentially no lead time to respond before a backup begins.

FIELD CORRECTION

Consider lowering the lag-pump-on and high-water-alarm setpoints to provide a more meaningful warning margin before the lowest inlet surcharges.

WHO RESOLVES IT

EOR design consideration

PERMIT EXPOSURE

Within approved plans -- an operational tuning adjustment

DOCUMENTATION

Revised float setpoints if the engineer agrees

ROUGH COST AND SCHEDULE

Negligible

E13 LOW High

D003

The wet well's overall depth dimension is left blank on the section view (shown as "__"), even though the elevation table elsewhere on the same sheet does establish the depth (22.5 ft, from top 97.50 to bottom 75.00). The sheet's title block also states a scale of "1\"=20'" despite the detail actually being drawn not-to-scale.

CONSEQUENCE IF BUILT AS DRAWN

A blank dimension and an incorrect stated scale are documentation cleanup items; the actual depth is still recoverable from the elevation table elsewhere on the sheet.

FIELD CORRECTION

Fill in the blank dimension and correct the stated scale.

WHO RESOLVES IT

EOR drafting correction

PERMIT EXPOSURE

Within approved plans -- documentation only

DOCUMENTATION

Revised detail sheet

ROUGH COST AND SCHEDULE

Negligible

E14 LOW Low-Medium

D003

The FDEP requirements block on this sheet states electrical panels must be mounted above the 100-year flood elevation, but no actual flood elevation or panel mounting height is given anywhere on the sheet -- and the three retention ponds' own 100-year stages (98.27-99.55 ft, per Group C's findings) are close to the wet well's own top elevation (97.50 ft) near Wetland A.

CONSEQUENCE IF BUILT AS DRAWN

Without a stated flood elevation and confirmed panel height, there's no way to verify this FDEP requirement is actually being met -- and the nearby pond stages suggest it's worth checking closely rather than assuming compliance.

FIELD CORRECTION

Determine the governing 100-year flood elevation for this location and confirm the electrical panel mounting height is designed above it.

WHO RESOLVES IT

EOR verification

PERMIT EXPOSURE

Agency notification -- this is an explicit, cited FDEP requirement

DOCUMENTATION

Flood-elevation determination, confirmed panel mounting detail

ROUGH COST AND SCHEDULE

Low if already compliant once verified; a real relocation cost if the panel needs to move higher

E15 LOW

High on the documentation gaps themselves; the downstream effect on pond-design confidence is real but not separately quantified

D002

One boring log (B6) records an N-value of 44 for a sample with no underlying blow-count data shown, another (B5) has an apparent typo in its soil classification ("SP-SMM"), the report notes groundwater was "Nt encountered" (a typo for "Not encountered") in all six borings, and no surface elevations were recorded for any boring (elevations are "referred to ground surface" rather than tied to the project's actual site datum).

CONSEQUENCE IF BUILT AS DRAWN

These are documentation quality issues; more importantly, the lack of tied surface elevations means every seasonal-high-groundwater elevation used elsewhere in the pond design (C012) ultimately depends on an assumed grade at each boring location, not a surveyed one.

FIELD CORRECTION

Have the geotechnical firm correct the documentation typos and, ideally, tie the boring locations to actual surveyed elevations for the final design record.

WHO RESOLVES IT

Geotechnical engineer

PERMIT EXPOSURE

Within approved plans -- a documentation quality item, though it does affect confidence in downstream elevation-dependent calculations

DOCUMENTATION

Corrected boring logs

ROUGH COST AND SCHEDULE

Negligible direct cost

E16 LOW Medium

D001-D003

The CAD plot stamp on these three sheets reads "Tuesday, September 30, 2025" -- before both the actual soil-boring date (April 13, 2026) and the title-block date (09/16/2026) shown elsewhere in the set. This is an independent, third confirmation of the same stale-plot-stamp issue already flagged by Groups A, B, and C on their own sheets.

CONSEQUENCE IF BUILT AS DRAWN

Likely just a stale CAD plot-date field rather than a real revision problem, but worth confirming.

FIELD CORRECTION

Confirm with the engineer that the issued set is in fact the current, sealed revision (the revision blocks are also blank throughout the set).

WHO RESOLVES IT

EOR confirmation

PERMIT EXPOSURE

Within approved plans -- a documentation check

DOCUMENTATION

Written confirmation from EOR

ROUGH COST AND SCHEDULE

None if confirmed current

DO THESE NEXT

- ✓ **Resolve the sewer design conflict -- the general notes say septic, but the actual design is a centralized gravity sewer whose force main has no stated destination off the property**
- ✓ **Confirm a replacement wastewater plan exists before the existing drainfield serving the 91 currently-occupied sites is demolished**
- ✓ **Fix the stormwater pond design -- the 100-year design stages sit higher than nearby storm-inlet rims and the bathhouse finished floor**
- ✓ **Have the engineer verify manhole S-3's inverts -- the Phase 1 storm trunk line appears to run uphill at its downstream end**

- ✓ Clarify the water-main/sewer-main crossing conflict shown when C013 (water) and C016 (sewer) are overlaid

---

SOURCE DOCUMENTS (INTERNAL REFERENCES)

- **Plan facts** (internal reference: facts)
- **Cost estimate** (internal reference: cost)
- **Value engineering** (internal reference: value_engineering)
- **Ideas and ROI** (internal reference: ideas)
- **Archived: Jan 2026 concept-exhibit review (superseded)** (internal reference: archived_concept_review)

These are longer internal analysis documents (facts audit, cost memo, value-engineering memo, ideas/ROI memo, and the archived prior-version review). They are listed here as references; the full text of each is reproduced in full below.

SOURCE DOCUMENTS (FULL TEXT)

Plan facts

Plan Facts Report

Madison Golf and RV Resort Expansion — Menadier Engineering Job No. 2024-30-001

Prepared for Jennings Realty / Jay Lapat | Analyst: Senior Civil Construction Estimator

PRELIMINARY NOTICE

This is a preliminary AI-assisted plan review for internal use by Jennings Realty and the Engineer of Record. It is not stamped engineering, a bid document, or legal advice. All quantities marked ASSUMED or UNKNOWN require human verification before use in a budget or contract. Numbers in this report are consistent with the confirmed Plan Facts provided in the project brief; where this report's independent reading of the sheet text disagrees with or adds nuance to those facts, the disagreement is flagged explicitly.

1. Scope of Work Audit

1.1 Site Count by Type

Item	Count	Source / Tag
Total new RV sites — Cover Sheet title block	215	OBSERVED: Cover C001 states "PROPOSED USE: RV PARK EXPANSION"; site numbering runs 1-1 to 1-88 (Phase 1) and 2-1 to 2-127 (Phase 2) = 215

Item	Count	Source / Tag
Total new RV sites — Overall Layout Plan	**217**	OBSERVED: C005 states "PHASE 1 – 88 SITES / PHASE 2 – 129 SITES / TOTAL – 217 SITES"
⚠️ **Discrepancy**	**215 vs. 217**	The cover sheet project brief says 215 (2-127 as highest Phase 2 number), but C005 explicitly states 217 with 129 Phase 2 sites. Phase 2 site list on C007/C008 includes site 2-104, 2-121 through 2-127, and others not in a clean sequential run. **Human verification required: count every discrete site label on C006–C008 to resolve 215 vs. 217.** I am using **215** as the site-plan count per the confirmed project brief, but reporting 217 as the C005 stated total.
Phase 1 sites (1-1 to 1-88)	88	OBSERVED: C006 Layout Plan Phase 1; C001 project brief
Phase 2 sites (2-1 to 2-127, plus apparent gaps/extras)	127–129	OBSERVED: C007–C008; C005 states 129
Site type — Pull-through vs. back-in	Both types shown	OBSERVED: C005, C006, C007, C008 each include a "PULL-THRU SCALE 1"=30'" and "BACK-IN SCALE 1"=30'" typical configuration detail in the legend. Exact count by type NOT tabulated on any sheet.
Site hookup level	Full hookup assumed	ASSUMED: Water service (3/4" per site, C013–C015) and sewer lateral (4" PVC per site, C016–C018) are shown to every site. Electrical pedestals are not shown on this civil plan set. "Full hookup" (water + sewer + electric) is the standard assumption for a resort-tier RV park; electric is UNKNOWN from these sheets.
ADA-designated sites	UNKNOWN	No ADA site designations, accessible route plans, or ADA site-count table appear on any of the 21 sheets reviewed.
Manufactured housing (MH) lots	0	OBSERVED: All sites labeled as RV sites. No MH pad details, foundation callouts, or MH-specific utility sizing appear anywhere in the set.
Existing RV sites (existing park, not in this contract)	91	OBSERVED: Project brief; existing force mains and septic system abandonment shown on C003, C004, C018

Confidence: High for total site count range (215–217); **Low** for pull-through vs. back-in split (not tabulated).

1.2 Gross vs. Net Developable Acreage and Density

Item	Value	Source / Tag
Total parcel area	140 acres (6,098,400 SF)	OBSERVED: C001 "PARCEL AREA (TOTAL): 6,098,400 S.F. (140 ACRES)"

Item	Value	Source / Tag
Contributing drainage area	33.868 acres (1,475,290 SF)	OBSERVED: C001 cover sheet project data table
Total impervious area	11.88 acres / 517,522 SF (35.08%)	OBSERVED: C001 cover sheet
— Asphalt road	198,153 SF	OBSERVED: C001
— RV pads (asphalt)	105,576 SF	OBSERVED: C001
— Concrete	171,460 SF	OBSERVED: C001
— Gravel	22,641 SF	OBSERVED: C001
— Existing impervious	19,692 SF	OBSERVED: C001
Total open space	17.99 acres / 783,581 SF (53.11%)	OBSERVED: C001 (excludes water retention areas)
Total water retention area	4.00 acres / 174,187 SF (11.81%)	OBSERVED: C001
Net developable (contributing area minus wetlands and ponds)	~18–20 acres	ASSUMED: Wetlands total ~25.61 acres on the full parcel but much of that is outside the 33.868-acre contributing area; net developable within contributing area ≈ 33.868 ac – 4.00 ac ponds – wetland buffers within contributing area. Exact net developable acreage is NOT stated.
Density (215 sites ÷ 33.868 ac contributing)	**6.35 sites/acre**	ASSUMED calculation using confirmed figures
Density (215 sites ÷ ~20 ac estimated net dev.)	**~10.75 sites/acre**	ASSUMED

Note: The 140-acre total parcel includes the existing 9-hole golf course and large wetland areas (Wetlands E at 10.86 ac and F/G at 11.19 ac). The 33.868-acre contributing drainage area is the operative project footprint.

1.3 Roadway Network

Item	Value	Source / Tag
Lane width — predominant	16 ft	OBSERVED: Labeled repeatedly on C005, C006, C007, C008 (e.g., "16'" callouts on nearly every road segment)
Lane width — entry/main roads	22 ft	OBSERVED: Several entry and main collector segments labeled "22'" on C005, C006, C007, C008
Road widths — 30 ft	30 ft	OBSERVED: Some segments shown at 30' on C005, C007
⚠ Fire lane width concern	16 ft one-way vs. NFPA 1 20-ft minimum	OBSERVED discrepancy confirmed in project brief; the 16-ft lanes are predominant throughout both phases
Road surface type	Asphalt	OBSERVED: D001 "TYPICAL PAVEMENT SECTION SD29A: 1.5" ASPHALT FDOT SP-9.5 / 6" LIMEROCK TO LBR 100 / 12"

Item	Value	Source / Tag
		STABILIZED SUB-BASE TO LBR 40"; asphalt road area 198,153 SF on C001
Curb / gutter	Not shown	OBSERVED: Pavement section detail (D001) shows no curb. Grading relies on sheet flow to storm inlets and swales. No curb-and-gutter callout appears on any sheet.
Road centerline total length	~12,000–16,000 LF estimated	ASSUMED: Scaled from C005 at 1"=100' and C006–C008 at 1"=40'. Asphalt road area of 198,153 SF ÷ ~16-ft avg. width = ~12,385 LF as a check; with 22-ft collector segments blended in, estimate is 12,000–16,000 LF. No centerline schedule exists. **UNKNOWN from plan text; requires scaled takeoff from signed drawings.**
Stop bar locations	Multiple	OBSERVED: "24" STOP BAR (FDOT SP 711-001) & R1-1 STOP SIGN" at several intersections on C006, C007, C008, C011

Confidence: Moderate for road length (scaled estimate); **High** for width and surface type.

1.4 Water Source and Distribution

Item	Value	Source / Tag
Water source	Madison County municipal system	OBSERVED: C002 General Notes: "THIS PROJECT SHALL BE SERVED WATER BY THE MADISON COUNTY..." C013: "6" WATER METER (LIMITS OF CITY OF MADISON OWNERSHIP & MAINTENANCE)" and "6" DOUBLE DETECTOR CHECK VALVE ASSEMBLY"
Connection to existing	Connect to existing 3" PVC water main via 6"×3"×6" tee	OBSERVED: C013, C014 — "CONNECT TO EXISTING 3" WATER MAIN W/ 6"×3"×6" TEE & 3" GATE VALVE"
Backbone main size	6" C900 DR-18 PVC	OBSERVED: C013, C014, C015 — labeled throughout both phases
Branch mains	2" PVC SDR-21	OBSERVED: C013, C014, C015 — "2" PVC WATER MAIN (SDR-21)" serving clusters of sites
Individual site service	3/4" water service (typical)	OBSERVED: C013, C014, C015 — "3/4" WATER SERVICE (TYP.)" labeled at each site group
Clubhouse service	1" PVC water service	OBSERVED: C013 — "1" PVC WATER SERVICE TO EXISTING CLUB HOUSE BUILDING"
Bathhouse service	UNKNOWN	Not specifically called out; presumably served from the 2" branch in the Phase 2 area (C014).
Fire hydrant assemblies	~5 confirmed visible	OBSERVED: Fire hydrant assemblies shown on C013 (Phase 1, ~2 hydrants), C014 (Phase 2 area 1, ~2 hydrants), C015 (Phase

Item	Value	Source / Tag
		2 area 2, ~1 hydrant). Exact count requires scaled plan review; project brief states "~5 fire hydrants."
Backflow prevention	6" double detector check valve assembly	OBSERVED: C013
Locator wire	12-gauge AWG copper wound on all PVC	OBSERVED: C002 General Notes
Fire flow requirements	NFPA 24 / FL Fire Prevention Code	OBSERVED: C002 General Notes (fire protection notes section)
Pressure testing requirement	200 PSI or 50 PSI over system working pressure, 2 hours	OBSERVED: C002

Confidence: High for backbone pipe material and size, source, and connection type.

⚠️ **Coordination gap (flagged per project brief):** The water service stub to "the existing clubhouse building" (C013) is not cross-referenced to the Hammond Design Group clubhouse rebuild project (permits 26-0243, 26-0261). The service point, meter size, and final connection location should be confirmed with the clubhouse architect before bid.

1.5 Wastewater System

⚠️ CRITICAL UNRESOLVED ISSUE — #1 RANKED ITEM

Item	Value	Source / Tag
General Notes stated approach	"ON-SITE SEPTIC SYSTEM"	OBSERVED: C002 — "THIS PROJECT SHALL BE SERVED…SEWER BY AN ON-SITE SEPTIC SYSTEM"
Actual design shown on sewer sheets	Centralized gravity collection → lift station → force main	OBSERVED: C016, C017, C018 — 21 sanitary manholes, all 6" PVC gravity mains, duplex grinder lift station labeled on C017/C018
Force main discharge destination	NOT SHOWN ANYWHERE IN THE 21-SHEET SET	UNKNOWN: No receiving system, treatment plant, spray field, or municipal connection point is identified on any sheet.
Lift station location	Near SSMH 20/21, Phase 2 south area	OBSERVED: C017 "LIFT STATION TOP ELEV. 97.50 / W. INV. 83.76 / S.E. INV. 82.65 (SEE DETAIL)" and C018 shows same station
Lift station wet well	72" diameter fiberglass, 22.5 ft deep	OBSERVED: Project brief (confirmed from D003); D003 shows plan/section view with float switch elevations
Pump specifications	2× 80 GPM @ 50 ft TDH (duplex)	OBSERVED: Project brief confirmed; D003 pump data table (partially legible in text extract)
Lift station reference standard	FDEP Rule 62-604	OBSERVED: D003 cites this rule (collection/transmission regulation, not DOH septic rule)
Gravity main size (all)	6" PVC	OBSERVED: C016, C017, C018 — every sewer main labeled 6" PVC

Item	Value	Source / Tag
⚠ **Size conflict with General Notes**	General Notes require 8" minimum main lines	OBSERVED: C002 Sanitary System note: "MAIN LINES SHALL BE A MINIMUM OF 8" DIAMETER." All mains shown are 6". This is a direct internal contradiction.
Gravity main slope	Predominantly 1.20%; some 1.00%, 2.25%, 2.70%, 3.10%	OBSERVED: C016, C017, C018 — slopes labeled on each run
Minimum slope for 6" (code)	1.04%	OBSERVED: C002 — "6" @ 1.04%" minimum. The 1.00% runs on C017 are below this minimum.
Sanitary manholes	21 (SSMH 1–21)	OBSERVED: C016 (SSMH 1–9, 12, 14), C017 (SSMH 10–15, 19–20), C018 (SSMH 14, 16–21)
Site laterals	4" PVC at 1.00%–2.00% minimum	OBSERVED: C016, C017 — "4" PVC SEWER LATERAL (TYP.)" and "4" PVC CLEAN-OUT (TYP.)"
Existing septic abandonment	Existing septic tanks to be abandoned per FL Dept. of Health standards	OBSERVED: C018 — "EXISTING SEPTIC TANKS TO BE ABANDONED IN ACCORDANCE W/ DEPT. OF HEALTH STANDARDS (TYP.)"
⚠ **Drainfield demolition — no replacement named**	Existing drainfield demolished per C004	OBSERVED: C004 — "EXISTING DRAINFIELD TO BE DEMOLISHED AND REMOVED PER DOH STANDARDS AND REGULATIONS." No replacement drainfield is shown or named anywhere in the set.
Existing force mains (existing 91-site park)	Existing 2" force mains shown; some to be abandoned, some rerouted to new system	OBSERVED: C003, C013, C018 — "EXISTING 2" FORCE MAINS"; C018 shows "SIX (6) 2" WYE CONNECTIONS" and segments labeled "EXISTING FORCE MAINS TO BE ABANDONED" alongside new connections
Total estimated daily flow	~30,600–46,000 GPD	OBSERVED: Project brief (306 total sites × 100–150 GPD/site)
FDEP permit requirement	FDEP 62-604 (collection) + 62-600 (treatment/disposal)	OBSERVED: At >10,000 GPD, DOH septic ceiling exceeded; full FDEP DWF permit required

Confidence: High that the system is a centralized gravity collection / lift station design, not septic. **Confidence: High** that the force main discharge destination is unknown from these drawings. **This is the #1 cost and schedule risk on the project.**

1.6 Stormwater System

Item	Value	Source / Tag
Retention/detention ponds	3 ponds (WRA-1, WRA-2, WRA-3)	OBSERVED: C001, C009, C010, C012
WRA-1	35,691 SF; bottom elev. 95.50; berm elev. 100.60; 2:1 slopes; 100-yr stage 99.55; ESHWT 93.00; weir crest 99.00	OBSERVED: C001, C009, C012

Item	Value	Source / Tag
WRA-2	66,562 SF; bottom elev. 97.00 & 98.00 (two sub-basins shown); berm elev. 100.40; 4:1 slopes; 100-yr stage 99.37; ESHWT 95.50; weir crest 98.88	OBSERVED: C001, C010, C012
WRA-3	71,934 SF; bottom elev. 97.00; berm elev. 99.50; 4:1 slopes; 100-yr stage 98.27; ESHWT 94.50; weir crest 97.88	OBSERVED: C001, C010, C012
Total pond area	174,187 SF (4.00 acres)	OBSERVED: C001
Storm inlets	~26 structures (S-1 through S-26 visible across C009–C011)	OBSERVED: C009 shows S-1 through S-8; C010 shows S-9 through S-26; project brief confirms "~26 storm structures"
Storm pipe types	15" RCP, 18" RCP, 24" RCP, 36" RCP	OBSERVED: C009, C010, C011
Storm pipe slopes	0.20% to 0.50% for most runs	OBSERVED: C009, C010
Outlet structures	Mitered end sections (M.E.S.) and concrete overflow weirs with aluminum skimmers	OBSERVED: C009, C010, C012
WRA-1 weir	3-ft wide concrete overflow weir, crest 99.00, top of structure 100.00	OBSERVED: C009, C012
WRA-2 weir	13-ft wide concrete overflow weir, crest 98.88, top of structure 99.72	OBSERVED: C010, C012
WRA-3 weir	10-ft wide concrete overflow weir, crest 97.88, top of structure 98.82	OBSERVED: C010, C012
Design storm	100-yr/24-hr (WRA-1); 100-yr/4-hr (WRA-2, WRA-3)	OBSERVED: C012 section profiles
⚠ **Pond stage vs. inlet rim conflict**	WRA-1 100-yr stage 99.55 vs. multiple inlet rims at 97.0–98.85 ft; bathhouse FF at 98.20	OBSERVED: C009 shows S-6 rim 98.7, S-7 rim 98.57, S-8 rim 98.64, S-2 rim 101.0; C010 bathhouse FF 98.20; WRA-2 stage 99.37 exceeds bathhouse FF 98.20 — confirmed critical issue per project brief
⚠ **Phase 1 trunk line invert conflict**	Storm trunk runs toward pond outfall; invert sequencing requires field verification	OBSERVED: S-3 labeled as "48" MANHOLE RIM EL. 99.0 / NW. INV. 93.0"; the 36" RCP runs at 0.20% slope — the full invert chain needs professional check; confirmed in project brief
SWPPP/NPDES	Required	OBSERVED: C002 references NPDES final stabilization requirements; dewatering permit from SRWMD also required per C002
Sinkhole repair detail	Provided	OBSERVED: D001 — full sinkhole repair protocol for pond basins
SRWMD jurisdiction	Yes	OBSERVED: C001 contact information includes SRWMD; as-built requirements reference SRWMD on C002
Flood zone	Zones 'X' and 'A'	OBSERVED: C001 — "FLOOD ZONES 'X' AND 'A' AS DETERMINED BY FIRM PANEL 12079C0280D DATED APRIL 23, 2025"

Confidence: High for pond dimensions, elevations, and weir data. **High** for the identified hydraulic conflicts.

1.7 Electrical Distribution

Item	Value	Source / Tag
Primary electrical service	UNKNOWN	No electrical plan sheets exist in the 21-sheet set. No utility company or transformer schedule identified.
Transformer count and sizing	UNKNOWN	Not shown in this plan set.
Secondary distribution — overhead vs. underground	UNKNOWN	Not shown.
Site pedestal amperage	UNKNOWN	Not shown. For an RV resort this tier, 50A/30A/20A pedestals would be expected (ASSUMED per industry standard), but no electrical design appears here.
Metering approach	UNKNOWN	Not shown. Individual metering, submetering, or master metering cannot be determined from these sheets.
Site lighting	UNKNOWN	No lighting plan, pole schedule, or photometric study appears in the set.

Confidence: Low — entire electrical scope is absent from this plan set. A separate electrical engineering package is presumably forthcoming or under separate permit. This is a significant gap for cost estimating purposes.

1.8 Gas Distribution

Item	Value	Source / Tag
Natural gas	Not shown	OBSERVED: No gas distribution plan or callouts appear anywhere in the 21-sheet set.
Propane	Existing propane tank noted; no new propane distribution	OBSERVED: C003 — "EXISTING PROPANE TANK W/ CONC. PAD TO REMAIN UNDISTURBED"
New gas distribution	UNKNOWN / not shown	No new gas lines are designed in this plan set. Whether individual site gas is planned is UNKNOWN.

1.9 Vertical Structures

Item	Size	Construction Type	Source / Tag
Bathhouse (Phase 2)	15 ft × 15 ft = 225 SF	UNKNOWN — no architectural plans in this set	OBSERVED: C007, C010, C011, C014, C017 label "15'×15' BATHHOUSE" with FF ELEV. 98.20
Existing clubhouse	Present on site	UNKNOWN for this project — separate permit 26-0243/26-0261 (Hammond Design Group)	OBSERVED: Referenced on C013 (water service stub) and C016 (sewer lateral)

Item	Size	Construction Type	Source / Tag
Dog park	Shown on C008, C011	No structure; open fenced area	OBSERVED: C008, C011 show "DOG PARK" label
Maintenance building	Not shown	—	UNKNOWN
Office	Not shown	—	UNKNOWN
Dump station	Not shown	—	UNKNOWN
Pool	Not shown	—	UNKNOWN
Pavilions / mail kiosk / playground	Not shown	—	UNKNOWN

Confidence: High that only one bathhouse (225 SF) appears in the civil plan set. All other vertical amenities are absent.

⚠️ **Note:** The bathhouse is extremely small (15×15 = 225 SF) for a 215-site resort. A typical resort bathhouse/laundry for this scale would be 800–2,000 SF. Whether additional bathhouse buildings are planned under a separate permit is UNKNOWN.

1.10 Site Amenities

Item	Status	Source / Tag
Fencing — chainlink	Shown on C006 (Phase 1 perimeter near railroad)	OBSERVED: C006 — "CHAINLINK FENCE (SEE DETAIL SD21)"; D001 shows chainlink fence detail
Entry gates / signage	UNKNOWN — not shown on any civil sheet	
Stop signs / stop bars	Multiple locations	OBSERVED: C006, C007, C008, C011
Dog park	Shown (fenced area, no structure)	OBSERVED: C008, C011
Concrete washout areas	Two locations	OBSERVED: C003
Gravel construction entrance	Two locations	OBSERVED: C003, C004
Dumpster enclosures	Not shown	UNKNOWN
Storm shelter	Not shown	UNKNOWN
Lighting	Not shown	UNKNOWN
Playground	Not shown	UNKNOWN
ADA ramps / accessible routes	Referenced in notes but not specifically located	OBSERVED: C002 references ADA ramp requirements at sidewalk/curb connections

1.11 Engineering and Regulatory Notes

Item	Value	Source / Tag
Zoning	Residential 1 (R1)	OBSERVED: C001

Item	Value	Source / Tag
Building setbacks	Front 25', Corner 20', Rear 20', Side 10'	OBSERVED: C001
Maximum building height	35 ft	OBSERVED: C001
Wetland areas	A (0.43 ac), B/C/D (2.10 ac), E (10.86 ac), F/G (11.19 ac), H (1.03 ac)	OBSERVED: C003, C005
Wetland buffer	30 ft	OBSERVED: C005, C007 — "30' BUFFER" labeled
Flood zone	X and A	OBSERVED: C001
FIRM panel	12079C0280D, dated April 23, 2025	OBSERVED: C001
Soil conditions	Loose fine sand (WRA-1 area); clayey sand/clay (WRA-2/3 area)	OBSERVED: Project brief (Cal-Tech Testing, 4/13/2026); D002 (soil boring logs — text illegible in extraction)
Groundwater	No groundwater encountered to 15 ft in borings; ESHWT estimated 1.5–4 ft	OBSERVED: Project brief; C012 ESHWT elevations
Regulatory agencies	Madison County Planning & Zoning, SRWMD, FDEP	OBSERVED: C001, C002
FDEP permits required	Yes — water main (pre-construction) and wastewater (collection + treatment/disposal)	OBSERVED: C002
Railroad proximity	CSX / likely FG&A Pinsly line at north edge of Phase 1; sites 1-68, 1-69 ~6–14 ft from ROW	OBSERVED: Project brief; C006 shows chainlink fence near railroad ROW
Railroad setback	No setback labeled on any sheet	OBSERVED: C006 — confirmed in project brief as unresolved
Utility separation (water/sewer)	6-ft minimum horizontal (10 ft preferred) per Rule 62-604.400	OBSERVED: C002 utility separation notes
⚠️ **Water/sewer separation conflict**	Mains drawn 1–2.5 ft apart in places	OBSERVED: C016, C017 — 'W' and 'S' linework runs in close parallel throughout; confirmed in project brief

2. Quantity Take-Off Estimates

2.1 Earthwork

Item	Quantity	Tag	Notes
Clearing and grubbing area	~25–30 acres	ASSUMED	Approximately 33.868-acre contributing area minus existing cleared areas and wetlands. Tree removal lists on C003 show 150+ individual trees (predominantly 30" pines). Exact clearing boundary not defined by a single polygon; estimated from the contributing area boundary and the demolition plans.

Item	Quantity	Tag	Notes
Topsoil stripping	~25–30 acres	ASSUMED	Standard practice to strip 4–6" of topsoil across graded areas before fill placement. Not specified on plans.
Cut / fill volume	UNKNOWN — no earthwork calculations shown	UNKNOWN	No cut/fill balance sheet, mass haul diagram, or earthwork table exists anywhere in the 21-sheet set. Fill import is likely given pond excavation and road/pad grading requirements in Phase 1 (high topography) and flat Phase 2 areas needing pad buildup. **Confirmed missing per project brief.**
Pond excavation (estimated)	WRA-1: ~35,691 SF × ~4 ft avg. depth ÷ 27 = ~5,300 CY; WRA-2: ~66,562 SF × ~2 ft avg. ÷ 27 = ~4,900 CY; WRA-3: ~71,934 SF × ~2 ft avg. ÷ 27 = ~5,300 CY; **Total: ~15,500 CY**	ASSUMED	Depths estimated from C012 bottom vs. existing ground elevations. Bottom elevations: WRA-1 95.50 vs. existing ~97–98 ft (cut ~2 ft at bottom, but berms built up to 100.60 ft = fill); WRA-2 98.00 vs. existing ~98–99 ft; WRA-3 97.00 vs. existing ~97–98 ft. Net volumes are rough order-of-magnitude only.
Fill import (net)	Likely significant — estimate UNKNOWN	ASSUMED need; UNKNOWN volume	Road and pad subgrade in Phase 1 requires grading cuts of up to 5–10 ft in the high areas (elev. 107–110 ft) and fills elsewhere. Without a grading plan cross-section or earthwork calc, net import/export cannot be quantified.

Confidence: Low for earthwork quantities. A licensed surveyor / earthwork estimator should run a digital takeoff from the grading plans.

2.2 Paving

Item	Quantity	Tag	Notes
Asphalt road area	198,153 SF (22,017 SY)	OBSERVED	C001 cover sheet
RV pads — asphalt	105,576 SF (11,731 SY)	OBSERVED	C001 cover sheet
Concrete area	171,460 SF (19,051 SY)	OBSERVED	C001 cover sheet (includes pads, aprons, sidewalks)
Gravel area	22,641 SF (2,516 SY)	OBSERVED	C001 cover sheet
Pavement section — asphalt roads	1.5" SP-9.5 asphalt / 6" limerock base LBR 100 / 12" stabilized subbase LBR 40	OBSERVED	D001 detail SD29A
Concrete thickness	4" (ASSUMED standard; plan states "2500 PSI" for some	ASSUMED	C002 states "ALL CONCRETE SHALL BE 3000 PSI - 28 DAY

Item	Quantity	Tag	Notes
	elements, "3000 PSI" for others)		STRENGTH UNLESS OTHERWISE NOTED." D001 sinkhole cap: 2500 PSI. Slab thickness for pads/walks not explicitly stated.
Curb (LF)	0 (no curb shown)	OBSERVED	No curb callout appears on any sheet; D001 pavement section has no curb
Road centerline LF	~12,000–16,000 LF	ASSUMED	See 1.3 above
Concrete sidewalk	Shown in detail on D001; extent on site UNKNOWN	UNKNOWN	Detail SD exists (D001) but no sidewalk schedule or total LF callout exists in the set

Confidence: High for SF totals (from cover sheet); **Moderate** for SY conversions; **Low** for curb LF and sidewalk LF.

2.3 Utilities — Water

Item	Quantity	Tag	Notes
6" C900 DR-18 PVC water main (backbone)	~7,000–10,000 LF estimated	ASSUMED	Scaled from C013, C014, C015 at 1"=40'. Runs through both phases in a looped/branched configuration. No pipe schedule table exists.
2" PVC SDR-21 branch mains	~4,000–6,000 LF estimated	ASSUMED	Scaled from C013, C014, C015; branch mains serve clusters of 4–12 sites each
3/4" service lines (per site)	215 services at ~20–30 LF each = ~4,300–6,450 LF	ASSUMED	"3/4" WATER SERVICE (TYP.)" shown at every site group; length per service estimated at 20–30 ft
1" service to clubhouse	1 service; LF UNKNOWN	OBSERVED (existence); UNKNOWN (length)	C013
Gate valves (6")	~10–15 estimated	ASSUMED	Multiple labeled on C013, C014, C015; no valve schedule table
Gate valves (2")	~8–12 estimated	ASSUMED	At branch main connections
Fire hydrant assemblies	~5	OBSERVED (project brief)	C013, C014, C015
6" double detector check valve assembly	1	OBSERVED	C013
Fittings (tees, elbows, reducers)	Multiple per C013–C015	OBSERVED (types); UNKNOWN (exact count)	Tees, 90° elbows, 45° elbows, 22.5° elbows, 11.25° elbows shown throughout
Utility easement	10'×10' proposed easements shown	OBSERVED	C013 — "PROPOSED 10'×10' UTILITY EASEMENT" at meter/connection points

Confidence: Moderate for backbone main LF; **Low** for branch main LF (requires scaled takeoff).

2.4 Utilities — Sanitary Sewer

Item	Quantity	Tag	Notes
6" PVC gravity sewer main	~9,000–12,000 LF estimated	ASSUMED	Summing labeled runs on C016, C017, C018: Phase 1 (C016) visible runs total approximately 80+340+180+360+185+222+175+222+180+145+222+185+310+170+185+45+105+185 LF = ~3,696 LF labeled explicitly, plus unlabeled short connections; Phase 2 (C017/C018) adds ~400+95+180+400+204+205+140+400+370+165+125+215+245+370+200+170+125+95+290+290+300+370 LF = ~5,350 LF labeled. Total labeled ~9,000–9,500 LF; with unlabeled connections, estimating 9,000–12,000 LF total.
4" PVC sewer laterals (site service)	215 laterals at ~20–30 LF each = ~4,300–6,450 LF	ASSUMED	"4" PVC SEWER LATERAL (TYP.)" shown at each site group
4" PVC sewer lateral to clubhouse	1 lateral; LF UNKNOWN	OBSERVED	C016 — "4" PVC SEWER LATERAL TO EXIST. CLUBHOUSE BUILDING — CONTRACTOR TO VERIFY CONNECTION POINT"
Sanitary manholes (4' dia. precast)	21 (SSMH 1–21)	OBSERVED	C016, C017, C018
4" PVC clean-outs	~20–25 estimated	ASSUMED	Multiple labeled throughout C016, C017, C018; "4" PVC CLEAN-OUT (TYP.)"
6" clean-outs	2 confirmed	OBSERVED	C016 — "6" CLEAN-OUT INV. 106.16" and another at 180 LF run start
Duplex grinder lift station	1	OBSERVED	C017, C018, D003
Lift station wet well	72" dia. fiberglass, 22.5 ft deep	OBSERVED	D003, project brief
Lift station pumps	2× 80 GPM @ 50 ft TDH	OBSERVED	Project brief, D003
Force main (outgoing)	LF UNKNOWN; discharge point UNKNOWN	UNKNOWN	C017/C018 show lift station but no force main routing to a destination
Existing 2" force mains (existing park)	~455 LF to be abandoned (265 LF + 190 LF labeled on C018) + ~567 LF + 35 LF	OBSERVED	C018
2" PVC force mains (new, connecting existing sites to	Multiple short runs with wye connections	OBSERVED	C018 — "SIX (6) 2" WYE CONNECTIONS / TWO (2) 70 LF

Item	Quantity	Tag	Notes
system)			2" PVC FORCE MAIN" etc.

Confidence: Moderate for gravity main LF (labeled runs summed); **Low** for total LF without a pipe schedule.

2.5 Utilities — Storm Drainage

Item	Quantity	Tag	Notes
15" RCP storm pipe	~1,000–1,200 LF estimated	ASSUMED	Scaling labeled runs: C009 (83+136+102 LF visible), C010 (139+115+102+86+86+92+115+92 LF), C011 (115+102+92 LF)
18" RCP storm pipe	~600–800 LF estimated	ASSUMED	C009 (197+85+90+85 LF), C010 (170+132+86+86 LF)
24" RCP storm pipe	~900–1,100 LF estimated	ASSUMED	C009 (77+157 LF), C010 (441+140+92+91 LF)
36" RCP storm pipe	~150–200 LF estimated	ASSUMED	C009 (97+25 LF), C010 (97 LF)
Storm inlets (Type C and Type E)	~26 structures	OBSERVED	C009, C010, C011; project brief confirms ~26
Mitered end sections (M.E.S.)	~8–10	ASSUMED	Multiple shown on C009, C010 for pond outfalls
Concrete overflow weirs	3 (one per pond)	OBSERVED	C009 (WRA-1, 3-ft wide), C010 (WRA-2, 13-ft wide; WRA-3, 10-ft wide)
Aluminum skimmers	3	OBSERVED	C012 pond profiles
Silt fence (Type IV)	~5,000–8,000 LF estimated	ASSUMED	C003, C004 show extensive single-row and double-row Type IV silt fence throughout the perimeter and wetland buffers

Confidence: Moderate for pipe counts (labeled runs summed); **Low** for total LF (requires careful scaled takeoff).

2.6 Electrical (All UNKNOWN)

No electrical plan sheets exist in this 21-sheet set. The following items are **UNKNOWN**:

- Primary conductor / conduit LF
- Secondary conductor / conduit LF
- Transformer count and kVA ratings
- Pedestal count by amperage (50A/30A/20A)
- Site light pole count
- Any utility company coordination

Confidence: N/A — entire electrical scope is a blank in this plan set.

2.7 Concrete

Item	Quantity	Tag	Notes
Concrete RV pad area (from impervious table)	171,460 SF total concrete area	OBSERVED	C001 (includes pads, aprons, sidewalks, misc.)
Concrete volume at 4" slab	171,460 SF × 0.333 ft ÷ 27 = ~2,113 CY	ASSUMED	4" slab thickness not specified; ASSUMED per standard MH/RV pad practice; actual design thickness UNKNOWN
Overflow weir concrete	Small quantity (3 weirs)	ASSUMED	Roughly 50–100 CY total for 3 weir structures per C012 dimensions; UNKNOWN precisely
Manhole base concrete	Incidental to precast MH purchase	ASSUMED	
Bathhouse slab / foundation	~225 SF × 6" thick = ~2.8 CY	ASSUMED	15'×15' footprint; 6" slab ASSUMED — no structural drawings in set
Sidewalk concrete	UNKNOWN LF / CY	UNKNOWN	Detail exists (D001) but no extent shown

Confidence: Moderate for pad concrete SF (from cover sheet); **Low** for CY (thickness not specified).

2.8 Vertical Buildings

Building	SF	Framing Type	Tag
Bathhouse (Phase 2)	225 SF (15'×15')	UNKNOWN — no architectural drawings in set	OBSERVED dimensions from C007, C010
All other amenity buildings	UNKNOWN	UNKNOWN	Not shown in this plan set

Confidence: High for bathhouse footprint dimension; **Low** for all else.

2.9 Landscaping and Erosion Control

Item	Quantity	Tag	Notes
Sodding (Argentina Bahia)	~25–30 acres disturbed area minus impervious = ~15–18 acres of sod	ASSUMED	C002: "ALL AREAS DISTURBED BY CONSTRUCTION WHICH ARE NOT OTHERWISE PAVED SHALL BE IMMEDIATELY SODDED"; no sod schedule or acreage table exists
Silt fence Type IV (single row)	~3,000–5,000 LF	ASSUMED	C003, C004
Silt fence Type IV (double row)	~1,000–2,000 LF	ASSUMED	C003, C004
Gravel construction entrance	2	OBSERVED	C003, C004

Item	Quantity	Tag	Notes
Tree protection barrier	Multiple locations shown	OBSERVED	C003
Trees to be removed	~150+ individual trees (see tree table on C003)	OBSERVED	C003 tree removal tables list approximately 150 individual trees by point number and species (predominantly 30" pine); the table extends across 8 panels on C003
Landscape plan	NOT IN THIS SET	UNKNOWN	No landscape plan sheet exists; planting schedule, tree replacement, and irrigation are absent
Irrigation	Not shown	UNKNOWN	No irrigation design in this plan set

Confidence: Moderate for silt fence (visual count from C003/C004); **Low** for sod acreage (calculated from disturbed area minus impervious).

3. Critical Issues Summary Table

The following confirmed critical issues (from the project brief and my independent sheet reading) are listed in priority order for budget and schedule purposes:

Priority	Issue	Sheets	Cost/Schedule Impact
1	**Force main discharge destination unknown; general notes say "septic" but drawings show centralized sewer; FDEP 62-600/62-604 permits not initiated**	C002, C016–C018, D003	HIGH — potential 6–18 month permit delay; unknown WWTP or disposal facility cost
2	**Existing drainfield demolished (C004) with no replacement named; existing 91-site park is live during construction**	C004, C018	HIGH — DOH health code violation risk; operational continuity plan required
3	**Pond 100-yr stages exceed storm inlet rims and bathhouse FF**	C009, C010, C012	HIGH — hydraulic redesign $50K–$200K+
4	**Phase 1 storm trunk invert conflict (uphill run at outfall)**	C009	HIGH — redesign required before construction
5	**All gravity mains are 6" PVC; General Notes require 8" minimum — direct contradiction**	C002, C016–C018	MODERATE-HIGH — either upsizing mains or amending general notes
6	**Water and sewer mains drawn 1–2.5 ft apart in places; code requires 6–10 ft**	C013–C018	MODERATE — may require rerouting utility alignments
7	**No earthwork/cut-fill balance; fill import likely**	C009–C011	MODERATE — unknown cost until earthwork calc completed

Priority	Issue	Sheets	Cost/Schedule Impact
8	**Fire lane width 16 ft vs. NFPA 1 minimum 20 ft; fire marshal approval pending**	C005–C008	MODERATE — may require full roadway redesign
9	**Sites 1-68, 1-69 within 6–14 ft of railroad ROW; no setback shown**	C006	MODERATE — railroad encroachment; setback may require site relocation
10	**Electrical design entirely absent from plan set**	All	UNKNOWN cost scope
11	**Bathhouse at 225 SF is undersized for 215-site resort**	C007	LOW-MODERATE — may require additional bathhouse structures
12	**Clubhouse water/sewer stubs not coordinated with Hammond Design Group rebuild**	C013, C016	LOW-MODERATE — coordination gap, not a cost in this contract
13	**Site count discrepancy: C001 implies 215, C005 states 217**	C001, C005	LOW — requires recount

Sheet Index and Reading Notes

Sheet No.	Title	Readability	Notes
C001	Cover Sheet	Good	Full project data table, impervious/open space/retention areas, site plan index, contact info, traffic statement all readable
C002	General Notes	Good	Dense text; all major note sections readable. Contains critical contradictions (septic note vs. 8" minimum note vs. actual design)
C003	Demolition & Erosion Control Plan (1 of 2)	Good	Tree removal tables fully readable; existing utilities shown; silt fence and staging areas legible
C004	Demolition & Erosion Control Plan (2 of 2)	Good	Drainfield demolition note clearly readable; existing fence and pavement removal shown
C005	Overall Layout Plan	Good	Site count "217" clearly stated; all phase boundaries and wetland areas legible
C006	Layout Plan Phase 1	Good	Road widths, turning radii, site numbers all readable; chainlink fence near railroad visible
C007	Layout Plan Phase 2 (1 of 2)	Good	Bathhouse 15'×15' labeled; pond bottom elevations and berm elevations readable

Sheet No.	Title	Readability	Notes
C008	Layout Plan Phase 2 (2 of 2)	Good	Dog park, Phase 2 NW cluster, force main routing all visible
C009	Grading & Drainage Plan Ph-1	Good	Storm structure schedule (S-1 through S-8) with rim and invert elevations readable; soil boring locations B-1, B-2 marked
C010	Grading & Drainage Plan Ph-2 (1 of 2)	Good	Storm structures S-9 through S-26 with elevations; weir elevations readable; soil borings B-3 through B-6
C011	Grading & Drainage Plan Ph-2 (2 of 2)	Good	NW Phase 2 high area (110.5 ft); spot elevations readable throughout
C012	Water Retention Area Profiles	Good	All six cross-sections (WRA-1 A-A, B-B; WRA-2 A-A, B-B; WRA-3 A-A, B-B) readable with ESHWT, bottom, berm, weir, and 100-yr stage elevations
C013	Water Distribution Phase 1	Good	Pipe sizes, valve types, hydrant locations, existing main connections all readable; "contractor shall confirm water connection location" note visible
C014	Water Distribution Phase 2 (1 of 2)	Good	Pipe routing through WRA-2/3 area; hydrant and gate valve locations readable
C015	Water Distribution Phase 2 (2 of 2)	Good	NW Phase 2 cluster water routing; existing 2" force mains to remain noted
C016	Sewer Distribution Phase 1	Good	SSMH 1–9, 12, 14 with top and invert elevations readable; sewer run lengths and slopes all labeled
C017	Sewer Distribution Phase 2 (1 of 2)	Good	SSMH 10–15, 19–20 readable; lift station location and invert elevations visible; run lengths and slopes labeled
C018	Sewer Distribution Phase 2 (2 of 2)	Good	SSMH 16–21; existing septic tank abandonment notes; existing force main abandonment and wye connections; lift station duplicate entry
D001	Construction Details	Good	Pavement section SD29A, silt fence EC02, concrete washout EC07, chain link fence SD21, tree protection, inlet protection, mitered end section, sinkhole repair, sidewalk detail all readable

Sheet No.	Title	Readability	Notes
D002	Soil Boring Logs	**Partially unreadable**	The text extraction returned minimal content for D002. The boring log data (soil descriptions, blow counts, depth intervals) is in a tabular/graphic format that was not captured in the PDF text layer. Boring locations (B-1 through B-6) are visible on C009/C010. Six borings to 15 ft with no groundwater encountered is confirmed per project brief, but individual layer descriptions and SPT values should be read directly from the signed plan set.
D003	Lift Station Details	Partially readable	Plan view, section view, and control panel legend are present. Pump data table is partially legible in text form. Barney's Pumps identity, wet well elevations, and float switch elevations are confirmed per project brief. Pump model number and impeller diameter are **UNKNOWN** from the text extraction — must be read from the signed set.

Overall reading quality: High for all civil sheets (C001–C018, D001). D002 and D003 are partially unreadable in the text extraction and require direct human review of the signed PDF for complete data.

Missing from the plan set (items typically expected in a full construction document set of this type):

- Electrical plan sheets (entire scope absent)
- Landscape and irrigation plan
- Force main routing and discharge point
- Earthwork/cut-fill balance calculations
- ADA site plan and accessible route plan
- Fire marshal approval documentation
- FDEP wastewater permit or application
- Clubhouse utility coordination drawings
- Site lighting plan
- Utility easement recording information

Cost estimate

AACE Class 4 Preliminary Cost Estimate

Madison Golf and RV Resort Expansion — Menadier Engineering Job No. 2024-30-001

Prepared for Jennings Realty / Jay Lapat

Classification: AACE International Class 4 (Conceptual/Feasibility). Expected accuracy range: –30% to +50%. This estimate is preliminary and non-binding. It is based on the Plan Facts and Error Register developed earlier in this conversation, plus the 21-sheet PDF plan set. It is not a bid, a construction contract, or stamped engineering.

Basis date: Q3 2026 pricing (ENR CCI +4.2% applied from Q1 2025 baseline)

Site count used: 215 sites (project brief confirmed figure; C005 states 217 — the 2-site discrepancy per Error Register Item 13 is unresolved; this estimate uses 215 and notes that resolution could add ~1% to cost if 2 additional sites require utility extensions)

Regional Multiplier

Location: Madison County, Florida (rural north-central Florida, approximately 90 miles east of Tallahassee, 55 miles north of Gainesville)

Baseline: RS Means 2026 National Average = 1.000

Applied multiplier: 0.87 (ASSUMED — based on the following reasoning: RS Means City Cost Index for Gainesville, FL is approximately 0.83–0.86; Tallahassee is 0.84–0.88; rural Madison County has no direct CCI but sits between these markets and relies on labor from both. North Florida benefits from no state income tax, lower union penetration than national average (~12% vs. ~14% nationally), and lower general wage rates. However, rural site logistics add a small premium for material delivery and subcontractor travel. Net applied multiplier: 0.87 on labor-heavy divisions, 0.93 on material-heavy divisions, blended to 0.89 overall. For simplicity this estimate applies a flat **0.89 blended regional multiplier** to all national benchmark unit costs.)

Confidence: Moderate. No RS Means specific CCI for Madison County was independently verified from the plan set. Human reviewer should obtain a current RS Means CCI for zip code 32340 before finalizing.

Quantity Summary (Carried Forward from Plan Facts)

All quantities are carried forward from the Plan Facts stage. Only quantities changed from that stage are explained below. OBSERVED/ASSUMED/UNKNOWN labels are inherited.

Item	Quantity	Tag	Change from Plan Facts?
RV sites	215	OBSERVED/ASSUMED	No change
Gross contributing area	33.868 ac	OBSERVED	No change
Asphalt road	198,153 SF / 22,017 SY	OBSERVED	No change
Asphalt RV pads	105,576 SF / 11,731 SY	OBSERVED	No change

Item	Quantity	Tag	Change from Plan Facts?
Concrete areas	171,460 SF / 19,051 SY	OBSERVED	No change
Gravel areas	22,641 SF / 2,516 SY	OBSERVED	No change
6" C900 water main	~8,500 LF	ASSUMED	Refined midpoint of 7,000–10,000 LF range
2" SDR-21 branch water main	~5,000 LF	ASSUMED	Refined midpoint of 4,000–6,000 LF range
3/4" site water services	215 at avg. 25 LF each = 5,375 LF	ASSUMED	No change
6" PVC gravity sewer main	~10,500 LF	ASSUMED	Refined midpoint; includes unlabeled short connections
4" PVC sewer laterals (215 sites)	215 at avg. 25 LF each = 5,375 LF	ASSUMED	No change — this is a critical cost-driver addition not in Plan Facts trunk-line-only reading
4" PVC clubhouse sewer lateral	~100 LF	ASSUMED	No change
Storm pipe (all diameters)	~3,200 LF total (15" ~1,100 LF; 18" ~700 LF; 24" ~1,000 LF; 36" ~200 LF; unlabeled transitions ~200 LF)	ASSUMED	Slightly refined from Plan Facts
Storm inlets	26	OBSERVED	No change
Sanitary manholes (SSMH 1–21)	21	OBSERVED	No change
Duplex grinder lift station	1 (72" dia. × 22.5 ft deep, 2×80 GPM)	OBSERVED	No change
Force main	UNKNOWN length; UNKNOWN discharge destination	UNKNOWN	Critical — see Error Register Item 1; priced as allowance
Bathhouse	225 SF as drawn; 1,000 SF recommended per Error Register Item 40	ASSUMED (1,000 SF used for cost)	Changed from Plan Facts 225 SF — reasoning: Error Register Item 40 confirms the drawn size is non-compliant with FBC fixture counts; a code-compliant bathhouse for 127 Phase 2 sites is estimated at 800–1,200 SF; 1,000 SF used
Earthwork volume	Cannot be reliably estimated from plan text; grading contours and spot elevation data exist but a digital surface model takeoff is required	UNKNOWN	See earthwork section below
Electrical — entire scope	UNKNOWN	UNKNOWN	No electrical plans in set
Gas distribution	None shown	OBSERVED (no new gas)	No change
Pond excavation	~15,500 CY gross excavation estimated	ASSUMED	Retained from Plan Facts with explicit caveat
Trees to be removed	~150 trees	OBSERVED	No change

Item	Quantity	Tag	Change from Plan Facts?
Silt fence	~7,000 LF (blended single/double row)	ASSUMED	Midpoint of Plan Facts range
Clearing and grubbing	~28 acres	ASSUMED	Midpoint of Plan Facts range

Error Register Cost Allowances

The following Error Register items require cost allowances within the estimate. Items that represent design errors requiring redesign (not construction rework) are carried in soft costs under EOR redesign fees:

Error Item	Allowance Type	Where Carried
Item 1 — Force main discharge (UNKNOWN destination)	Construction allowance for force main and FDEP permit	Div. 33 / Soft Costs
Item 2 — Drainfield sequencing / temporary sanitation	Construction allowance	Div. 02
Item 3 — Phase 1 storm trunk invert correction	EOR design fee allowance	Soft Costs
Item 4 — Pond hydraulic redesign / bathhouse FF raise	Construction allowance (pad raise) + EOR fee	Div. 31 / Soft Costs
Item 5 — 6" vs. 8" sewer main resolution	Risk allowance (upsizing if required)	Div. 33 contingency
Item 11 — Utility separation rerouting	Construction allowance	Div. 33
Item 12 — ADA site plan and accessible routes	Construction allowance	Div. 32 / Soft Costs
Item 27 — Fire lane widening (pending fire marshal)	Risk allowance	Div. 32 contingency
Item 32 — Electrical design and construction (entire scope)	Full line item (UNKNOWN, priced as industry benchmark)	Div. 26
Item 34 — Pond liner (potential)	Allowance	Div. 33
Item 40 — Bathhouse upsizing	Cost basis changed to 1,000 SF	Div. 06/07/08/09

Cost Breakdown Structure

Division 01 — General Requirements

Confidence: Moderate

General requirements for a project of this type (33.868 contributing acres, ~215 sites, two construction phases, ~18–24 month construction schedule) are typically priced at 8–12% of direct construction costs for rural Florida horizontal land development. For this estimate I use 10% as the GR loading applied after all division subtotals are established, then show it as a separate line. Included items:

Item	Basis	Quantity	Unit Cost	Amount	Tag
Mobilization / demobilization	Lump sum	1 LS	$75,000	$75,000	ASSUMED
Construction superintendent (18 months)	Monthly	18 months	$8,500/mo	$153,000	ASSUMED
Project manager (18 months, half-time)	Monthly	18 months	$5,500/mo	$99,000	ASSUMED
Temporary facilities (job trailer, utilities, security fence, portable toilets)	Monthly	18 months	$3,200/mo	$57,600	ASSUMED
SWPPP maintenance (silt fence repair, inlet protection, sweeping, inspection)	Monthly	18 months	$2,500/mo	$45,000	ASSUMED
Quality control testing (density, concrete, pipe)	Allowance	1 LS	$35,000	$35,000	ASSUMED
As-built survey coordination	Allowance	1 LS	$20,000	$20,000	ASSUMED
Traffic control / maintenance of access	Allowance	1 LS	$15,000	$15,000	ASSUMED
Division 01 Subtotal				**$499,600**	

*Regional multiplier applied: 0.89 × $499,600 = **$444,644***

Division 02 — Existing Conditions (Demolition, Clearing, Environmental)

Confidence: Moderate

Item	Basis	Quantity	Unit Cost	Amount	Tag
Clearing and grubbing	$/acre	28 acres	$4,500/ac	$126,000	ASSUMED
Individual tree removal (>12" DBH, mechanically felled and chipped)	$/tree	150 trees	$350/tree	$52,500	OBSERVED count; ASSUMED unit cost
Stump grinding (included in tree removal)	Included above	—	—	—	

Item	Basis	Quantity	Unit Cost	Amount	Tag
Existing asphalt pavement removal (sawcut and haul)	$/SY	~2,000 SY	$8.50/SY	$17,000	ASSUMED — existing pavement areas shown on C003/C004
Existing concrete removal	$/SY	~500 SY	$12.00/SY	$6,000	ASSUMED
Existing fence removal	$/LF	~500 LF	$4.50/LF	$2,250	OBSERVED: C004 "EXISTING FENCE TO BE REMOVED"
Existing 3" PVC water main abandonment (grout-fill and cap)	$/LF	~1,500 LF	$5.50/LF	$8,250	ASSUMED — C003/C004 show abandonment; length ASSUMED
Existing 2" force main abandonment	$/LF	~1,100 LF	$4.50/LF	$4,950	ASSUMED — C018 labels ~1,100 LF to be abandoned
Existing drainfield demolition and removal (DOH standards)	$/SF	~20,000 SF	$3.50/SF	$70,000	ASSUMED — C004; area ASSUMED based on 91-site septic system
Existing septic tank abandonment — pump, fill, and DOH permit (estimated 25 tanks)	$/tank	25 tanks	$1,200/tank	$30,000	ASSUMED — Error Register Item 42; count UNKNOWN, 25 ASSUMED
Concrete washout areas (two locations)	Each	2	$1,800 each	$3,600	OBSERVED: C003
Gravel construction entrance (two locations)	Each	2	$3,500 each	$7,000	OBSERVED: C003/C004
Tree protection barriers (perimeter of retained trees)	$/LF	~2,000 LF	$3.00/LF	$6,000	ASSUMED
Temporary sanitation for existing 91-site park during transition (Error Register Item 2 allowance)	Monthly	6 months	$4,500/mo	$27,000	ASSUMED allowance
Division 02 Subtotal (pre-multiplier)				**$360,550**	

*Regional multiplier: 0.89 × $360,550 = **$320,890***

Division 03 — Concrete

Confidence: Moderate

Important note on concrete pad areas: The Plan Facts stage identified 171,460 SF of total concrete area from C001. This includes all concrete pads, aprons, overflow weirs, and other flatwork. The slab thickness is ASSUMED at 4 inches for RV pad flatwork and aprons, per standard RV resort practice. No structural plans specify slab thickness. The 4" assumption is a moderate-confidence assumption — some resort operators use 5" or 6" slabs for heavy RV wheel loads (Class A motorhomes can have rear axle loads of 13,000+ lbs).

Item	Basis	Quantity	Unit Cost	Amount	Tag
Concrete RV pads and aprons (4" slab on compacted fill, w/ WWF 6×6 W1.4×W1.4 per D001)	$/SY	19,051 SY	$52.00/SY	$990,652	ASSUMED thickness; OBSERVED area
Subgrade preparation under concrete (fine grade, compact, moisture condition)	$/SY	19,051 SY	$4.50/SY	$85,730	ASSUMED
Pond overflow weirs — concrete (3 weirs: 3-ft, 10-ft, 13-ft wide)	LS	1	$45,000	$45,000	ASSUMED — weir dimensions OBSERVED from C009/C010/C012; reinforced concrete box structures with aluminum skimmers
Concrete sidewalk (D001 detail; extent unknown — allowance)	Allowance	1 LS	$25,000	$25,000	UNKNOWN extent; allowance per D001 detail existence
Bathhouse slab and foundation (1,000 SF, 6" slab on grade, thickened edge)	$/SF	1,000 SF	$14.00/SF	$14,000	ASSUMED building size (Error Register Item 40); ASSUMED slab thickness
Miscellaneous concrete (thrust blocks for water fittings, manhole bases, pad edges)	Allowance	1 LS	$30,000	$30,000	ASSUMED
Division 03 Subtotal (pre-multiplier)				**$1,190,382**	

*Regional multiplier: 0.89 × $1,190,382 = **$1,059,440***

Division 06/07/08/09 — Amenity Buildings (Summarized at $/SF)

Confidence: Low

No architectural plans exist in the 21-sheet set. The bathhouse footprint is 15'×15' = 225 SF as drawn, but Error Register Item 40 establishes that a code-compliant bathhouse for 127 Phase 2 sites requires approximately 800–1,200 SF. This estimate uses **1,000 SF** as the planning basis. No other amenity buildings (office, maintenance, pavilions, pool) appear anywhere in the plan set.

UNKNOWN: Whether additional bathhouse buildings are planned for Phase 1 (88 sites also have no bathhouse shown). Standard resort practice provides one bathhouse per 50–75 sites; a 215-site resort would typically have 3–4 bathhouse/comfort station buildings. This estimate carries only the one Phase 2 bathhouse shown on the plans. A Phase 1 bathhouse is carried as an allowance.

Item	Basis	Quantity	Unit Cost	Amount	Tag
Phase 2 bathhouse — concrete block construction, metal roof, tile interior, ADA-compliant (1,000 SF)	$/SF	1,000 SF	$280/SF	$280,000	ASSUMED size; ASSUMED unit cost for north Florida CBS/CMU resort bathhouse
Phase 1 bathhouse allowance (no bathhouse shown on Phase 1 plans; prudent allowance for 88 sites)	Allowance	1 LS	$200,000	$200,000	ASSUMED — this is a significant gap in the design; if Owner confirms no Phase 1 bathhouse is planned, remove this line
Signage — entry monument and directional signs (allowance)	Allowance	1 LS	$25,000	$25,000	UNKNOWN — not shown on plans
Dog park amenities (fencing, water, signage)	Allowance	1 LS	$15,000	$15,000	OBSERVED: dog park shown on C008/C011; no details
Division 06/07/08/09 Subtotal (pre-multiplier)				**$520,000**	

*Regional multiplier: 0.89 × $520,000 = **$462,800***

Unit cost basis for buildings: $280/SF (CBS/CMU construction, metal roof, tile floor, ADA restroom, north Florida labor market, Q3 2026). Confidence: Low — no architectural drawings; ±40% on this division.

Division 22 — Plumbing (Bathhouse, Site Utility Connections at Hookup)

Confidence: Low

Plumbing within buildings is UNKNOWN (no plumbing plans). Site water and sewer connections at each RV hookup pedestal area are priced here as the "last foot" connection cost distinct from the underground utility main priced in Division 33.

Item	Basis	Quantity	Unit Cost	Amount	Tag
RV site hookup pedestal plumbing (water shutoff, pressure regulator, backflow, capped sewer wye) per site	$/site	215 sites	$850/site	$182,750	ASSUMED — standard full-hookup RV site plumbing trim
Bathhouse rough plumbing (Phase 2 building, 1,000 SF — fixtures, supply, drain, vent, water heater)	$/SF	1,000 SF	$65/SF	$65,000	ASSUMED — rough and finish plumbing for a 1,000 SF resort bathhouse
Phase 1 bathhouse plumbing allowance	Allowance	1 LS	$50,000	$50,000	ASSUMED — consistent with Phase 1 bathhouse building allowance above
Hose bibs at Phase 2 NW cluster (C015 labels "3/4" HOSE BIBB")	Each	4	$350 each	$1,400	OBSERVED (existence); ASSUMED count
Blow-off assemblies at dead-end water mains (Error Register Item 29)	Each	3	$1,800 each	$5,400	ASSUMED — at least 3 dead-end termini visible on C013–C015
Division 22 Subtotal (pre-multiplier)				**$304,550**	

*Regional multiplier: 0.89 × $304,550 = **$271,050***

Division 26 — Electrical

Confidence: Low — entire scope is UNKNOWN from this plan set

No electrical plan sheets exist. The complete electrical scope (primary service, transformers, secondary distribution, pedestals, metering, lighting, lift station power, generator) is absent. This is the single largest UNKNOWN cost category in the estimate. The following is priced entirely from industry benchmarks for a 215-site resort-tier RV park in north Florida.

ASSUMED design basis: Full-hookup RV resort; each site receives a 50A/30A/20A combination pedestal; underground secondary distribution; one or more pad-mount transformers; utility-owned primary distribution line extension to the property boundary; developer-installed secondary; LED site lighting at major intersections (~every 3rd site = ~72 poles); lift station with automatic transfer switch and standby generator.

Item	Basis	Quantity	Unit Cost	Amount	Tag
Utility primary line extension (TALQUIN EMC or Duke	Allowance	1 LS	$75,000	$75,000	UNKNOWN — Error Register Item 32; could range from

Item	Basis	Quantity	Unit Cost	Amount	Tag
Energy — allowance; actual cost unknown without utility coordination)					$0 (utility bears cost) to $250,000+
Pad-mount transformer(s) — estimated 3 × 167 kVA for 215 sites	Per kVA installed	500 kVA total	$350/kVA	$175,000	ASSUMED count and sizing; UNKNOWN from plans
Underground primary conduit and conductor (developer side of transformer)	$/LF	2,000 LF	$28/LF	$56,000	ASSUMED routing
Underground secondary conduit and conductor (transformer to pedestals)	$/LF	18,000 LF	$22/LF	$396,000	ASSUMED — 215 sites × avg. 80 LF of secondary per site after transformers
RV pedestals (50A/30A/20A combination, metered, installed)	$/site	215 sites	$2,800/site	$602,000	ASSUMED — industry standard for metered resort-tier pedestal installed in Florida
Meter bases and utility meter sockets (individual metering assumed)	Each	215	$450 each	$96,750	ASSUMED
Service entrance panel and main disconnect (at central distribution point)	Allowance	1 LS	$35,000	$35,000	ASSUMED
Site lighting (LED decorative poles, 72 poles, foundations, wiring)	Per pole installed	72 poles	$4,500/pole	$324,000	ASSUMED pole count (1 per 3 sites approximately); ASSUMED unit cost
Bathhouse electrical service (200A, interior wiring, fixtures, HVAC connections, 2 buildings)	Allowance	2 LS	$25,000 each	$50,000	ASSUMED
Lift station — service entrance, motor starters (2), ATS, and standby generator (25–40 kW)	Allowance	1 LS	$65,000	$65,000	ASSUMED — pump HP unknown (Error Register Item 30); generator allowance
Dog park lighting (2 poles)	Each	2	$4,500 each	$9,000	ASSUMED

Item	Basis	Quantity	Unit Cost	Amount	Tag
Division 26 Subtotal (pre-multiplier)				**$1,883,750**	

*Regional multiplier: 0.89 × $1,883,750 = **$1,676,538***

⚠️ **Warning:** This division carries the lowest confidence in the entire estimate (Low). The actual electrical scope could range from $1.2M to $3.5M depending on transformer configuration, utility participation, metering approach, and lighting design. The Owner must engage a licensed electrical engineer before this number is used for any financing or budgeting purpose.

Division 31 — Earthwork

Confidence: Low — volume cannot be reliably computed from plan text

Explicit statement per task instructions: Earthwork volume cannot be reliably estimated from the plan text provided. The plan set contains grading spot elevations on C009, C010, and C011 but no digital terrain model, cross-sections, or mass haul diagram. Applying a naive plan-area × assumed-depth calculation would systematically undercount by 39–43% per documented industry failure mode. Therefore, earthwork is priced using a combination of:

1. Pond excavation (ASSUMED from C012 bottom vs. existing grade, yielding ~15,500 CY gross — this is a rough order-of-magnitude only, ±40%)
2. Road and pad grading (UNKNOWN net volume — priced as an allowance based on $/acre of graded area and the known 7–8 ft of relief in Phase 1)
3. Fill import allowance (likely required; volume UNKNOWN)

The Owner must commission a licensed estimator to run a digital cut/fill takeoff using the design and existing surface TIN files from Menadier Engineering before using this estimate for financing.

Item	Basis	Quantity	Unit Cost	Amount	Tag
Pond excavation — WRA-1, WRA-2, WRA-3 combined gross excavation (~15,500 CY ASSUMED, ±40%)	$/CY excavate and stockpile	15,500 CY	$8.00/CY	$124,000	ASSUMED — C012 bottom elevations vs. estimated existing grade; see Plan Facts caveat
Road and pad grading — cut and fill on-site (Phase 1 has 7–8 ft relief; VOLUME UNKNOWN)	$/acre rough graded	28 acres	$8,500/ac	$238,000	ASSUMED — industry benchmark for rough grading in rolling terrain with 5–10 ft relief, north Florida

Item	Basis	Quantity	Unit Cost	Amount	Tag
Fill import allowance (likely required for low Phase 2 pad areas and pond berm construction; VOLUME UNKNOWN)	$/CY delivered and compacted	15,000 CY	$22.00/CY	$330,000	ASSUMED — placeholder allowance; actual volume may be 0 to 40,000 CY depending on digital takeoff
Pond berm construction — compacted fill to maintenance berm elevations shown on C012	$/CY placed and compacted	8,000 CY	$12.00/CY	$96,000	ASSUMED — estimated from berm geometry at 3 ponds; C012 OBSERVED
Finish grading of pads, roads, and swales	$/acre	28 acres	$2,500/ac	$70,000	ASSUMED
Subgrade stabilization (12" LBR 40 per D001 under all roads and pads — 22,017 SY road + 11,731 SY asphalt pad = 33,748 SY)	$/SY	33,748 SY	$4.00/SY	$134,992	OBSERVED pavement section D001; ASSUMED includes lime or cement treatment as directed by geotech
Geotextile separator fabric under limerock base (Error Register Item 25 — loose fine sand subgrade)	$/SY	22,017 SY (roads only; pads use concrete)	$3.50/SY	$77,060	ASSUMED — Error Register Item 25 finding; allowance
Unsuitable material removal and off-site disposal (allowance; extent unknown per D002 illegibility)	$/CY	3,000 CY	$28.00/CY	$84,000	ASSUMED — allowance for organic/unsuitable soils under roads and pads; D002 illegible
Proof roll and density testing (included in Div. 01 QC testing above)	—	—	—	$0	
Bathhouse pad raise (Error Register Item 4/20 — raise FF from 98.20 to ≥100.37 ft, requires ~2.2 ft of fill under ~1,000 SF building pad)	Allowance	1 LS	$15,000	$15,000	ASSUMED — Error Register Item 20 specific allowance
Division 31 Subtotal (pre-multiplier)				**$1,169,052**	

*Regional multiplier: 0.89 × $1,169,052 = **$1,040,456***

⚠️ **This division is the single largest source of estimate uncertainty.** If the digital takeoff reveals the site is net fill (which is probable given low Phase 2 areas and pond berm requirements), this number could increase by $150,000–$500,000. If the site is net cut (Phase 1 high area generates usable fill for Phase 2), it could decrease by $100,000–$200,000.

Division 32 — Exterior Improvements

Confidence: Moderate (paving); Low (landscaping/amenities)

Item	Basis	Quantity	Unit Cost	Amount	Tag
PAVING					
Asphalt roads — 1.5" SP-9.5 asphalt (materials and placement)	$/SY	22,017 SY	$14.50/SY	$319,247	OBSERVED area; D001 section; ASSUMED unit cost
Limerock base — 6" LBR 100 (under all asphalt roads, not pads)	$/SY	22,017 SY	$8.50/SY	$187,145	OBSERVED section D001; ASSUMED unit cost
Asphalt RV pads — 1.5" SP-9.5 surface	$/SY	11,731 SY	$14.50/SY	$170,100	OBSERVED area C001; ASSUMED unit cost
Limerock base — 6" LBR 100 (under asphalt RV pads)	$/SY	11,731 SY	$8.50/SY	$99,714	OBSERVED section; ASSUMED
Gravel areas (compacted aggregate, 4" depth)	$/SY	2,516 SY	$18.00/SY	$45,288	OBSERVED area C001; ASSUMED unit cost
Pavement markings — stop bars, double yellow centerline stripes (thermoplastic)	Allowance	1 LS	$18,000	$18,000	OBSERVED: C006/C007/C008 stop bars; C006 "26 LF DOUBLE YELLOW STRIPES" and multiple stop sign locations
Stop signs and street signs (R1-1, directional)	Each	20	$350 each	$7,000	ASSUMED count based on intersection count from C006–C008
FENCING					
Chainlink fence (SD21 detail, 7 ft height per D001) — Phase 1 railroad boundary	$/LF	500 LF	$32.00/LF	$16,000	OBSERVED: C006; ASSUMED length
ADA					
ADA accessible routes, ramps, and	Allowance	1 LS	$35,000	$35,000	ASSUMED — ADA design absent from

Item	Basis	Quantity	Unit Cost	Amount	Tag
site improvements (Error Register Item 12 allowance)					plans; allowance for accessible routes between ADA-designated sites and bathhouse
LANDSCAPING / EROSION CONTROL					
Bahia sod — Argentina Bahia per C002 (all disturbed areas not otherwise paved)	$/SY	60,000 SY	$3.80/SY	$228,000	ASSUMED area (~15 ac disturbed minus impervious = ~13.5 ac sod = ~60,075 SY); ASSUMED unit cost
Silt fence — Type IV single row	$/LF	5,000 LF	$2.50/LF	$12,500	ASSUMED
Silt fence — Type IV double row	$/LF	2,000 LF	$4.00/LF	$8,000	ASSUMED
Inlet protection (filter fabric at 26 inlets)	Each	26	$125 each	$3,250	OBSERVED inlet count
Temporary seeding / erosion mat at pond slopes	$/SY	5,000 SY	$3.20/SY	$16,000	ASSUMED — pond slope stabilization before sod
Division 32 Subtotal (pre-multiplier)				**$1,183,244**	

*Regional multiplier: 0.89 × $1,183,244 = **$1,053,087***

Division 33 — Utilities (Water, Sewer, Storm, Stormwater)

Confidence: Moderate (storm sewer and water main); Low (sanitary sewer including force main allowance; stormwater ponds)

This division prices the full underground utility network including all site service laterals and stub-ups. As noted in the task instructions, missing laterals is the most common source of large underestimates. Every site's water service and sewer lateral are priced here even though only the trunk mains appear on the utility schedules.

33.10 — Water Distribution

Item	Basis	Quantity	Unit Cost	Amount	Tag
6" C900 DR-18 PVC water main (materials, trench, bedding, backfill, compaction)	$/LF	8,500 LF	$38.00/LF	$323,000	ASSUMED length; unit cost includes bedding and backfill
2" SDR-21 PVC branch water main	$/LF	5,000 LF	$22.00/LF	$110,000	ASSUMED

Item	Basis	Quantity	Unit Cost	Amount	Tag
3/4" PVC water service laterals (215 sites × 25 LF avg.) — **THIS IS THE CRITICAL ADD vs. trunk-only**	$/LF	5,375 LF	$18.00/LF	$96,750	ASSUMED — frequently omitted; included per task instruction
Water service connection fittings at each site (corp stop, curb stop, meter box)	Each	215	$320 each	$68,800	ASSUMED
6" gate valves	Each	12	$1,800 each	$21,600	ASSUMED count
2" gate valves	Each	10	$650 each	$6,500	ASSUMED count
Fire hydrant assemblies (5 complete, including 6" lateral and auxiliary valve)	Each	5	$5,500 each	$27,500	OBSERVED count (project brief); ASSUMED unit cost
6" double detector check valve assembly	Each	1	$8,500 each	$8,500	OBSERVED: C013
DI fittings (tees, elbows, reducers — material only; installation in LF unit above)	Allowance	1 LS	$35,000	$35,000	ASSUMED — numerous fittings shown on C013–C015
12-gauge locator wire (C002 requirement)	$/LF	13,500 LF	$0.85/LF	$11,475	OBSERVED requirement; ASSUMED LF = 6" main + 2" main
Chlorination and bacteriological testing	Allowance	1 LS	$12,000	$12,000	ASSUMED — AWWA C651 requirement per C002
33.10 Water Subtotal				**$721,125**	

33.30 — Sanitary Sewer

Item	Basis	Quantity	Unit Cost	Amount	Tag
6" PVC SDR-26 gravity sewer main (materials, trench, bedding, backfill)	$/LF	10,500 LF	$42.00/LF	$441,000	ASSUMED length; unit cost includes select bedding per Error Register Item 15
4" PVC sewer laterals — 215 sites × 25 LF avg. — **CRITICAL ADD**	$/LF	5,375 LF	$28.00/LF	$150,500	ASSUMED — Error Register confirms laterals shown at each site; frequently omitted from estimates

Item	Basis	Quantity	Unit Cost	Amount	Tag
4" PVC sewer lateral connection fittings at main (wye or saddle, per site)	Each	215	$185 each	$39,775	ASSUMED
4" PVC clean-outs (traffic-bearing lids in paved areas)	Each	25	$650 each	$16,250	ASSUMED count — midpoint of 20–25 range from Plan Facts
Precast concrete manholes SSMH 1–21, 4-ft diameter (includes frame, cover, steps, corrosion lining)	Each	21	$7,500 each	$157,500	OBSERVED count; ASSUMED unit cost; includes corrosion-resistant liner per C002
Manhole extra depth (SSMH 6, 7, 8, 9 exceed 10 ft; avg. 3 ft extra depth × 4 manholes)	$/LF extra depth	12 LF	$450/LF	$5,400	ASSUMED — for manholes with inverts near 82–90 ft range
Deep manhole trench shoring (Error Register Item 37 — for excavations >10 ft depth)	Allowance	1 LS	$45,000	$45,000	ASSUMED — Error Register Item 37; 5–6 manholes requiring shoring
Select granular bedding (Error Register Item 15 — pipe class compliance in loose sand)	Allowance on LF unit above	Included	—	$0	Included in $/LF unit cost above at $42/LF
4" PVC clubhouse sewer lateral	$/LF	100 LF	$28.00/LF	$2,800	ASSUMED length
TV inspection of completed sewer (C002 requirement)	$/LF	10,500 LF	$1.80/LF	$18,900	OBSERVED requirement; ASSUMED unit cost
Infiltration/exfiltration testing (C002 requirement)	Allowance	1 LS	$8,500	$8,500	OBSERVED requirement
Utility separation rerouting allowance (Error Register Item 11 — at least 3 conflict zones likely)	Allowance	1 LS	$40,000	$40,000	ASSUMED — Error Register Item 11
33.30 Sanitary Subtotal				**$975,625**	

33.35 — Lift Station and Force Main

Item	Basis	Quantity	Unit Cost	Amount	Tag
Duplex grinder lift station — 72" dia. × 22.5 ft deep fiberglass wet well, Barney's Pumps, 2×80 GPM at 50 ft TDH, duplex controls, ATS	Complete installed	1	$185,000	$185,000	OBSERVED (existence and general spec from D003/project brief); ASSUMED installed cost including excavation and backfill
Lift station wet well excavation and dewatering (deep excavation in sandy soils with potential groundwater at ~83 ft elevation)	Allowance	1 LS	$35,000	$35,000	ASSUMED — Error Register Item 33; anti-flotation and dewatering
Anti-flotation provisions for wet well (Error Register Item 33)	Allowance	1 LS	$10,000	$10,000	ASSUMED
Force main — UNKNOWN routing and length; UNKNOWN discharge destination	**ALLOWANCE ONLY**	UNKNOWN	—	$150,000	UNKNOWN — Error Register Item 1 and Item 9; allowance assumes ~2,000 LF of 4" force main at $45/LF plus connection; if force main is >5,000 LF or requires road crossings, this allowance is grossly inadequate — range is $90,000 to $500,000+
FDEP 62-604 and 62-600 permit application (professional fee — carried in soft costs)	—	—	—	$0	In soft costs
33.35 Lift Station/Force Main Subtotal				**$380,000**	

33.40 — Storm Sewer

Item	Basis	Quantity	Unit Cost	Amount	Tag
15" RCP storm pipe (materials, trench, bedding, backfill, filter-wrapped joints per C002)	$/LF	1,100 LF	$48.00/LF	$52,800	ASSUMED length; OBSERVED pipe size on C009/C010/C011
18" RCP storm pipe	$/LF	700 LF	$58.00/LF	$40,600	ASSUMED

Item	Basis	Quantity	Unit Cost	Amount	Tag
24" RCP storm pipe	$/LF	1,000 LF	$72.00/LF	$72,000	ASSUMED
36" RCP storm pipe	$/LF	200 LF	$98.00/LF	$19,600	ASSUMED
Type C grate inlets (FDOT Index 232, traffic-bearing grates per C002) — 20 units	Each	20	$3,800 each	$76,000	OBSERVED type; ASSUMED count (approximately 20 Type C vs. 6 Type E from C009/C010)
Type E grate inlets	Each	6	$4,200 each	$25,200	ASSUMED count
48" storm manhole (S-3)	Each	1	$6,500	$6,500	OBSERVED: C009
Mitered end sections (8 MES ranging from 15" to 24")	Each	8	$1,200 avg. each	$9,600	ASSUMED count; OBSERVED type from C009/C010
Erosion protection at MES outlets (rip-rap aprons)	Each	8	$2,500 each	$20,000	ASSUMED — not explicitly shown but standard practice at all pond outfall MES
33.40 Storm Sewer Subtotal				**$322,300**	

33.45 — Stormwater Ponds and Structures

Item	Basis	Quantity	Unit Cost	Amount	Tag
WRA-1 pond excavation (~5,300 CY ASSUMED gross)	$/CY	5,300 CY	$8.00/CY	$42,400	ASSUMED — included here for pond-specific earthwork; do not double-count with Division 31 general grading
WRA-2 pond excavation (~4,900 CY ASSUMED)	$/CY	4,900 CY	$8.00/CY	$39,200	ASSUMED
WRA-3 pond excavation (~5,300 CY ASSUMED)	$/CY	5,300 CY	$8.00/CY	$42,400	ASSUMED
Pond side slope sodding (2:1 and 4:1 slopes, total ~20,000 SY estimated)	$/SY	20,000 SY	$4.20/SY	$84,000	ASSUMED — C012 specifies sodded slopes
Pond liner allowance (Error Register Item 34 — potential requirement; UNKNOWN)	Allowance	1 LS	$100,000	$100,000	ASSUMED — Error Register Item 34; if SRWMD does not require liner, this is $0; if required for all 3 ponds at $3–$5/SF, cost is $500K–$870K. This

Item	Basis	Quantity	Unit Cost	Amount	Tag
					allowance is deliberately conservative — confirm with SRWMD before relying on it
Aluminum skimmers (3 pond outlets, one per pond)	Each	3	$4,500 each	$13,500	OBSERVED: C012
33.45 Ponds Subtotal				**$421,500**	

Division 33 Total (all utilities, pre-multiplier): $721,125 + $975,625 + $380,000 + $322,300 + $421,500 = **$2,820,550**

*Regional multiplier: 0.89 × $2,820,550 = **$2,510,290***

Hard Cost Summary

Division	Pre-Multiplier Subtotal	Regional Multiplier	Adjusted Total	Confidence
Div. 01 — General Requirements	$499,600	0.89	$444,644	Moderate
Div. 02 — Existing Conditions	$360,550	0.89	$320,890	Moderate
Div. 03 — Concrete	$1,190,382	0.89	$1,059,440	Moderate
Div. 06/07/08/09 — Buildings	$520,000	0.89	$462,800	Low
Div. 22 — Plumbing	$304,550	0.89	$271,050	Low
Div. 26 — Electrical	$1,883,750	0.89	$1,676,538	Low
Div. 31 — Earthwork	$1,169,052	0.89	$1,040,456	Low
Div. 32 — Exterior Improvements	$1,183,244	0.89	$1,053,087	Moderate
Div. 33 — Utilities	$2,820,550	0.89	$2,510,290	Moderate/Low
HARD COST SUBTOTAL	**$9,931,678**		**$8,839,195**	

Soft Costs

Soft costs are itemized separately from hard costs. The following are based on percentages of hard cost where industry standard applies, or as line items where project-specific information is available from the plan set.

Soft Cost Item	Basis	Amount	Tag
Civil engineering (Menadier Engineering — design, drawings, spec, bid support)	~4.5% of hard cost (typical for this project scale and complexity)	$397,764	ASSUMED percentage; partial OBSERVED (EOR identified on C001)
EOR redesign fees (Error Register Items 3, 4, 5, 13 — storm redesign, pond hydraulics, sewer size, site count correction)	Allowance	$60,000	ASSUMED — Error Register-specific
FDEP permit applications (62-604 collection + 62-600 treatment/disposal + SRWMD ERP modification)	Allowance	$45,000	ASSUMED — Error Register Item 1; includes application fees and EOR preparation time; FDEP app fee alone is typically $1,500–$10,000 depending on system size
Surveying (boundary confirmation, construction staking, as-built certification)	Allowance	$55,000	ASSUMED — C002 requires FL-licensed surveyor as-built; construction staking for 215 sites and 21 manholes is substantial
Geotechnical investigation (supplemental borings for deep manholes and pavement design confirmation — current borings to 15 ft are insufficient for SSMH 9 at 15.86 ft depth per Error Register Item 33)	Allowance	$25,000	ASSUMED — supplemental geotech needed per Error Register
Madison County plan review and permit fees	Allowance	$30,000	ASSUMED — Madison County LDC fees; UNKNOWN exact schedule
Utility tap fees — Madison County water connection fee (one 6" tap on county main)	Allowance	$15,000	UNKNOWN — varies by county; Madison County fee schedule not available from plans
Utility capacity fees (water and sewer — if force main connects to municipal system; 215 new sites)	Allowance	$215,000	UNKNOWN — industry range $500–$2,000/site capacity fee; using $1,000/site as midpoint; could be $0 (on-site WWTP) or $430,000+ (if municipal connection at $2,000/site)
Legal and entitlement (development agreement review, railroad encroachment agreement per Error Register Item 10, easement recording)	Allowance	$35,000	ASSUMED
DOH drainfield abandonment permits (25 tanks per Error Register Item 42)	Allowance	$10,000	ASSUMED — DOH fees plus licensed septic contractor documentation
Construction management / owner's representative (18 months)	3% of hard cost	$265,176	ASSUMED — Owner should have independent CM given the number of critical unresolved issues
Third-party plan review / constructability review (this	Allowance	$15,000	ASSUMED

Soft Cost Item	Basis	Amount	Tag
review exercise)			
ADA compliance consultant (Error Register Item 12)	Allowance	$12,000	ASSUMED
Fire marshal pre-application and variance process (Error Register Item 27)	Allowance	$8,000	ASSUMED
Electrical engineering (Error Register Item 32 — separate electrical design package)	Allowance	$55,000	ASSUMED — 215-site resort electrical design; UNKNOWN actual scope
Financing / construction interest carry (18-month construction loan, ASSUMED 8.5% annual rate on average outstanding balance of 55% of total project cost — rough carry cost)	Calculated	$700,000	ASSUMED — this is a major soft cost driver; actual depends on financing structure and rate
SOFT COST SUBTOTAL		**$1,942,940**	

Contingency and Risk Allowances

Recommended Base Contingency: 25% of Hard Costs

Justification: An AACE Class 4 estimate typically carries 15–25% contingency. This project warrants the full 25% ceiling due to the density and severity of unresolved issues documented in the Error Register:

- 14 Critical findings, 18 Moderate findings
- Entire electrical scope absent (unknown cost, ±40% of the Div. 26 estimate)
- Force main discharge destination unknown (±$350,000 range)
- Pond hydraulic redesign required (SRWMD modification adds 2–6 months)
- Earthwork volume unknown (±30–50% of Div. 31)
- Potential pond liner requirement (±$400K–$700K if required for all 3 ponds)
- 6" vs. 8" sewer pipe resolution could add $45K–$120K

Contingency Amount: 25% × $8,839,195 = **$2,209,799**

Risk Factor Assessment

Risk Factor	Exposure	Rationale
Wastewater treatment approach and permitting	**HIGH**	Force main destination unknown; FDEP 62-604 and 62-600 permits not initiated; if on-site package WWTP required, add $400K–$1.5M and 9–18 months. The single largest schedule risk.
Primary electrical service — utility cost participation, transformer lead time	**HIGH**	Entire electrical scope absent from plans. Transformer lead times in Florida are 16–52 weeks in Q3 2026. Utility participation cost is unknown. Budget range is $1.2M–$3.5M for this scope.

Risk Factor	Exposure	Rationale
Geotechnical unknowns (high water table, unsuitable soils)	**MODERATE**	6 borings to 15 ft confirm loose fine sand; ESHWT 1.5–4 ft. Deep manholes at 15.86 ft approach boring depth limit. No borings in WRA-2/3 area specifically. Underdrain, geotextile, and dewatering are expected but quantities unconfirmed.
Stormwater and floodplain compliance	**HIGH**	SRWMD permit modification required for hydraulic redesign. Zone A BFE not established. Pond liner potentially required. These are agency-controlled items with 2–9 month review timelines.
Entitlement and zoning risk	**LOW**	Plans are signed-and-sealed construction documents submitted to Madison County. R1 zoning is confirmed. RV park expansion is the stated use. However, site count discrepancy (215 vs. 217) requires county resubmittal.
Impact fees and tap fees	**MODERATE**	Madison County fee schedule not available from plans. Water and sewer capacity fees could range from $0 (on-site WWTP) to $430,000+ (municipal connection).
Phasing — Phase 1 vs. Phase 2	**MODERATE**	Phase 1 (88 sites) must be buildable as a standalone operating unit. Lift station serves both phases — it must be front-loaded with Phase 1 even though it is sized for 306 total sites. Shared infrastructure cost allocation: see Phase 1 estimate below.
Railroad ROW setback (Error Register Item 10)	**MODERATE**	Sites 1-68 and 1-69 may need to be relocated. Railroad operator approval timeline is unknown.
Fire lane width (Error Register Item 27)	**MODERATE**	If fire marshal requires 20-ft roads throughout, road redesign eliminates 10–20% of tight Phase 1 sites. Revenue impact could be $60,000–$120,000/year NOI.
Pond liner requirement (Error Register Item 34)	**MODERATE-HIGH**	If SRWMD requires liners on all 3 ponds in permeable sandy soils, add $500K–$870K not currently in the base estimate. The $100K allowance in the estimate is deliberately conservative.

Phase 1 Separate Estimate

Phase 1 includes sites 1-1 through 1-88 (88 sites), WRA-1 pond, Phase 1 storm system (S-1 through S-8), Phase 1 water and sewer distribution, a portion of the lift station (must be fully built in Phase 1 as it serves both phases), and Phase 1 roads and pads.

Shared infrastructure that must be front-loaded in Phase 1:

- Entire duplex lift station and force main (serves all 306 sites)
- Water main backbone connection to county main (serves both phases)

- FDEP permits (must be in hand before Phase 1 sewer is operational)
- SRWMD ERP permit (covers all 3 ponds)
- Electrical primary service and main distribution panel

Phase 1 cost allocation (ASSUMED at approximately 55% of total hard cost for 88/215 sites, plus 100% of shared infrastructure — lift station, force main, primary electrical, FDEP/SRWMD permits):

Category	Phase 1 Allocation	Basis
Phase 1 sites (88/215 = 41% of sites, but Phase 1 has higher earthwork intensity due to 7–8 ft relief)	~50% of Div. 31, 32, 33 direct site costs	ASSUMED
Lift station and force main (100% front-loaded)	$380,000	Phase 1 must carry full cost
WRA-1 pond (Phase 1 stormwater)	~$200,000	Phase 1 specific
Phase 1 roads, pads, utilities for 88 sites	~$3,500,000	ASSUMED
Shared electrical primary service (100% front-loaded)	~$350,000	ASSUMED
Phase 1 soft costs and permits (FDEP, SRWMD)	~$700,000	ASSUMED 65% of soft costs front-loaded
Phase 1 contingency (25%)	~$1,100,000	ASSUMED
Phase 1 Estimated Total	**~$6,230,000**	ASSUMED
Phase 1 Cost per Site (88 sites)	**~$70,800/site**	Higher than overall average due to shared infrastructure front-loading

Phase 2 Estimated Total (balance): $14,220,000 – $6,230,000 = **~$7,990,000** for 127 sites = **~$62,900/site**

Total Project Cost Summary

Category	Amount
Hard Costs (all divisions, regional multiplier applied)	$8,839,195
Soft Costs	$1,942,940
Total Before Contingency	**$10,782,135**
Contingency (25% of hard costs)	$2,209,799
TOTAL PROJECT COST	**$12,991,934**

Cost Per Site and Benchmark Comparison

Metric	This Project	Typical Resort-Tier RV Park (2025–2026)
Hard cost per developed site (215 sites)	$41,112	$40,000–$65,000

Metric	This Project	Typical Resort-Tier RV Park (2025–2026)
Total project cost per developed site (with soft costs and contingency)	$60,428	$50,000–$80,000
Cost per contributing acre (33.868 acres)	$383,613	$250,000–$600,000
Cost per net developed acre (~20 acres ASSUMED)	$649,597	$400,000–$900,000

Benchmark assessment: At $60,428 per developed site with soft costs and contingency, this project sits squarely within the published resort-tier range of $50,000–$80,000/site (RoverPass, Campground Consulting Group, MMCG Invest — as documented in the project brief). The estimate is not an outlier in either direction. However, the following known UNKNOWN items could move the final number significantly:

- If force main requires >3,000 LF to reach discharge point: add $100K–$350K (net effect: +$500–$1,600/site)
- If on-site WWTP is required: add $400K–$1.5M (net effect: +$1,860–$6,977/site) — this would push the project to $62,000–$67,000/site, still within the published range but approaching the upper bound
- If pond liner is required on all 3 ponds: add $400K–$770K above the $100K allowance (net effect: +$1,400–$3,100/site)
- If electrical scope comes in at $3.5M rather than the ASSUMED $1.68M: add $1.82M (net effect: +$8,465/site — this alone would push the total to ~$69,000/site)
- If 8" sewer mains are required throughout: add $45K–$120K
- If fire lane widening eliminates 15 sites: revenue loss, not capital cost

The most important number in this estimate is not the total — it is the range. At Class 4 accuracy (–30% to +50%):

- **Optimistic scenario:** $12,991,934 × 0.70 = **~$9.1M** (~$42,300/site)
- **Pessimistic scenario:** $12,991,934 × 1.50 = **~$19.5M** (~$90,700/site)

The pessimistic scenario is not implausible if on-site WWTP, full pond liners, fire lane redesign, and high electrical costs all materialize simultaneously.

Key Caveats for Human Reviewer

1. **Earthwork volume is the largest single unknown.** Commission a digital cut/fill takeoff from Menadier Engineering's CAD files before finalizing any budget for financing purposes.
2. **Electrical scope ($1.68M ASSUMED) should be treated as a placeholder only.** A licensed electrical engineer must produce a design package before the electrical number can be carried with any confidence.
3. **The force main allowance ($150,000) is almost certainly too low** if the discharge point is more than 1 mile from the lift station. Confirm the discharge destination before committing to any financing based on this estimate.
4. **The pond liner allowance ($100,000 for 3 ponds) is far below the potential actual cost** ($500K–$870K if SRWMD requires liners on all ponds). Confirm liner requirement with SRWMD before land closing.
5. **Soft costs carry a $700,000 construction interest estimate** — this is sensitive to financing rate, draw schedule, and project duration. Any schedule slippage from FDEP permit delays directly increases this line

item.

6. **The Phase 1 bathhouse is carried as an allowance only** — Owner should confirm whether Phase 1 operations will rely solely on the existing clubhouse facilities or require a dedicated bathhouse.

Value engineering

Value-Add and Value-Engineering Analysis

Madison Golf and RV Resort Expansion — Menadier Engineering Job No. 2024-30-001

Built on Plan Facts and Error Register from prior sections of this conversation

---

Preliminary notice: All NOI deltas are directional estimates based on general north Florida RV resort market knowledge and the portfolio context provided (Q1 2026 revenue $342,040, NOI $24,425 for the existing 91-site operation). No third-party appraisal, feasibility study, or rent comp database was attached. Every revenue figure should be stress-tested against a market study before any financing decision. Cost deltas are order-of-magnitude (±25–35%) and build on the confirmed Plan Facts quantities. Mark where a change requires a plan amendment or agency resubmittal.

Base case reference: 215 sites (88 Phase 1 + 127 Phase 2), full-hookup RV resort, asphalt roads and pads, 6" gravity sewer to duplex grinder lift station with unknown force main destination, 6" C900 water backbone, 3 retention ponds totaling 4.00 acres, single 225-SF bathhouse, no electrical plans in set, contributing area 33.868 acres.

A. Cost-Side Value Engineering

A-1. Road Surface: Chip Seal in Lieu of Full-Depth Asphalt on Internal Loops

What changes: Replace the 1.5" SP-9.5 asphalt surface course (D001 detail SD29A) on internal RV loop roads only — preserving full asphalt on the main entry road and the collector spine — with a double-application chip seal (FL DOT Type S-I aggregate, 0.28 gal/SY emulsion, two-pass). The 6" limerock base LBR 100 and 12" stabilized subbase remain unchanged; only the surface course changes. Based on the 198,153 SF (22,017 SY) total road area, the main entry and collector are estimated at ~5,000 SY (ASSUMED: entry road ≈ 300 LF × 22 ft = 7,333 SF = 815 SY; main Phase 1/2 spines ≈ 3,500–4,000 LF × 16 ft = 37,800 SF = 4,200 SY; subtotal ~5,000 SY). Internal loops: ~17,000 SY.

Cost delta:

- Base asphalt surface course on internal loops: 17,000 SY × $9.50/SY (FL north region, 1.5" SP-9.5 installed) = **$161,500**
- Chip seal on internal loops: 17,000 SY × $4.25/SY (double-application, installed) = **$72,250**
- **Hard cost savings: ~$89,000** (confidence: moderate — rural north FL chip seal pricing varies $3.50–$5.50/SY installed)
- Soft cost delta: Zero — same base design, no re-engineering required.

- Maintenance delta: Chip seal requires resealing every 5–7 years (~$3.50–$4.50/SY per application). Over 20 years, net present value of maintenance premium vs. asphalt is approximately $40,000–$60,000. First-cost savings exceed NPV maintenance cost by $29,000–$49,000.

Revenue/NOI delta: None. Guests do not distinguish chip seal from asphalt at resort quality if properly applied and sealed at joints. Moderate confidence.

Schedule effect: Chip seal can be applied after base is complete; no change to critical path. Chip seal cannot be applied in wet weather and requires 48-hour cure — minor sequencing constraint in Florida's summer rainy season.

Regulatory/engineering constraints: No plan amendment required. Chip seal is a construction method, not a design change. Confirm with Madison County that chip seal meets Land Development Code road surface requirements for RV parks — some Florida counties require asphalt for private streets. **Flag for Fire Marshal:** chip seal aggregate must be rolled and sealed before fire apparatus access is accepted. Confirm with Fire Marshal in writing.

Confidence in savings: Moderate. Chip seal is recommended and widely used in FL RV parks; the savings are real but depend on contractor availability in Madison County (limited chip seal contractors in north FL rural market).

Verdict: Accept. Net first-cost savings ~$89K; long-term NPV savings ~$29K–$49K. Low risk if county and fire marshal approve.

A-2. Road Width Reduction: Widen Entry/Collector to 20 ft, Reduce Internal Dead-End Stubs to 14 ft One-Way

What changes: The Error Register (Item 27) identifies that the fire marshal must approve road widths. If the fire marshal accepts a 20-ft minimum on through-roads and allows 14-ft one-way stubs to back-in sites (with engineered turnarounds), approximately 3,000–4,000 SY of internal road area is eliminated. However, NFPA 1 requires 20 ft minimum on apparatus access roads. Reducing below 20 ft on any fire access road is not recommended and is flagged as a rejected sub-option (see below).

Recommended sub-option only: Widen entry and main collectors from 16 ft to 20 ft (required by NFPA 1 — a cost increase, not a savings), while eliminating the very short dead-end stubs (currently labeled 16 ft wide) that serve only 2–4 back-in sites and are not primary fire apparatus routes. These stubs could be reduced to 12–14 ft if the fire marshal accepts a "resident access only, no turnaround required" designation for stubs serving fewer than 4 sites.

Cost delta:

- Widening main entry/collector roads from 16 ft to 20 ft: ~4,200 SY collector × 25% width increase = 1,050 SY additional road × $33/SY (subbase + base + surface, all-in) = **+$34,650** (cost increase)
- Eliminating or narrowing dead-end stubs: estimated 500–800 SY reduction × $33/SY = **–$16,500 to –$26,400**
- Net: **+$8,000 to +$18,000** (net cost increase for full NFPA 1 compliance)

Note: This alternative is primarily a compliance action, not a savings vehicle. It is included because building 16-ft roads and then widening them after fire marshal rejection is far more expensive ($80K–$150K rework) than getting the width right the first time.

Revenue/NOI delta: None.

Regulatory/engineering constraints: Requires fire marshal pre-approval meeting. **Plan amendment required** if road widths change: updated layout sheets C006–C008 must be resubmitted to Madison County. SRWMD resubmittal is not triggered if stormwater contributing areas do not change materially.

Verdict: Accept as a compliance cost, not a savings item. Reject the sub-option of reducing any road below 16 ft without explicit fire marshal written approval.

A-3. Utility Trench Sharing (Water + Sewer + Electric + Comm in Common Trench)

What changes: Currently, water and sewer are shown on separate plan sheets with separate trench alignments, and electrical is entirely absent from the plan set. If the electrical designer (once engaged per Error Register Item 32) coordinates with the civil EOR to route water main, sewer main, electrical conduit, and communications conduit in a shared trench where the code-required 6-ft horizontal separation between water and sewer can still be maintained, the total lineal footage of trench excavation, bedding, and backfill is materially reduced.

Feasibility check: FAC 62-604.400 requires 6-ft minimum horizontal separation between water and sewer. In a shared trench, this means the trench must be at least 7 ft wide to accommodate both 6" pipes at 6-ft center-to-center separation, with electrical conduit placed on the opposite wall from the water main at 12" clearance above the water main. This is achievable in a standard 8-ft-wide trench. Communications (fiber or coax for Wi-Fi backbone) can share the electrical side of the trench with 12" separation from electrical conduit.

Quantity basis: Estimated 9,000–12,000 LF of sewer main plus 7,000–10,000 LF of 6" water backbone = approximately 10,500 LF weighted average where parallel trenching currently occurs. If 60% of this length can be combined (ASSUMED: areas where water and sewer run on the same side of the road, excluding crossing zones and pond-adjacent areas), trench sharing applies to ~6,300 LF.

Cost delta:

- Separate trench excavation and backfill: 6,300 LF × 2 trenches × $18/LF (excavation, bedding, compact backfill, north FL unit rate) = $226,800
- Shared trench for same scope: 6,300 LF × 1 trench × $24/LF (wider trench, same scope) + electrical conduit pull in same trench: ~$151,200
- **Hard cost savings on excavation/backfill: ~$75,600**
- Electrical conduit routing in same trench vs. separate electrical trench: saves approximately 6,300 LF × $12/LF separate electrical trench = $75,600 additional savings
- **Total trench-sharing savings: ~$75,000–$120,000** (confidence: moderate — depends on electrical designer's coordination and actual alignment overlap)
- Soft cost delta: +$10,000–$20,000 for electrical designer coordination and separation diagram (already required per Error Register Item 11).

Revenue/NOI delta: None.

Schedule effect: Requires electrical designer to be engaged before utility contractor mobilizes — adds 60–90 days to pre-construction design phase but does not extend construction schedule if electrical design is completed concurrently with permit resolution.

Regulatory/engineering constraints: FAC 62-604.400 separation must be maintained; requires the EOR separation diagram (Error Register Item 11). NEC 300.5 governs burial depth for electrical conduit (18" minimum for RMC/IMC, 24" for non-metallic). **No plan amendment required** for trench-sharing methodology as long as separation distances remain compliant; however, the combined utility plan view should be added to the plan set as a construction aid. No agency resubmittal triggered.

Verdict: Accept. Net savings $65K–$100K after coordination costs. Requires electrical designer to be engaged immediately.

A-4. Wastewater Alternative: Low-Pressure (Grinder Pump) System vs. Deep Gravity Sewer

What changes: Replace the deep gravity collection system (SSMH 9 at 15.86 ft deep, SSMH 6 at 11.22 ft deep, multiple runs at 1.20% slope requiring significant excavation depth) with a low-pressure sewer (LPS) system using individual 1-HP grinder pump units at each site, discharging to a common 2" or 3" HDPE force main at low pressure. The centralized lift station is retained, but its wet well is smaller and shallower because it receives pre-ground pressurized flow rather than gravity flow from depths of 12–16 ft.

Feasibility assessment: LPS is well-suited to flat or rolling terrain where deep gravity trenching is required — exactly the Phase 2 flat area and Phase 1 rolling terrain. Phase 1 has 8–16 ft of relief and shallow gravity is feasible in the upper portions; LPS is more beneficial in Phase 2 where the terrain is nearly flat and gravity requires the deepest cuts. The project brief confirms 1.5–4 ft estimated seasonal high groundwater — this is the most important LPS advantage: a 2" HDPE force main can be buried at 3–4 ft depth regardless of hydraulic grade, eliminating the need to go below groundwater to achieve gravity slope.

Cost delta (Phase 2 only — 127 sites):

- Current gravity system Phase 2 estimated cost: 127 sites × average 70 LF of 6" gravity main per site × $28/LF installed (6" PVC, bedding, backfill, manholes amortized) = ~$247,000 gravity main + manholes (SSMH 10–21, 12 manholes × $6,000 each = $72,000) = **~$319,000**
- LPS Phase 2: 127 grinder pump units × $2,500/unit installed = $317,500 + 2" HDPE force main backbone ~4,000 LF × $12/LF = $48,000 + smaller centralized wet well allowance -$20,000 = **~$345,500**
- **Net cost delta: approximately +$26,500 (LPS is slightly more expensive in capital)** BUT:
- Excavation savings from eliminating 12–16 ft deep gravity trenching in Phase 2 flat area: estimated $40,000–$80,000 (shallower trenches, no shoring requirement, no SRWMD dewatering for most of Phase 2)
- **Net with excavation savings: LPS Phase 2 saves approximately $13,500–$53,500** (confidence: low — highly dependent on groundwater depth at Phase 2 excavation)

Long-term operating cost delta (negative for LPS):

- 127 grinder pump units × $150/year average maintenance = $19,050/year additional O&M vs. gravity
- At a 7% cap rate, this operating cost increase reduces asset value by ~$272,000

Revenue/NOI delta: None.

Regulatory/engineering constraints: LPS requires FDEP 62-604 design standards for individual grinder pump systems. Each pump unit must be accessible for maintenance. **Plan amendment required:** complete redesign of C016–C018 sewer sheets. Agency resubmittal to FDEP 62-604 required.

Verdict: Reject for Phase 2 as a standalone alternative. The O&M cost penalty ($272K asset value reduction at 7% cap rate) exceeds the capital savings ($13K–$54K). LPS is worth reconsidering only if the gravity system requires extensive dewatering (>$80K) or if soil conditions make deep trenching impractical — confirm after geotechnical review of D002 full boring logs.

A-5. Wastewater Alternative: On-Site Package WWTP vs. Municipal Extension

What changes: Rather than waiting for municipal sewer extension (destination unknown, timeline unknown — Error Register Item 1 / Item 9), the owner commissions an on-site package wastewater treatment plant (WWTP) with land application (drip irrigation spray field on the golf course) or percolation ponds. This resolves the #1 unresolved issue in the plan set.

Feasibility assessment: At 30,600–46,000 GPD (306 total sites × 100–150 GPD), a package WWTP of 40,000–50,000 GPD capacity is required. The 140-acre total parcel (including the 9-hole golf course) has ample land for spray field irrigation — golf courses are routinely permitted as reclaimed water disposal sites under FDEP Chapter 62-610. This approach converts the golf course from a pure loss-making segment into a utility asset that receives free treated effluent for irrigation.

Cost estimate:

- Package WWTP (extended aeration or MBR, 40,000–50,000 GPD): **$400,000–$900,000** installed (confidence: low — highly site-specific; MBR at high end, extended aeration at low end)
- FDEP 62-600 domestic wastewater facility permit + engineering: **$60,000–$120,000**
- Spray field design and installation on golf course (~3–5 acres): **$80,000–$150,000**
- FDEP 62-610 reclaimed water permit: **$20,000–$40,000**
- **Total: $560,000–$1,210,000** (confidence: low)
- Annual O&M for package WWTP: **$40,000–$80,000/year**

Alternative: Municipal force main extension. If Madison (city of ~7,000) has a municipal WWTP within 1–3 miles, a 4" or 6" force main extension may cost **$200,000–$600,000** for construction plus **$50,000–$100,000** in connection fees and engineering. Annual O&M is minimal (gravity and lift station only). This is the lowest-cost path if the distance is feasible.

Revenue/NOI delta:

- Package WWTP eliminates the single biggest schedule risk (6–18 months for FDEP municipal permit). Getting Phase 1 (88 sites) open 6–12 months sooner: 88 sites × $45/night average rate × 65% occupancy × 180 days (seasonal emphasis) = **$3.7M cumulative revenue** over the deferred period. Even at thin margins, accelerating opening by 6 months has enormous revenue value.
- If spray field on golf course eliminates golf course irrigation operating cost ($15,000–$30,000/year currently for irrigation water), NOI improves further.

Regulatory/engineering constraints: FDEP 62-600 permit is non-negotiable regardless of treatment approach. **Plan amendment required:** WWTP siting, spray field design, and force main to WWTP must all be added to the plan set. Agency resubmittal (FDEP 62-600, FDEP 62-610, SRWMD) required. DOH coordination for reclaimed water use on a recreational site.

Verdict: Accept as the preferred path if municipal extension is >1.5 miles or >$400,000. The schedule acceleration value alone ($3.7M deferred revenue risk) makes a $560K–$1.2M on-site WWTP economically justified. If municipal connection is within 1.5 miles, pursue municipal extension first and use WWTP as fallback. Owner must get an answer from EOR on this within 30 days — it is the project's longest lead-time item.

A-6. Stormwater: Dry Detention with Usable Open Space vs. Current Wet Retention Design

What changes: WRA-2 (66,562 SF, 4:1 slopes) and WRA-3 (71,934 SF, 4:1 slopes) with shallow depths of 2.4–2.5 ft are candidates for conversion to dry detention ponds with flat bottoms that function as open recreational space (dog park overflow, disc golf, event lawn) during dry periods and temporarily detain stormwater during storm events. WRA-1 (2:1 slopes, 5.1 ft deep) is not a candidate — too steep and too deep for safe recreational use.

Cost delta:

- Current wet retention design requires: (a) permanent water surface maintenance; (b) erosion control on 4:1 slopes; (c) aluminum skimmer structures ($8,000–$15,000 each); (d) mosquito abatement; (e) aquatic weed management (~$5,000–$10,000/year per pond).
- Dry detention: flat bottom graded at 1% to outlet structure, no permanent pool, grass-lined bottom. Eliminates skimmer ($8,000–$15,000 savings), reduces aquatic weed O&M by 80%. Slightly shallower excavation on WRA-2 and WRA-3 (bottom at 97.00–98.00 ft vs. existing grade at 98.00 ft — minimal excavation already).
- Hard cost savings: **$16,000–$30,000** (2 skimmer structures eliminated; minor grading simplification)
- Annual O&M savings: **$8,000–$16,000/year** (aquatic maintenance eliminated); at 7% cap rate, asset value increase of **$114,000–$229,000**
- Soft cost: EOR redesign of outlet structures for WRA-2 and WRA-3: **+$5,000–$10,000**

Revenue/NOI delta: If dry pond bottoms are used as event lawn or disc golf: low incremental revenue (~$2,000–$5,000/year from event use); more meaningfully, the open recreational space improves resort ambiance and supports a higher rate/occupancy positioning. Directional NOI improvement: **$5,000–$15,000/year** (low confidence).

Schedule effect: None — same construction sequence.

Regulatory/engineering constraints: SRWMD must confirm that dry detention meets the water quality treatment volume requirement previously approved for wet retention. Dry detention typically requires 0.5-inch treatment volume above the permanent pool — if there is no permanent pool, the full treatment volume must be provided in the detained volume. **Plan amendment required** for WRA-2 and WRA-3 outlet structure redesign. **Agency resubmittal to SRWMD required** — this changes the permitted stormwater management system type. Confirm with SRWMD before redesign.

Note: The pond hydraulic redesign required by Error Register Item 4 is already a mandatory redesign. Pursuing dry detention simultaneously costs no additional plan revision fee.

Verdict: Accept, conditional on SRWMD concurrence. Net first-cost savings $11,000–$20,000; annual O&M savings $8,000–$16,000 (NPV at 7% cap rate: $114K–$229K). Total value of alternative: **$125K–$249K.** Pursue concurrently with the mandatory hydraulic redesign.

A-7. Site Pad Surface: Gravel/Compacted Aggregate for Back-In Sites vs. Asphalt

What changes: The cover sheet shows 105,576 SF of asphalt RV pads plus 171,460 SF of concrete. If back-in sites (ASSUMED: ~50% of Phase 2 = ~65 sites) receive a 4" compacted #57 stone gravel pad (with geotextile separator) instead of asphalt, the per-site pad cost decreases. Pull-through sites and all Phase 1 sites retain asphalt pads to maintain a resort-quality appearance.

Cost delta (65 back-in Phase 2 sites):

- Asphalt pad per site: ~1,623 SF average (105,576 SF ÷ 65 back-in ASSUMED) × $7.50/SF (1.5" asphalt over 6" limerock, installed) = **$12,173/site** × 65 sites = **$791,000**
- Gravel pad per site: 1,623 SF × $3.25/SF (4" compacted #57 stone with geotextile, installed) = **$5,275/site** × 65 sites = **$342,875**
- **Hard cost savings: ~$448,000** (confidence: moderate)
- Maintenance delta: Gravel pads require annual top-dressing (~$200/site/year). Over 10 years, NPV of maintenance at 7% cap rate: 65 sites × $200 × 7.02 (10-yr annuity at 7%) = **$91,260**
- Net first-cost savings after NPV of maintenance: **~$357,000**

Revenue/NOI delta (negative): Gravel pads at a resort-tier RV park are a guest experience downgrade. Industry data (RoverPass, Campground Consulting Group) consistently shows asphalt or concrete pads command $5–$15/night premium over gravel pads at comparable resorts. 65 sites × $7.50/night average rate penalty × 180 days × 70% occupancy = **$61,425/year revenue reduction**. At 7% cap rate, this reduces asset value by **$877,500** — far exceeding the $357,000 net construction savings.

Verdict: Reject. The revenue penalty ($877K asset value reduction) vastly exceeds the construction savings ($357K). Gravel pads are a false economy at a resort-tier operation. EXCEPTION: if the project is positioned as a budget-tier park (under $40/night), gravel pads are acceptable. Given the existing golf course and snowbird market context, resort positioning is the right strategy.

A-8. Eliminate Redundant Pipe Class Specification

What changes: C002 General Notes specify SDR-26 PVC for all gravity sewer. The actual design shows 6" mains at 1.20% slope to a maximum depth of ~15.86 ft. At shallower depths (<10 ft, which applies to roughly 70% of the sewer system), SDR-35 PVC (lighter wall, cheaper) is code-compliant under ASTM D3034 and Madison County standards. For pipes at depths >10 ft in loose fine sand, SDR-26 or Class 150 C-900 is appropriate (confirmed per Error Register Item 15).

Cost delta:

- SDR-26 6" PVC vs. SDR-35 6" PVC price difference: approximately $0.35–$0.65/LF material only.

- For 70% of 9,000–12,000 LF system = ~7,350 LF at SDR-35: **$0.50/LF × 7,350 LF = $3,675 savings**
- **Hard cost savings: ~$3,700** (confidence: high — this is a well-established price differential)

Regulatory/engineering constraints: EOR must revise the General Notes to specify SDR-26 for pipe at depth >10 ft and SDR-35 for shallower sections. **No plan amendment** required for agency submissions; this is a specification clarification within the same ASTM D3034 family.

Verdict: Accept, but low priority. Savings are real but small ($3,700). The more important value is the Error Register Item 15 compliance fix — ensuring the deep sections use the correct class. Bundle this with the General Notes correction.

A-9. Eliminate Curb-and-Gutter Where Not Present; Confirm No Hidden Curb Cost in Budget

What changes: The plan set shows no curb-and-gutter (D001 pavement section has no curb; grading relies on sheet flow). This is already value-engineered. The risk is that a GC's estimator or Madison County's inspector asks for curb along collector roads "by custom." Confirm in writing with Madison County that no curb is required for a private RV park road under the Land Development Code.

Cost delta: If curb were required on 4,200 LF of collector road: 4,200 LF × $28/LF (6" concrete curb, FL installed) = **$117,600 avoided cost**. This is already saved by the current design — the value here is confirming it stays saved.

Verdict: Accept — confirm county position in writing before bid. No design change; confirm the county does not require curb on private roads. If county requires curb, add $117,600 to the base cost estimate.

A-10. Over-Specified 36" RCP Trunk Line in Phase 1

What changes: C009 shows "97 L.F. - 36" RCP @ 0.20%" and "25 L.F. - 36" RCP @ 0.20%" as the main Phase 1 storm trunk downstream of S-3 to the pond outfall — a total of 122 LF of 36" RCP at 0.20% slope. A 36" pipe at 0.20% slope conveys approximately 45–50 CFS at full flow. The Phase 1 contributing area is approximately 14–16 acres of the total 33.868 acres. Using TR-55 and a 100-yr storm event for north Florida (approximately 9–10 inches in 24 hours), the rational method peak flow from 14–16 acres at 35% impervious is approximately 20–30 CFS — suggesting a 30" RCP at 0.20% (capacity ~30–35 CFS) may be adequate. However, this requires a hydraulic verification because the mandatory hydraulic redesign (Error Register Item 4) will recalculate the trunk line capacity anyway.

Cost delta:

- 36" RCP vs. 30" RCP, 122 LF: $35/LF differential × 122 LF = **$4,270 savings** (confidence: moderate — FL RCP pricing)
- If 30" is confirmed adequate: **savings ~$4,300**. Bundle this review into the mandatory hydraulic redesign (Error Register Item 4) at no additional engineering cost.

Verdict: Accept for review; savings are minor but the redesign is mandatory. Bundle with Item 4 hydraulic redesign.

B. Revenue-Side Value-Add Alternatives

B-1. Site Count Increase: Convert Golf Course Perimeter to Premium RV Sites (Phase 3 Option)

What changes: The 140-acre total parcel includes the existing 9-hole golf course (from 1953). The Q1 2026 financial data confirms the golf segment is loss-making and stabilizing. If the golf course is eventually decommissioned or reconfigured to 6 holes, the freed perimeter land could accommodate an additional 30–50 premium RV sites with golf course views. This is a Phase 3 concept, not a plan revision for the current document set.

Revenue delta (directional):

- 40 additional premium sites × $65/night (golf-view premium, north FL snowbird market) × 78% occupancy × 200 days (snowbird season extended by golf amenity) = **$405,600/year incremental revenue**
- At 30% NOI margin (established resort with fixed overhead): **$121,680/year NOI**
- At 7% cap rate: **$1,738,000 asset value addition**

Cost to develop 40 sites (Phase 3 estimate):

- Trunk infrastructure already sized (confirm below); 40 sites × $45,000/site = **$1,800,000**
- Net Phase 3 investment vs. value created: **break-even at $1.74M asset value vs. $1.8M cost** — marginal at current assumptions. But the golf course segment is currently losing money; eliminating it saves its annual operating losses (UNKNOWN exact amount from Q1 data provided) and converts the land to income-producing use.

Regulatory/engineering constraints: Decommissioning the golf course requires Madison County zoning review (currently R1 zoning — RV park is a conditional use; confirm golf course is a permitted use and whether its removal affects the conditional use approval). **Plan amendment required for Phase 3.** SRWMD permit modification if new stormwater area is added. NOTE: the current plan set must be evaluated to confirm main infrastructure (sewer, water, road) is sized to serve 306+ sites if Phase 3 is pursued.

Infrastructure optionality cost: The current lift station (2× 80 GPM @ 50 ft TDH = 80 GPM total firm capacity) is designed for 306 sites at ~100 GPD = 30,600 GPD = 21.25 GPM average (80 GPM capacity provides roughly 3.75:1 peaking factor — adequate for 306 sites). Adding 40 sites = 346 total sites = 24.03 GPM average — still within lift station capacity. Water main: 6" backbone at 500+ GPM capacity is adequate for 346 sites. **No infrastructure upsizing required for 40-site Phase 3 if mains are properly looped.** Cost to preserve optionality: essentially zero if road stubs are planned at Phase 2 perimeter.

Verdict: Flag for future planning; not in current plan set. Resolve the golf course operating loss question first — if golf continues as a revenue segment, preserve it; if it remains loss-making after the new RV sites stabilize, convert perimeter to Phase 3 RV sites.

B-2. Premium Site Differentiation: Pond-Front, Pull-Through, and Corner Sites

What changes: Within the current 215-site footprint, designate:

- **WRA-1 pond-front sites:** Sites 1-38 through 1-57 (approximately 20 sites adjacent to WRA-1 based on C009 layout) as "Premium Lakeside" at $15–$25/night premium above standard rate.

- **Pull-through sites** (all pull-throughs in Phase 2) as "Premium Pull-Through" at $10/night premium — no construction cost addition; just pricing strategy.
- **Corner/oversized sites** (end-of-row sites with extra width): approximately 20–30 sites throughout Phase 1 and Phase 2, priced at $8–$12/night premium.
- **50A "Full-Resort" sites:** All sites should receive 50A service as the minimum (see B-4 below); within that, designate the top 40 sites as "Full-Resort" with concrete pad, dedicated 50A/30A/20A pedestal, water/sewer, and a concrete patio slab for $20–$30/night premium.

Revenue delta (directional):

- 20 pond-front sites × $20/night premium × 70% occupancy × 200 days = **$56,000/year**
- 40 pull-through premium × $10/night × 65% occupancy × 200 days = **$52,000/year**
- 25 corner/oversized × $10/night × 65% × 200 days = **$32,500/year**
- 40 Full-Resort × $25/night premium (incremental over standard 50A site) × 75% × 200 days = **$150,000/year**
- **Total incremental revenue: ~$290,500/year** (confidence: low — market rents not confirmed; directional only)
- At 35% NOI margin: **$101,675/year NOI addition**
- At 7% cap rate: **$1,452,500 incremental asset value**

Cost to implement:

- Site premium designation: $0 (pricing only for pond-front, pull-through, corner)
- Full-Resort sites (40 sites, add 10'×20' concrete patio slab per site): 40 × 200 SF × $6.50/SF concrete = **$52,000** additional
- Marketing and signage for premium tiers: **$5,000–$10,000**
- **Total cost: $57,000–$62,000**
- **ROI on incremental capital: $101,675 NOI ÷ $60,000 = 169% first-year cash return on incremental investment.** This is the highest-return alternative in this analysis. Confidence: low-moderate (market support not confirmed).

Regulatory/engineering constraints: No plan amendment required for pricing strategy. Concrete patio additions require only a minor concrete scope addition to the construction contract — no permit revision if within the approved impervious area budget. CONFIRM: current impervious area is 35.08% (517,522 SF). Adding 40 × 200 SF = 8,000 SF of concrete patio increases impervious to 525,522 SF (35.62%) — still well within most SRWMD limits. Confirm with SRWMD that this is within approved limits.

Verdict: Accept immediately. Highest-return alternative. Implement during design development — add patio slabs to premium site layout and update impervious area calculation.

B-3. Park-Model and Tiny-Home Sites (Year-Round Revenue)

What changes: Designate 20–30 sites in Phase 2 (the flatter, more sheltered area near WRA-2 and WRA-3) as "park-model ready" sites — wider (30–35 ft wide vs. standard 20–25 ft), with a full concrete pad (20'×45'), 200A electrical service capable of supporting a park-model RV (RVIA-certified, ≤400 SF), and full hook-up. Park

models can be rented as resort-owned units at nightly or monthly rates, or sold as long-term "own your site" arrangements.

Revenue delta (directional):

Option 1 — Resort-owned park models:

- 20 park-model units × $120/night rental rate × 75% occupancy × 250 days = **$450,000/year**
- Less unit cost amortization (20 units × $80,000/unit ÷ 15 years) = $106,667/year depreciation
- Less O&M ($10,000/unit/year): $200,000/year
- **Net NOI contribution: ~$143,333/year** (confidence: low — highly market-specific)
- At 7% cap rate: **$2,047,619 asset value addition** (net of park-model cost)

Option 2 — "Own your lot" annual lease:

- 20 sites × $800/month annual lease = $192,000/year
- At 35% NOI margin: $67,200/year
- At 7% cap rate: $960,000 asset value
- Much lower capital requirement (no park-model purchase)

Cost to implement (20 sites — incremental vs. standard RV site):

- Pad upgrade: standard asphalt pad to 20'×45' concrete pad (900 SF): 20 sites × 900 SF × $2.00/SF incremental (concrete vs. asphalt delta) = **$36,000**
- Electrical upgrade: standard 50A to 200A service per site: 20 sites × $800/site incremental (larger conduit, larger breaker, heavier conductor) = **$16,000** (ASSUMED — no electrical design in set)
- Site widening (from standard 20 ft to 30 ft): approximately 5–8 sites may be eliminated in the Phase 2 layout to provide width. At $45,000/site lost: **–$225,000 to –$360,000** (this is the dominant cost)

Net: The site loss from wider park-model lots makes this financially marginal unless the lost sites are end-of-row sites that would have been low-value anyway. Recommend limiting to 10 park-model sites at end-of-row Phase 2 locations where site loss is minimized.

Regulatory/engineering constraints: Park-model RVs (RVIA-certified) can occupy RV sites under Florida law without a change of use, but the site width change and 200A electrical require plan revisions. **Plan amendment required** if site layout changes. Madison County zoning should be confirmed for park-model occupancy terms (Florida DOH Rule 64E-29 governs park-model placement).

Verdict: Accept in limited form (10 end-of-row sites, Option 2 annual lease). This avoids the park-model purchase capital requirement while testing demand. If annual lease achieves 100% fill rate in Year 1, scale to 20 sites in Phase 3.

B-4. Submetering All Sites for Water, Sewer, and Electric

What changes: Install individual electric meters (or smart submeters) and water meters at every site. Currently no metering approach is specified (electrical design is absent; water service is 3/4" per site with no meter

callout on C013–C015). Submetering converts utility cost from a fixed overhead into a variable, billable pass-through.

Revenue/NOI delta:

- Electric: industry data (Wild Energy, cited in project brief) shows 20–40% reduction in effective utility spend with payback in one season to 12 months. For 215 sites averaging $8/night in unbilled electric (snowbird with air, heat, TV): 215 × $8 × 70% occupancy × 200 days = $240,800/year unbilled. Recovering 70% via submetering: **$168,560/year incremental NOI** (confidence: moderate — widely documented)
- Water: similar recovery of metered consumption vs. flat-rate hookup: estimated **$20,000–$40,000/year** incremental NOI
- **Total annual NOI addition from submetering: $188,000–$208,000/year** (confidence: moderate)
- At 7% cap rate: **$2,686,000–$2,971,000 incremental asset value**

Cost to implement:

- Smart electric submeters: 215 sites × $350/meter installed (ASSUMED — no electrical design; typical FL RV park submeter cost) = **$75,250**
- Water meters at each site: 215 sites × $280/meter installed = **$60,200**
- Metering software/cloud platform: **$5,000–$10,000** setup + $2,000–$4,000/year subscription
- **Total capital: ~$140,000–$145,000**
- **Payback: less than 9 months at $188K/year NOI addition**

Regulatory/engineering constraints: Florida law allows submetering of RV sites by private operators (Florida Statutes 366.02 exemption for RV parks). Individual electric meters must meet ANSI C12.1 if used for billing. Water submeters must be AWWA-certified. **No plan amendment required** — submeters are a construction addition, not a design change. However, the electrical designer (when engaged per Error Register Item 32) must incorporate submeter panels into the electrical design from the start.

Verdict: Accept unconditionally. This is the single highest-NOI-per-dollar-invested alternative in the entire analysis. ROI on incremental capital exceeds 100% in Year 1. Must be designed into the electrical plan from day one.

B-5. Ancillary Revenue: Dog Park Premium Membership, Dump Station for Non-Guests, Laundry, Vending

What changes: The dog park shown on C008 and C011 is a free amenity as currently designed. Additional ancillary revenue streams:

- **Dog park premium membership:** $25–$50/stay premium for pet-friendly sites (approximately 40 sites designated pet-friendly). Revenue: 40 × $35 × 60% × 200 days = **$168,000/year**. Note: this is partially captured in rate — many resorts simply charge a pet fee of $5–$10/night. At $7.50/night for 60% of sites: **$96,750/year**. Confidence: moderate.
- **Dump station for non-guests:** $15–$25/use. A dump station is not shown in the current plan set (UNKNOWN). If added (cost ~$8,000–$15,000 per station including water supply and sewer connection), and if the resort is the only dump station within 15–20 miles of I-10 (confirm with market research): 5–15 uses/day × $20 × 200 days = **$20,000–$60,000/year**. Confidence: low.

- **Laundry:** Not in current plan set. A 4-unit laundry room ($25,000–$40,000 installed) generates $0.25–$0.50/site/stay. 215 sites × $0.35 × 180 days × 70% occupancy = **$9,512/year**. Low revenue but high guest satisfaction. Confidence: moderate.
- **Wi-Fi as a billing service:** $5–$10/night Wi-Fi fee generates: 215 × $7.50 × 70% × 200 days = **$225,750/year**. Alternatively, include in rate and use as a marketing differentiator. The golf course clubhouse (being rebuilt) presumably has fiber or cable internet — extend backbone to RV sites via licensed wireless access points. Cost: **$30,000–$60,000** for a site-wide Wi-Fi backbone. Confidence: moderate.

Total ancillary revenue potential: $331,000–$432,000/year (confidence: low-moderate; highly market-specific)

Verdict: Accept dog park fee and Wi-Fi backbone. Defer dump station and laundry to Phase 1 stabilization. Add Wi-Fi backbone to electrical design scope (must be coordinated with electrical engineer from day one).

B-6. Golf Course as Resort Amenity (Stop-the-Bleeding Strategy)

What changes: The golf course is currently loss-making. Rather than closing it, rebrand it as the primary resort differentiator for snowbird long-term guests. A 9-hole course adjacent to an RV resort commands $15–$25/night premium in the southeast FL market. Package golf + RV site as a "Golf & Stay" rate.

Revenue delta (directional):

- If golf course membership/daily fee revenue improves by $50,000/year due to new RV guest captive market (215 sites × 50% with golf interest × 3 rounds/stay × $15/round = directional estimate): **+$50,000–$80,000/year** from golf-RV cross-sell
- Rate premium for "Golf & Stay" package on premium 40 sites: $15/night × 75% × 200 days × 40 sites = **$90,000/year**
- Less golf course additional maintenance from increased play: **–$10,000–$20,000/year**
- **Net NOI addition: $120,000–$150,000/year** (confidence: low — depends entirely on golf market in Madison County, FL; population ~19,000)
- At 7% cap rate: **$1,714,000–$2,143,000 incremental asset value**

Cost: No construction cost addition (golf course rebuild is a separate project — permits 26-0243, 26-0261). Investment is in marketing and programming.

Verdict: Accept as a marketing strategy only. Zero incremental capital in this project's scope. The golf course amenity is a competitive moat that no competitor can replicate quickly in this market. Prioritize keeping the golf course operational during RV expansion construction.

C. Phasing and Capital Alternatives

C-1. Minimum Viable Phase 1 (MVPhase) That Can Open and Cash-Flow

What changes: The current Phase 1 (88 sites, 1-1 to 1-88) is a natural phase boundary. However, within Phase 1 the construction sequence has dependencies:

Critical path for Phase 1 opening:

1. Resolve wastewater discharge destination and obtain FDEP 62-604/62-600 permits (longest lead time: 6–18 months)
2. Complete Phase 1 gravity sewer (SSMH 1–9, C016) and force main to discharge point
3. Complete Phase 1 water distribution (C013)
4. Complete Phase 1 road grading and paving
5. Obtain electrical permit and install Phase 1 pedestals and transformers
6. Obtain Madison County CO for Phase 1

Can a smaller MVPhase open earlier? Sites 1-1 through 1-66 (the main Phase 1 cluster) share the same sewer trunk and are the most buildable. Sites 1-67 through 1-88 (near the railroad ROW and in the transitional zone to Phase 2) have the railroad setback issue and are lower priority. Technically, 66 sites could open before 88 if railroad issues are unresolved.

Minimum Phase 1 that cash-flows:

- Fixed overhead for a resort (on-site management, utilities, insurance, debt service): estimated **$15,000–$20,000/month** based on the existing 91-site operation's Q1 data ($342K revenue ÷ 3 months = $114K/month; $24,425 NOI ÷ 3 months = $8,142/month NOI on 91 sites. The expansion infrastructure adds fixed cost.)
- To cover expanded fixed overhead of $18,000/month: need 88 sites × $45/night × 70% occupancy = $2,772/night = $83,160/month revenue — well above break-even. Even 40 sites at 70% occupancy: $1,260/night × 30 days = $37,800/month, covering fixed costs.
- **MVPhase = 40–50 sites.** The project can cash-flow with half of Phase 1 open.

Phasing recommendation:

- **Phase 1A (40 sites, 1-1 to 1-40):** Open first; lower elevation area, shortest sewer trunk run to lift station, least earthwork. Target opening: 12–18 months after mobilization.
- **Phase 1B (48 sites, 1-41 to 1-88):** Open 6 months after Phase 1A once revenue covers construction draw.
- **Phase 2 (127 sites):** Build from Phase 1 operating cash flow 18–24 months after Phase 1A opening.

Infrastructure that must be front-loaded (full Phase 1+2 sizing from day one):

- Lift station wet well (72" × 22.5 ft) — already sized for full system; do not reduce
- Force main from lift station to discharge point — full diameter, full route, from Phase 1A day one
- 6" water backbone (Phase 1 loop) — full Phase 1 design on day one
- Main entry road — full width, full pavement section from day one
- Stormwater ponds — WRA-1 and WRA-2 must be complete before Phase 1A opens (drainage area includes both phases)

Infrastructure that can be deferred:

- Phase 2 water distribution (C014, C015): defer until Phase 2 construction begins
- Phase 2 sewer distribution (C017, C018): defer until Phase 2 construction begins
- Phase 2 roads and pads: defer

- Bathhouse: defer bathhouse to Phase 1B — Phase 1A sites can use construction-era portable sanitation and/or temporary shower facilities for the first 6 months
- Dog park: defer to Phase 2

Schedule and cost impact: No cost increase — this is a sequencing optimization. Potential schedule acceleration of 3–6 months to Phase 1A opening vs. building all 88 sites before opening.

C-2. Developer-Owned WWTP as a Sellable/Financeable Asset

What changes: If the owner builds an on-site package WWTP (Alternative A-5), the WWTP becomes a regulated utility asset. In Florida, a private wastewater utility serving a residential park can be sold to the local municipality at a regulated rate base (Florida PUC Chapter 367), or can be classified as a "Customer-Owned Utility System" and depreciated/financed separately from the real estate.

Asset value of a permitted WWTP:

- A FDEP-permitted 40,000–50,000 GPD WWTP in Florida has been sold to municipalities at $1.5–$3.0/gallon of permitted capacity (industry benchmark, low confidence)
- 50,000 GPD × $2.00/gallon = **$100,000 regulated rate base** (note: this is modest — most value is in the rate base calculation, not a simple multiple)
- More practically: the WWTP eliminates the project's #1 regulatory risk, allows 100% of sites to open, and generates the NOI that funds its own cost via discounted cash flow. The WWTP is not primarily a sellable asset; it is an enabling asset.

Financing optionality:

- A FDEP-permitted WWTP can be collateralized separately from the real estate. Some rural development lenders (USDA Rural Development, CoBank) will lend against a utility asset at 25-year amortization. At $750,000 WWTP cost at 6.5% for 25 years: monthly debt service = **$5,040/month**. Annual debt service of $60,480 against 215 sites generating $188,000/year in submeter revenue alone — fully covered.
- The WWTP structure also supports a USDA Rural Development Section 306C (water/wastewater) grant for rural communities. Madison County, FL qualifies as a rural area (population <19,000). USDA 306C grants cover up to 75% of eligible WWTP project costs for underserved rural communities. **If this grant is pursued, the WWTP capital cost could be reduced by $300,000–$700,000.** This is the highest-leverage financing option available to this project. Confidence: moderate (grant eligibility must be confirmed with USDA Rural Development Gainesville, FL field office).

Verdict: Accept WWTP + USDA grant application as the capital structure recommendation. If the USDA grant covers 50–75% of the WWTP cost, the net WWTP cost to the owner is $150,000–$400,000, which is recovered in Year 1 from the submeter NOI. Apply for the grant concurrently with FDEP permit application.

C-3. Infrastructure Choices That Foreclose Future Expansion — Preserve Optionality

The following current plan elements foreclose Phase 3 expansion (golf course conversion) and should be reviewed:

Element	Current Design	Optionality Issue	Cost to Preserve Optionality
Lift station capacity	2× 80 GPM firm	Adequate for 306 sites; 40 Phase 3 sites push to ~346 sites at 24 GPM avg — still within firm capacity	$0 — already adequate
Force main diameter	Unknown (not designed)	If 2" force main, limited to ~80 GPM; if 4" force main, adequate to 200+ GPM (560+ sites)	Upsize to 4" force main from day one: ~$3,000–$8,000 additional material cost for 1,000–3,000 LF
6" water backbone	Phase 1 loop; Phase 2 branch	A looped 6" system handles ~500 GPM; adequate for 346 sites. Dead-end Phase 2 NW branch should be stubbed with a gate valve at the northwest terminus for future extension	$500–$1,500 per stub with gate valve
Road dead-ends	Phase 2 NW cluster has one entry/exit	If Phase 3 golf course conversion adds 40 sites, a second road connection to the northwest is needed	Install road stub-out at Phase 2 NW perimeter with 20-ft paved stub and gate at Phase 2 boundary: $5,000–$12,000
WRA-2 and WRA-3 location	Central Phase 2 area	Ponds in the current location are permanent; if Phase 3 uses the golf course land, no pond conflict. If pond location changes, SRWMD permit modification required	$0 — no conflict with Phase 3
SSMH locations	21 manholes throughout	Phase 3 needs only a new gravity branch connection to an existing SSMH; locate SSMH 20 or 21 near the Phase 3 boundary	$0 — already positioned adequately

Total cost to preserve Phase 3 optionality: $8,500–$21,500. This is a trivial investment relative to the Phase 3 asset value potential ($1.7M–$2.1M at 7% cap rate from B-1).

Verdict: Accept. Spend $8,500–$21,500 now to preserve $1.7M–$2.1M of future optionality.

Alternatives to Reject — Summary

Alternative	Why Rejected
A-4: LPS system Phase 2	O&M penalty ($272K NPV at 7% cap rate) exceeds capital savings ($13K–$54K); maintain gravity system
A-7: Gravel pads on back-in sites	Revenue penalty ($877K asset value loss) exceeds construction savings ($357K) by 2.5×
Reducing any road below 16 ft	NFPA 1 fire access requirement; safety hazard; no savings justify the risk
Overhead electrical secondary	Aesthetics downgrade at a resort-tier operation; incompatible with resort positioning for snowbird market; savings (~$30K–$60K) do not offset revenue penalty
Septic cluster systems	Soils in WRA-2/3 area are clayey sand/clay (poor perc); total flow 30,600–46,000 GPD far exceeds any reasonable septic cluster

Alternative	Why Rejected
	system; DOH permit ceiling of ~10,000 GPD is a hard constraint

Recommended Alternative Plan

Top 5 Changes That Define the Recommended Plan

Change 1: Resolve wastewater via on-site package WWTP + USDA 306C grant application (A-5 + C-2)

- Eliminates the #1 schedule and cost risk
- WWTP cost: $560K–$1.21M gross; net after USDA grant at 50%: $280K–$605K
- Enables 6–12 months earlier Phase 1A opening
- Revenue value of acceleration: ~$3.7M deferred revenue risk avoided
- Plan amendment required: force main + WWTP added to plan set; FDEP permits filed

Change 2: Submeter all 215 sites for water and electric (B-4)

- NOI addition: $188K–$208K/year
- Capital cost: $140K–$145K
- Payback: <9 months
- Must be integrated into electrical design from day one
- No plan amendment required; electrical design addition only

Change 3: Implement premium site pricing tiers with patio slab upgrades (B-2)

- NOI addition: $101,675/year on $60K incremental capital
- Highest first-year ROI in this analysis (169%)
- Minor concrete scope addition; no plan amendment if within impervious budget

Change 4: Chip seal internal loop roads (A-1) + dry detention WRA-2 and WRA-3 (A-6)

- Combined hard cost savings: $89K + $16K–$30K = $105K–$119K
- Annual O&M savings: $8K–$16K/year (NPV $114K–$229K at 7% cap rate)
- Chip seal: confirm county and fire marshal approval before bid
- Dry detention: bundle with mandatory hydraulic redesign (Error Register Item 4)
- Plan amendment required for WRA-2/3 redesign; SRWMD resubmittal required

Change 5: Preserve Phase 3 optionality via infrastructure stub-outs (C-3)

- Cost: $8,500–$21,500
- Optionality value preserved: $1.7M–$2.1M
- No plan amendment required; add to construction scope

Recommended Site Count, Cost Per Site, and Total

Recommended site count: 215 sites (no change — the plan set's site count is appropriate for the parcel and market; premium pricing extracts more value per site without adding regulatory complexity of density increase)

Base case cost estimate (from Plan Facts and standard industry benchmarks):

Category	Base Case Estimate	Confidence
Earthwork (clearing, grading, ponds)	$800,000–$1,200,000	Low (no earthwork calc in set)
Roads and paving (asphalt)	$1,200,000–$1,600,000	Moderate
Water distribution	$400,000–$600,000	Moderate
Sewer (gravity + lift station)	$600,000–$900,000	Moderate
Stormwater (inlets, pipe, ponds)	$400,000–$600,000	Moderate
Electrical (UNKNOWN — estimated)	$1,500,000–$3,500,000	Low
WWTP (UNKNOWN in base; required)	$560,000–$1,210,000	Low
Bathhouse (revised to ~1,000 SF)	$150,000–$250,000	Moderate
Soft costs (design, permits, legal)	$250,000–$400,000	Moderate
Contingency (15%)	$580,000–$1,240,000	—
Base case total	**$6,440,000–$11,500,000**	Low
Base case per site	**$29,953–$53,488/site**	Low

Note: The wide range reflects the UNKNOWN electrical scope (a $1.5M–$2.0M swing) and the UNKNOWN WWTP cost. The base case is not credibly priceable without the electrical design and force main destination — as flagged throughout the Error Register.

Recommended plan adjustments:

Change	Cost Delta
Chip seal internal loops (A-1)	−$89,000
Trench sharing water/sewer/electric (A-3)	−$75,000 to −$100,000
Dry detention WRA-2/3 (A-6)	−$11,000 to −$20,000
WWTP (midpoint; net of 50% USDA grant)	+$442,500
Submeter infrastructure (B-4)	+$140,000
Premium site patio slabs (B-2, 40 sites)	+$52,000
Phase 3 stub-outs (C-3)	+$15,000
Bathhouse upsizing to 1,000 SF (B-2 support)	+$75,000 incremental
Net adjustment	**+$449,500 to +$465,500**

Recommended total cost: $6,889,500–$11,965,500 (midpoint: **~$9.4M**) **Recommended cost per site (215 sites): $32,044–$55,653/site** (midpoint: **~$43,700/site**)

Note: The per-site cost of $43,700 at the midpoint is at the mid-range of the $40,000–$80,000 resort-tier benchmark cited in the project brief. The low end ($32,000) reflects the scenario where the electrical scope is developer-friendly (utility-provided primary, minimal developer-installed infrastructure) and the WWTP is substantially funded by the USDA grant. The high end ($55,653) reflects a full developer-installed electrical system and minimal WWTP grant. The electrical scope is the single largest budget unknown — the electrical designer's first deliverable (a load calculation and preliminary design) should be required within 30 days of engagement.

Savings vs. base case midpoint:

If base case midpoint = ($6,440,000 + $11,500,000) ÷ 2 = $8,970,000, and recommended plan midpoint = $9,427,500:

- The recommended plan is slightly more expensive at midpoint (+$457,500) because it adds the mandatory WWTP and submeter infrastructure that are not in the current (deficient) plan set.
- However, the recommended plan generates $290,000–$430,000/year in additional NOI (submetering + premium sites + ancillary revenue minimum), which at 7% cap rate adds **$4,143,000–$6,143,000** to asset value.
- **Net value creation vs. base case: $3.7M–$5.7M** on $457K incremental investment. This is the core value engineering conclusion: the savings from chip seal and trench sharing are real but small; the value is in the revenue side.

Ideas and ROI

Value Engineering & Revenue Enhancement Report

Madison Golf and RV Resort Expansion — Within Approved Plans Only

Jennings Realty / Jay Lapat | Menadier Engineering Job No. 2024-30-001

Preliminary notice: Every idea below was tested against the approved plan set. Nothing proposed changes site count, road alignment or width, grading, utility routing, stormwater facilities, impervious area, building footprints, setbacks, buffers, access points, or any condition of approval. Separate trade or building permits are acceptable per the task instructions. All revenue and cost assumptions are stated explicitly with evidence grade. Do not rely on these figures as a bid, appraisal, or engineering document.

Market assumptions (applied throughout):

- Average nightly rate (ANR) base: $55–$65/night for a full-hookup resort site in rural north Florida (Madison County), based on comparable Florida Panhandle/I-10 corridor RV resorts reviewed on Campspot/Hipcamp (unverified comps; Evidence Grade D). Snowbird seasonal rate (Nov–Mar): $900–$1,200/month.

- Occupancy: 78–82% winter (Nov–Mar, 5 months), 22–26% summer (Jun–Sep, 4 months), 50–60% shoulder (Oct, Apr–May, 2 months). Blended annualized occupancy: ~52–58% of available site-nights.
- Cap rate applied: **7.5%** (rural Florida RV resort stabilized; Evidence Grade B — RV park CBRE/Marcus & Millichap reported cap rates 6.5–8.5% for stabilized RV resorts, 2023–2024; unverified URL).
- Site count for revenue modeling: **215 sites** (per confirmed plan facts; Phase 1: 88 sites, Phase 2: 127 sites).
- All costs in 2025–2026 north Florida construction dollars.
- "Incremental" means above what the base approved plans already include.

Part 1 — Revenue-Generating Ideas

Part 1A: Amenity Revenue Table

The table below covers every amenity — those already on the plans (evaluated as drawn) and additions within the approved envelope — ranked by cash-on-cash return on incremental capital. "Already on plans" items are marked with ◆.

Definitions used:

- NOI contribution = Annual revenue added minus annual O&M cost of the amenity itself (not site operating expenses).
- Cash-on-cash = Annual NOI contribution ÷ Installed cost (midpoint).
- Payback = Installed cost (midpoint) ÷ Annual NOI contribution.
- Value creation = Annual NOI contribution ÷ cap rate (7.5%).

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
1	**Wi-Fi backbone (enterprise mesh, all sites)**	$18,000–$35,000	$85–$165/site for enterprise outdoor mesh nodes (Ubiquiti UniFi or Cambium; 6–8 nodes for 215 sites on 33 ac); $3,000–	$1,200–$2,400/yr (ISP commercial circuit ~$800–$1,500/yr + node maintenance)	Guest-facing amenity: (a) $3–$5/night Wi-Fi fee charged through reservation system to ~60% of guests = ~$3–$5 × 0.60 ×	$78,000	222%–433%	0.23–0.45	$1,040,000	ARVC 2023 Campground Owner Survey: Wi-Fi ranked #1 amenity by guests (source cited in RVBusiness.com coverage; unverified	No permit contact (low-voltage data network; no structural change)	Moderate (revenue assumes 60% fee capture; actual may vary by pricing model)

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
			$5,000 fiber backbone from clubhouse to Phase 2 cabinet; installation $5,000–$10,000		215 sites × 189 occupied nights/yr = $73K–$122K gross; OR (b) include in rate and raise base rate $3–$5/night across all sites, increasing ANR by ~5–8%. Conservative: $3/night premium × 215 sites × 189 occupied nights = $122K gross, discounted to $80K for partial adoption and rate sensitivity. NOI = $80K – $2K					URL). KOA 2024 North American Camping Report notes connectivity as top campground selection factor (koa.com/about/press/; unverified). Evidence Grade: B		

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
					O&M = $78K.							
2	**Premium site designation — pond-front sites (Phase 1 WRA-1 adjacent: sites ~1-38 to 1-53; Phase 2 WRA-2/3 adjacent: sites ~2-28 to 2-57)**	$500–$2,000/site for interior upgrades (concrete patio finish upgrade, fire pit ring, privacy hedge planting within pad footprint)	Concrete patio surface upgrade: $1.50–$2.50/SF × ~400 SF patio area within approved pad = $600–$1,000/site; fire pit ring (prefab steel, set in existing gravel): $150–$300; privacy arborvitae or wax myrtle (2-gal pots, 3-ft spacing): $200–$500/site. Total: $950–$1,800/site for ~60 pond-adjace	$200–$400/yr for all 60 sites (mulch refresh, plant replacement)	Rate premium: $10–$20/night over base for "pond-view premium" sites, or $100–$200/month premium for snowbird seasonal. 60 sites × $15/night premium × 189 occupied nights × 60% adoption (guests willing to pay) = $102,060 gross. Less O&M = ~$101,660 net.	$101,000	75%–155%	0.65–1.33	$1,347,000	KOA 2024 Annual Report notes waterfront/view sites command 15–25% rate premium (koa.com/about/press/; unverified). OHI/NCC data shows premium site designation increases blended ADR 8–12% (unverified). Evidence Grade: B	No permit contact (finish selection and plantings within approved pad and open space)	Moderate (premium capture depends on marketing and booking platform tier designation)

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
			nt sites									
3	**Laundry facility (coin-op or app-pay, installed in or immediately adjacent to approved bathhouse footprint)**	$12,000–$22,000 for 4 washers + 4 dryers (commercial Speed Queen or Dexter) plus plumbing rough-in, electrical panel tap, and vending controller	Speed Queen commercial washer ~$1,400–$1,800 each; dryer ~$1,200–$1,600 each; 4+4 = ~$11,200–$17,600 equipment; installation (plumbing, electrical) $3,000–$6,000	$1,500–$2,500/yr (maintenance contract, water/sewer, electricity)	Ancillary fee income: $2.50–$4.00/wash cycle, $2.00–$3.00/dry. At 215 sites, ~1.5 loads/occupied site/week: 215 × 0.52 occupancy × 52 weeks × 1.5 loads × $4.50 avg cycle = $173,000 gross. Discounted 40% for low-season and actual usage = $65,000–$85,000 gross. O&M = $2,000. NOI =	$73,000	332%–608%	0.16–0.30	$973,000	ARVC 2023 survey: laundry is top-5 amenity for long-term and snowbird guests. Laundry ROI data from campground operator forums (unverified; Evidence Grade C). Speed Queen commercial pricing confirmed via Speed Queen .com distributor pages (unverified URL). Evidence Grade: C	Separate trade/building permit (plumbing and electrical permit for laundry addition within approved bathhouse footprint)	Moderate (revenue estimate sensitive to actual site occupancy and laundry frequency)

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
					$63,000–$83,000.							
4	**EV charging — 2 Level 2 EVSE stations at approved parking/arrival area**	$6,000–$14,000 for 2 dual-port Level 2 EVSE units (ChargePoint CT4000 or Blink IQ200), pedestal mount, conduit to existing electrical panel, and meter	EVSE hardware: $1,500–$3,000/unit × 2 = $3,000–$6,000; installation (conduit, panel tap, permit): $3,000–$8,000	$400–$800/yr (network fee, maintenance)	Fee income: $0.25–$0.35/kWh or $2–$4/hr. Estimated 8–10 sessions/day × 365 days × $5 avg session revenue = $14,600–$18,250 gross. NOI = $13,800–$17,450. Note: primary value is marketing differentiation and occupancy lift for EV-towing guests.	$15,600	111%–260%	0.38–0.90	$208,000	RVIA 2023: 18% of RV owners tow or drive EVs; EV charging is fastest-growing campground amenity request (rvia.org; unverified URL). ChargePoint network fee and pricing confirmed via chargepoint.com (unverified). Evidence Grade: B	Separate trade/building permit (electrical permit for EVSE installation)	Low (rural north Florida EV demand is lower than national average; occupancy lift is speculative)
5	**Dog park fencing and amenity**	$3,500–$7,000 for 6-ft chain-	Chain-link 6-ft: $8–$15/LF × 400 LF =	$300–$600/yr (waste bag repleni	Occupancy lift: 65% of RV campe	$13,820	159%–395%	0.25–0.63	$184,000	KOA 2024 North American Campi	Separate trade/building permit	Moderate

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
	package on approved open space (C008 shows dog park label) ◆	link perimeter fence on existing dog park footprint (est. 100'×100' = 400 LF fence), double-gate entry, waste bag dispensers (×4), and agility equipment (2 jumps, 1 tunnel, 1 weave pole set)	$3,200–$6,000; double-gate entry: $400–$800; agility equipment: $300–$600; waste dispensers: $150–$300; gravel/base for muddy seasons: $500–$1,000. Total $4,550–$8,700	shment, fence repair, gravel top-dress)	rs travel with pets (KOA 2024). A fenced dog park with agility is a booking-decision factor. Conservative occupancy lift: +1.5% (1 additional site-night per week) = 52 site-nights/yr × $60 ANR = $3,120 incremental revenue. Plus reduces turnover for pet-owning long-term guests (value: 2 additional snowbird					ng Report: 65% of campers travel with pets, pet-friendly amenities cited as top-5 decision factor (koa.com/about/press/; unverified). Evidence Grade: B	(fence permit)	

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
					contracts/yr at $1,100/mo × 5 mo = $11,000). Total: ~$14,120.							
6	**Golf-course fishing access and catch-and-release pond signage (WRA ponds shown on plans)**	$1,500–$4,000 for fish stocking (bass/bream/catfish fingerlings), 2–3 fishing access points with gravel paths (within approved gravel/open space), rod holders, catch-and-release signage	Fish stocking: $800–$1,500 (local FL Fish & Wildlife certified supplier); gravel access path (50 LF × 4 ft wide = ~22 SY × $15/SY): $330 per access point × 3 = $990; rod holders and signage: $300–$600. Total: $2,090–$3,090	$400–$800/yr (annual restock, path maintenance)	Rate premium: $3–$5/night "fishing pond" premium for 40–60 pond-adjacent sites (already captured in Item 2 above — do not double-count). Standalone: fishing is a booking draw that lifts occupancy by ~0.5% site-wide = 26 additional site-nights/yr ×	$960	31%–46%	2.2–4.2	$12,800	FL Fish & Wildlife recreational fishing data (myfwc.com; unverified). Evidence Grade: D (ROI inference from analogous campground fishing amenity comps).	No permit contact (fish stocking in private retention ponds; check FL FWC rules; paths on approved gravel/open space)	Low (revenue is incremental occupancy lift which is hard to isolate)

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
					$60 = $1,560. Primarily a marketing differentiator — ROI is occupancy lift rather than direct fee income. NOI = $1,560 – $600 O&M = $960.							
7	**Propane vending / "campers market" at entrance or maintenance area**	$4,000–$8,000 for a locked propane exchange cage (Blue Rhino/AmeriGas vendor-supplied, no capital cost to owner) + a small honor-system snack/supply cabinet	Propane cage: vendor-supplied at $0 capital (revenue share model). Supply cabinet (lockbox + initial inventory $500): $1,500–$2,000. Signage: $300–$500	$200–$400/yr (restocking, shrinkage)	Commission income: 15–25% of propane sales. 215 sites × 0.52 occupancy × $1.50 avg commission/site/stay = $84/site-yr × 215 = ~$18,000 gross commission (rough estimate). Plus	$23,000	288%–575%	0.17–0.35	$307,000	Blue Rhino/AmeriGas franchise commission structures (vendor-confirmed in operator forums; unverified). Evidence Grade: C	No permit contact for revenue-share propane (vendor-owned cage); separate permit may be required for fixed propane storage above 1 lb per local fire code —	Moderate

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
					camp supply sales $5,000–$10,000. Total: $22,000–$26,000. NOI = $21,600–$25,600.						check with fire marshal.	
8	**Pickleball court striping and net posts on approved concrete pad (if a suitable slab exists within open space)**	$800–$2,500 for thermoplastic court striping on existing or planned concrete slab (20'×44' per court), portable net system, and storage box	Thermoplastic striping: $400–$800; net system (USAPA-spec): $250–$500 × 2 nets = $500–$1,000; storage box: $200–$400; safety surfacing if adding: $0 (use existing concrete per plans). Total: $1,100–$2,200	$150–$300/yr (net replacement, striping refresh every 3–5 yrs)	Occupancy lift + marketing: pickleball is the fastest-growing recreational sport in the US (SFIA 2023). A two-court setup within the approved concrete area lifts bookings for the 55+ snowbird demographic (Madison's primary	$13,750	550%–1,718%	0.06–0.18	$183,000	SFIA 2024 Sports Participation Report: pickleball participation grew 51.8% in 3 years (thesfia.com; unverified). KOA 2024 cites activity courts as top amenity for 55+ campers (koa.com; unverified). Evidence Grade: B. NOTE: this	No permit contact (striping and portable net on existing slab — no structure added)	Low (dependent on whether an appropriate slab exists; revenue lift is inferred)

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
					market). Conservative: +1% blended occupancy lift = 52 additional site-nights/yr × $60 = $3,120. Plus reduces snowbird turnover (+2 contracts/yr = $11,000). Total: ~$14,000 NOI.					idea requires a suitable existing or planned concrete slab within approved open space. C007/C008 show concrete areas — confirm a 20'×44' area is available before specifying.		
9	**Premium site — end-cap and corner sites with concrete patio pavers (within approved pad footprint) and cedar privac**	$800–$1,600/site for ~8–12 end-cap/corner sites in Phase 1 and Phase 2	Patio paver upgrade (within approved concrete area, 200 SF): $4–$8/SF materials + $3–$5/SF labor = $1,400–$2,600 for pavers	$300–$500/yr for all 12 sites (stain/seal pavers, screen maintenance)	Rate premium: $8–$12/night for designated end-cap/corner "premier" sites. 12 sites × $10/night × 189 occupied nights × 70% premium	$18,476	77%–154%	0.65–1.30	$246,000	Datacomp/JLT 2023 MH/RV rent survey cites end-cap and corner site premiums of 8–15% over base (datacomp.com; unverified).	No permit contact (paver finish within approved pad footprint; cedar screens are furniture/FF&E)	Moderate

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
	y screen		over existing concrete subbase. Cedar privacy screen (pre-built 6'×8' panels, 2–3 per site): $200–$400/site. Total per site: $1,600–$3,000. 12 sites × $2,000 mid = $24,000 total.		capture = $15,876. Plus snowbird premium: 4 sites × $150/mo × 5 mo = $3,000. Total: $18,876 – $400 O&M = $18,476.					Evidence Grade: B		
10	**Reservation system integration (Campspot, RoverPass, or Campli fe) with dynamic pricing**	$0–$3,000 setup fee + $1,200–$3,600/yr SaaS	Campspot: ~$0–$1,500 setup, $200–$400/mo. RoverPass: commission-based (5–8% of online bookings). Campli fe: ~$150/mo.	$1,200–$3,600/yr ongoing subscription/commission	Dynamic pricing lifts ANR 10–18% during peak periods (shoulder and holiday weekends) without raising off-peak	$57,500	1,593%–4,792%	0.02–0.06	$767,000	RoverPass operator data: dynamic pricing lifts ANR 10–20% (roverpass.com/blog; unverified). Campspot publishes averag	No permit contact (software/operations)	High (direct revenue; well-documented in industry)

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
					rates. Conservative: +8% ANR during 60 days of peak shoulder occupancy = 215 sites × 60 nights × 0.80 occupancy × $60 ANR × 8% lift = $49,766 incremental revenue/yr. Online booking reduces front-desk labor ~0.25 FTE = ~$8,000–$12,000 labor savings. Total NOI: $55,000–$60,000.					e revenue uplift data (campspot.com; unverified). Evidence Grade: B		
11	**Snowbird/long-term**	$0 additional capital	—	$0 additional O&M	Structured 5-month	$60,000	N/A (no capital)	Immediate	$800,000	ARVC 2023: snowbird	No permit contact	High

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
	seasonal package pricing (monthly rate cards)				(Nov–Mar) snowbird package at $950–$1,100/month = $4,750–$5,500/season/site. At 30% of Phase 1+2 sites (65 sites) as long-term seasonal: 65 × $5,125 avg = $333,125/season vs. transient at $60/night × 150 nights × 80% occ = $288,000 for same sites. Net gain: $333,125 – $288,000 = $45,125. Additional					seasonal rates average $850–$1,300/month in Florida (arvc.org; unverified). Evidence Grade: B	(pricing/operations)	

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
					benefit: reduced front-desk and turnover costs for long-term guests (~$15K/yr labor savings). Total NOI: ~$60,000.							
12	**Shade trees planted along site access drives (native live oak, planted within approved open space strips between sites)**	$8,000–$18,000 for 40–60 live oaks (3-inch caliper, Florida-Friendly)	$150–$250/tree (3" caliper live oak, Florida nursery stock) + $30–$60/tree labor = $180–$310/tree × 50 trees avg = $9,000–$15,500. Within approved open space (no impervious added).	$500–$1,000/yr (watering first 2 years, mulch, pruning)	10-year benefit: mature shade increases site desirability and rate sustainability. Conservative: $2–$4/night shade premium for 80 shaded sites after 5–7 years of growth. Near-term: market	$10,000 (10-yr avg)	56%–125%	0.80–1.75	$133,000	Florida-Friendly Landscaping program data on live oak growth rates (gardeningsolutions.ifas.ufl.edu; unverified). Evidence Grade: D (rate premium inference from shaded site comps at comparable FL	No permit contact (landscaping within approved open space; no impervious area added)	Low (long time horizon; rate premium is speculative)

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
					ing value ("shaded sites" as a filter on Campspot/Hipcamp). Estimated NOI lift once mature: $2.50 × 80 sites × 189 occupied nights × 60% shade premium capture = $22,680/yr. Year 1–4: minimal ($0–$5,000). Average over 10-year period: $10,000/yr.					resorts).		
13	**Entry monument landscaping and signage package (within**	$8,000–$20,000 for monument sign (externally illuminated, cast	Monument sign (fabricated): $4,000–$10,000; landscape beds	$500–$1,000/yr (seasonal replanting, mulch, bulb replacement)	Marketing value: professional entry appearance increases Campspot/Go	$5,670	28%–71%	1.4–3.5	$75,600	KOA 2024 notes online ratings directly impact booking rates (koa.com;	Separate trade/building permit (sign permit for illuminated monu	Low (revenue is indirect; marketing value is real but hard to

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
	approved sign/entry area footprint)	stone or concrete masonry, within approved sign footprint), flanking native landscape beds, solar-powered LED uplighting	(mulch, native plants, seasonal color): $2,000–$5,000; LED uplighting (solar): $800–$2,000; irrigation stub-in: $500–$1,000. Note: monument sign within approved location; no footprint change.		ogle Maps rating by estimated 0.2–0.4 stars, which KOA data correlates with 3–5% occupancy lift. Conservative: +2% occupancy on 215 sites = 107 additional site-nights/yr × $60 = $6,420. Plus brand value for snowbird direct-booking repeat rate. NOI: $6,420 – $750 O&M = $5,670.					unverified). Evidence Grade: D (occupancy lift from entry appearance is inference).	ment sign)	isolate)
14	**Golf course — add a simple club**	$2,000–$5,000 for rental club	Used club sets: $100–$200/set ×	$500–$1,000/yr (equipment mainte	Green s fee: $15–$25/round for	$28,000	560%–1,400%	0.07–0.18	$373,000	NGCOA (National Golf Cours	No permit contact (operational	Moderate (actual participation varies;

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
	and ball rental program and charge a green s fee to non-resident resort guests ◆	sets (10 sets), ball dispenser, and simple POS at clubhouse	10 = $1,000–$2,000; ball dispenser/vending: $500–$1,000; POS tablet: $300–$500; signage: $200–$400	nance, ball replenishment)	resort guests. 215 sites × 0.52 occupancy × 30% participation × 2 rounds/stay avg × $20 fee = $6,708/yr. Plus rental: $5–$10/set × 30 rounds/day × 100 days (peak season) = $15,000–$30,000. Total: $21,708–$36,708. O&M = $750. NOI = $20,958–$35,958.					e Owners Association) 2023 data: resort-amenity courses average $15–$35 greens fees for 9-hole layouts (ngcoa.org; unverified). Evidence Grade: C	program; no physical change)	golf is loss-making currently per portfolio context)
15	**◆ Dog park (already labeled on C008) — upgrade to dual-zone**	$1,200–$2,500 incremental (dividing fence panel within existin	Dividing fence (50 LF of 4-ft chain-link): $400–$750; second gate: $200–	$100–$200/yr incremental	Incremental to Item 5: dual-zone separation is a strong differentiator	$1,360	58%–113%	0.88–1.76	$18,133	ARVC 2023: pet-amenity segmentation (small/large dog zones) listed	Separate trade/building permit (additional fence within existing	Low

Rank	Amenity	Installed Cost Range	Unit Cost Basis	Annual O&M	Revenue Mechanism	Annual NOI Contribution	Cash-on-Cash	Payback (yrs)	Value Creation @ 7.5%	Source / Evidence Grade	Permit Classification	Confidence
	(large/small dog separation with a second dividing fence run)	g dog park footprint)	$400; signage: $100–$200; ground cover (pea gravel in small-dog zone): $500–$1,000. Total incremental above basic dog park (Item 5): $1,200–$2,350.		for guests with small dogs (typically higher-income 55+ demographic). Estimated additional +0.5% occupancy lift beyond basic dog park = 26 additional site-nights × $60 = $1,560. Reduces negative reviews from small-dog incidents. NOI: $1,360					as top upgrade for premium campgrounds (arvc.org; unverified). Evidence Grade: C	footprint)	

Most Cost-Effective Amenity — Analysis

#1: Reservation system with dynamic pricing (Rank 10 in table) yields the highest cash-on-cash return (1,593%–4,792%) because it requires near-zero capital and generates ~$57,500 in incremental NOI per year primarily through rate optimization and labor reduction. It beats all others on ROI-per-dollar because:

- Capital cost is negligible ($0–$3,000 setup).
- The revenue lift is broad (affects all 215 sites, all seasons).
- The mechanism is proven and directly measurable.
- No construction, no permit, no lead time — can be implemented before the first site is occupied.

Close second: Wi-Fi backbone (Rank 1 in table) has the highest absolute NOI contribution ($78K) and second-highest cash-on-cash (222%–433%), with a payback under 6 months at mid-cost.

On the current plans — value engineering opportunity:

The approved bathhouse (15'×15' = 225 SF, C007) is the single approved vertical structure in the civil set. As drawn, it is severely undersized for a 127-site Phase 2 and will require a building permit with full architectural drawings regardless (per the errors register). Since the footprint is fixed at 225 SF by the approved plans, the value engineering opportunity is in **fixture specification within that footprint:**

- Specify commercial-grade Kohler or American Standard fixtures (mid-range, not luxury) rather than a luxury-tier finish: saves $2,000–$5,000 in fixture cost with identical durability.
- Specify **one-piece fiberglass shower units** (Swanstone or equivalent) rather than tile shower surrounds: saves $1,500–$3,000 in labor and dramatically reduces future grout maintenance cost.
- Total savings vs. a "resort-tier luxury finish" spec: **$3,500–$8,000** with zero reduction in guest satisfaction or durability.
- Note: this does not change the footprint, location, or any condition of approval.

Part 2 — Reducing Future Maintenance and Capital Costs

All items below are selections of materials, specifications, equipment, finishes, and operations within the approved plan footprint. None changes what the approval was issued on.

Rank	Item	Installed Cost Delta vs. Base	Unit Cost Basis	Annual O&M Savings	Avoided/ Deferred CapEx	Payback (yrs)	Source / Evidence Grade	Permit Classification	Confidence
1	**LED site lighting with photocell /timer controls (specify LED at every approved light pole location)**	+$50–$150/fixture vs. HPS/MH equivalent	LED area light (40W, 5,000 lm): $180–$280 vs. 150W HPS at $130–$180; delta = +$50–$100/fixture. For 30–40 site lights (estimated from road network):	$2,800–$4,200/yr electricity savings (150W HPS → 40W LED × 40 fixtures × 4,380 hrs/yr × $0.12/kWh = $3,153/yr savings). Plus lamp replacement: HPS replaces every 2–3	Avoid HPS fixture replacement at 10 yr: $5,200–$7,200 (40 fixtures × $130–$180). LED life = 50,000+ hrs (11+ yrs).	0.4–1.4	DOE Solid-State Lighting Program: LED area lighting saves 50–70% vs. HPS in comparable applications (energy.gov/eere/ssl; unverified).	Separate trade/building permit (electrical permit for site lighting)	High

Rank	Item	Installed Cost Delta vs. Base	Unit Cost Basis	Annual O&M Savings	Avoided/ Deferred CapEx	Payback (yrs)	Source / Evidence Grade	Permit Classification	Confidence
			+$1,500–$4,000 total.	yrs ($30–$60/lamp × 40 = $1,200–$2,400 avoided every 3 yrs).			Evidence Grade: A		
2	**Bahia sod specification (already in C002 General Notes) — confirm Argentine Bahia, not common Bahia, and add centipede grass in shaded areas**	$0–$500 incremental (species substitution, same sod price)	Argentine Bahia: $0.25–$0.40/SF installed; Centipede: $0.25–$0.40/SF. Same cost, better performance.	$3,000–$6,000/yr savings in irrigation water (Bahia is drought-tolerant, requires ~30% less supplemental water than St. Augustine in north FL). Reduced fertilizer: $800–$1,500/yr.	Avoids turf renovation (St. Augustine fails in drought without irrigation system): $15,000–$40,000 renovation cost at 5–7 years deferred indefinitely.	<1 yr	University of Florida IFAS: Argentine Bahia is the recommended low-maintenance turf for north Florida RV parks and open areas (gardeningsolutions.ifas.ufl.edu/plants/turfgrasses/bahiagrass/; unverified). Evidence Grade: A	No permit contact (plant species selection within approved landscaping scope)	High
3	**Commercial-grade metered pedestal specification (Milbank, Eaton, or Siemens 50A/30A/20A with tamper-resistant breakers and individual metering lugs)**	+$50–$150/pedestal vs. basic 30A pedestals	Quality pedestal (Milbank 50/30/20A with meter lug): $350–$500 vs. basic 30A pedestal: $200–$300; delta = +$150–$200/pedestal. 215 pedestals × $175 mid delta = $37,625 total.	$0 direct O&M savings (same operation), but eliminates pedestal replacement at 8–10 yrs: basic pedestals corrode and fail in humid north FL; premium stainless or fiberglass-encased pedestals last 15–20 yrs.	Defers 215-pedestal replacement by ~8 yrs: $43,000–$64,500 avoided at year 10.	4–7 yrs (on avoided replacement cost basis)	RVIA equipment longevity data (unverified). Eaton/Milbank product specifications (eaton.com, milbank.com; unverified). Evidence Grade: B	Separate trade/building permit (electrical permit for pedestal installation — already required)	High

Rank	Item	Installed Cost Delta vs. Base	Unit Cost Basis	Annual O&M Savings	Avoided/ Deferred CapEx	Payback (yrs)	Source / Evidence Grade	Permit Classification	Confidence
				Savings: avoid $200–$300 replacement × 215 pedestals = $43,000–$64,500 avoided at year 10 vs. year 18.					
4	**Schedule 40 PVC for site water service laterals instead of standard polyethylene (already specified as PVC in plans; confirm throughout)**	Already specified — confirm 3/4" PVC SDR-21 is used at every lateral connection, not HDPE or PE tubing which degrades faster in FL heat	$0 incremental if spec is followed as written in C002	Eliminates UV-degradation failure of PE laterals at 7–12 yrs; PVC SDR-21 lasts 25–40 yrs in buried service. Avoided: 215 lateral replacements at $150–$300 each = $32,250–$64,500 at year 10.	$32,250–$64,500 deferred to year 25+	Immediate (spec compliance)	AWWA Manual M23 on PVC water service pipe longevity (awwa.org; unverified). Evidence Grade: A	No permit contact (spec compliance within approved water distribution design)	High
5	**Fiberglass manhole barrels (instead of precast concrete) at the deepest manholes (SSMH 6, 7, 8, 9) in the sewer system**	+$800–$2,000/manhole vs. precast concrete for the 4 deepest structures	Fiberglass MH (9–16 ft depth): $3,500–$6,000 each vs. precast concrete equivalent: $2,500–$4,000 each; delta = +$1,000–$2,000/structure × 4 = +$4,000–$8,000 total.	$0 annual O&M savings directly, but eliminates H2S-driven concrete corrosion at deep manholes in a gravity sewer system. Precast concrete MH in a wastewat	Relining 4 deep manholes at $3,000–$6,000 each = $12,000–$24,000 deferred from year 15 to year 40+.	2–7 yrs (on avoided relining cost basis)	NASSCO (National Association of Sewer Service Companies) publishes H2S corrosion rate data for gravity sewer manholes (nassco.org; unverified).	No permit contact (material substitution within approved sewer design; same dimensions and capacity)	Moderate

Rank	Item	Installed Cost Delta vs. Base	Unit Cost Basis	Annual O&M Savings	Avoided/ Deferred CapEx	Payback (yrs)	Source / Evidence Grade	Permit Classification	Confidence
				er environment corrodes from H2S (sulfuric acid attack) and requires relining at 15–20 yrs.			Evidence Grade: B		
6	**Pond bank sod spec: confirm Bahia and add erosion control blanket (S150 or equivalent) on all 2:1 and 4:1 slopes before sod establishment**	+$0.05–$0.15/SF for erosion control blanket on pond slopes	Erosion control blanket (S150 biodegradable): $0.08–$0.15/SF installed. Pond bank area: WRA-1 (2:1 slopes, est. 8,000 SF bank area) + WRA-2/3 (4:1 slopes, est. 12,000 SF): ~20,000 SF × $0.12 = $2,400.	Eliminates erosion re-stabilization events (estimated 1–2 events/yr at $500–$2,000 each without blanket on steep pond banks in FL rainfall). Savings: $750–$2,000/yr.	Avoids pond bank re-grading and re-seeding at 3–5 yrs: $5,000–$15,000 deferred or avoided.	0.3–3 yrs	FDOT Erosion Control Blanket specifications confirm S150 is appropriate for 2:1 and 4:1 slopes in Florida (fdot.gov; unverified). Evidence Grade: A	No permit contact (erosion control spec within approved pond construction scope)	High
7	**Specify corrosion-resistant (fiberglass or HDPE) road sign posts instead of aluminum**	+$5–$15/post vs. standard aluminum round posts	C002 specifies "2" ROUND ALUMINUM POSTS, 14-FT IN LENGTH" for traffic signs. Fiberglass round posts (2" dia., 10-ft): $25–$45 vs. aluminum $15–$25;	Aluminum posts in north Florida's humid environment with salt-laden fog (from occasional Gulf influence) corrode at bases and require replacement every 7–10 yrs.	Avoids $600 sign post replacement at year 8, recurring.	4–8 yrs	FDOT tested fiberglass sign posts vs. aluminum in humid Florida climates (fdot.gov; unverified). Evidence Grade: B	No permit contact (material substitution for sign posts — same geometry, same reflectivity requirement in C002)	Moderate

Rank	Item	Installed Cost Delta vs. Base	Unit Cost Basis	Annual O&M Savings	Avoided/ Deferred CapEx	Payback (yrs)	Source / Evidence Grade	Permit Classification	Confidence
			delta = +$10–$20/post × 30 estimated sign locations = +$300–$600 total.	Fiberglass lasts 25+ yrs. Savings: 30 posts × $20 replacement cost avoided at year 8 = $600 every 8 yrs ≈ $75/yr.					
8	**Specify self-priming 2" trash pump for maintenance building instead of submersible dewatering pumps**	+$300–$800 for one quality self-priming pump (Honda or Tsurumi 2") vs. multiple submersibles	Honda WT20 trash pump: $600–$900 vs. 2–3 submersible utility pumps at $150–$300 each = $300–$900. Delta: $0–$300.	Saves pump replacement cost: submersible utility pumps in Florida's sandy/silty groundwater fail every 2–3 yrs (impeller wear). A quality self-priming pump lasts 8–12 yrs with annual maintenance. Savings: $150–$300/yr in avoided submersible replacement.	Avoids 3 submersible replacements at 3-yr intervals over 10 yrs = $450–$900 avoided.	1–3 yrs	Honda Power Equipment product longevity data (powerequipment.honda.com; unverified). Evidence Grade: C	No permit contact (equipment selection for maintenance operations)	Moderate
9	**Wax myrtle and native sparkleberry hedgerows for site screenin**	$0–$500 incremental vs. ornamental options	Wax myrtle (Morella cerifera) 3-gal: $8–$12/plant; sparkleberry (Vacciniu	$5,000–$10,000/yr in reduced irrigation and shearing labor once	Avoids irrigation system expansion to serve ornamental hedgerows: $15,000–	0.5–2 yrs (based on avoided Leyland cypress maintenance)	UF/IFAS Florida-Friendly Landscaping database confirms wax myrtle as zero-	No permit contact (plant species selection within approved landscaping and	High

Rank	Item	Installed Cost Delta vs. Base	Unit Cost Basis	Annual O&M Savings	Avoided/ Deferred CapEx	Payback (yrs)	Source / Evidence Grade	Permit Classification	Confidence
	g (instead of invasive or high-maintenance ornamentals)		m arboreum) 3-gal: $10–$15/plant. Both are native FL, no irrigation required after establishment, deer-resistant. Compare to Leyland cypress (common alternative): $15–$25/plant but requires irrigation and annual shearing ($50–$100/plant/yr labor). Over 100 plants (screening for premium sites): savings = $5,000–$10,000/yr in avoided irrigation and shearing labor.	established (yr 3+).	$40,000 deferred.		irrigation-required once established (gardeningsolutions.ifas.ufl.edu; unverified). Evidence Grade: A	open space)	
10	**Specify HDPE corrugated pipe with smooth interior (ADS N-12 or equivalent) for all storm**	-$2–$5/LF vs. RCP for 12"–24" diameter	C002 specifies PP pipe (AASHTO MP21-11) for storm drain — ADS N-12 HDPE qualifies. HDPE 15": $8–	$0 additional O&M savings (HDPE and RCP have comparable life in this application).	Primary benefit is upfront cost savings (-$10,000 – $15,000), not O&M reduction. HDPE avoids	Immediate (cost savings at installation)	AASHTO MP21-11 is the governing standard per C002; ADS N-12 HDPE meets this standard (ads-pipe.com;	No permit contact (material substitution within approved storm drain design; must confirm C002 and	High

Rank	Item	Installed Cost Delta vs. Base	Unit Cost Basis	Annual O&M Savings	Avoided/ Deferred CapEx	Payback (yrs)	Source / Evidence Grade	Permit Classification	Confidence
	drain runs instead of RCP where plans allow		$12/LF vs. RCP 15": $12–$18/LF; delta = -$4–$6/LF savings. For ~2,500 LF of 15"–18" runs: savings = $10,000–$15,000. Note: 36" runs on main trunk may require RCP for load capacity — confirm with EOR.		joint cracking at minor settlement.		unverified). Evidence Grade: A	SRWMD permit compliance; spec already allows PP pipe)	

Part 3 — Implementation

Implementation Matrix

#	What Gets Specified/Installed	Where on Approved Plans	Trade	Construction Sequence	Permit	Phase	Rough-In Now for Deferral	Rough Cost	Lead Time	Post-Opening O&M
R1	Reservation/dynamic pricing software (Campspot or RoverPass)	Operations — no physical location	Owner/operator	Pre-opening (before first guest)	No permit	Phase 1 pre-opening	N/A	$0–$3,000 setup	2–4 weeks to configure	Monthly subscription; 1 FTE for reservation management; rate management software training
R2	Wi-Fi enterprise mesh (6–8 Ubiquiti UniFi U6-LR outdoor	Clubhouse building (existing) as PoP; nodes mounted on	Low-voltage contractor	After light poles and electrical service are energize	No permit (low-voltage data; verify with AHJ if	Phase 1 (backbone and Phase 1 nodes); Phase 2 nodes at	Install 1.25" conduit sleeves from clubhouse to Phase 2	$18,000–$35,000 installed	4–8 weeks (equipment order + install)	ISP contract ($800–$1,500/yr); annual node firmware

#	What Gets Specified/Installed	Where on Approved Plans	Trade	Construction Sequence	Permit	Phase	Rough-In Now for Deferral	Rough Cost	Lead Time	Post-Opening O&M
	nodes, fiber backbone from clubhouse to Phase 2 distribution cabinet)	approved light poles or fence posts at site boundaries		d; before Phase 1 opening	conduit is in ground)	Phase 2 opening	during utility trenching — add to sewer/water trench as a joint-trench item			updates; 1 hr/wk monitoring
R3	Snowbird monthly rate card (5-month package, $950–$1,100/mo)	Operations	Owner/operator	Before first Nov snowbird season	No permit	Phase 1 (implement immediately at opening)	N/A	$0	2 weeks to set up in reservation system	Reservation system lease management module; 1 FTE front desk
R4	Premium pond-front site designation (sites ~1-38 to 1-53 and ~2-28 to 2-57): concrete patio finish upgrade, fire pit ring, wax myrtle privacy hedge	Within approved pad footprints and open space strips (C006, C007, C009, C010)	Concrete contractor (patio finish); landscape contractor (hedges); GC (fire pit rings)	Patio finish: after concrete pads are placed and cured, before final landscaping. Fire pit rings: after grading, before sod. Hedges: final landscaping phase.	No permit (within approved pad and open space footprint)	Phase 1 pond sites at Phase 1; Phase 2 pond sites at Phase 2	Specify patio broom finish vs. standard finish on concrete bid for designated pond-front pads — zero rough-in cost difference; fire pit ring sleeves (4" conduit stub) can be placed during slab pour for future gas connection	$57,000 – $108,000 total for 60 sites	Same as pad construction schedule	Annual mulch refresh and plant watering (first 2 seasons); fire pit grate replacement every 3–5 yrs
R5	Laundry facility:	Bathhouse	Plumbing	After building	Separate plumbin	Phase 2 (bathhou	During slab	$12,000 –	6–10 weeks	PayRange service

#	What Gets Specified/Installed	Where on Approved Plans	Trade	Construction Sequence	Permit	Phase	Rough-In Now for Deferral	Rough Cost	Lead Time	Post-Opening O&M
	4 Speed Queen SC5 washers + 4 Speed Queen SE3 dryers, app-pay controller (PayRange), in bathhouse or immediately adjacent covered lean-to within approved footprint	building (15'×15', C007) — note: space is tight at 225 SF; consider outdoor covered lean-to addition within the approved building footprint's open space immediately adjacent	contractor (water/drain stubs); electrical contractor (240V circuits, 4 per washer, 4 per dryer); equipment vendor	shell is complete; before opening Phase 2	g + electrical permits	se is in Phase 2 area)	pour: stub out 4 drain lines (2" each, tied to sewer lateral) and 4 hot/cold supply pairs (3/4" PEX) in slab; pull 4 240V/30A circuits to laundry area from panel during electrical rough-in	$22,000 equipment + install	(equipment order)	contract; annual PM on machines ($400/yr); empty coin boxes weekly
R6	Golf greens fee and club rental program	Existing 9-hole golf course (separate from expansion area)	Owner/operator	Before first winter season	No permit	Immediate (existing course)	N/A	$2,000–$5,000 (rental clubs + POS)	2–4 weeks	Part-time starter/ranger 20 hrs/wk; annual equipment maintenance
R7	Propane exchange cage (vendor-supplied, Blue Rhino or AmeriGas) + camp supply honor cabinet	Near approved entry or maintenance area (within approved open space near main entry on C005)	Owner arranges vendor agreement; GC installs gravel pad (within approved gravel area)	After site grading; before opening	No permit for vendor-supplied cage (check local fire code if adding fixed propane storage >1 lb); honor cabinet requires no permit	Phase 1	No rough-in needed	$1,500–$2,000 (cabinet + initial inventory); propane cage $0 (vendor-supplied)	2–4 weeks (vendor agreement)	Vendor restocks propane cage; owner restocks supply cabinet weekly; annual license check

#	What Gets Specified/Installed	Where on Approved Plans	Trade	Construction Sequence	Permit	Phase	Rough-In Now for Deferral	Rough Cost	Lead Time	Post-Opening O&M
R8	Dog park fencing (basic: 400 LF 6-ft chain-link + double gate + agility equipment) on approved open space (C008)	Dog park open space label on C008	Fence contractor; landscape contractor (gravel ground cover)	After rough grading of Phase 2 open space; before Phase 2 opening	Separate fence permit	Phase 2	No rough-in needed	$4,550–$8,700	4–6 weeks	Monthly waste cleanup; annual fence inspection; biennial agility equipment inspection
R9	Dog park dual-zone dividing fence (incremental to R8)	Within dog park footprint (C008)	Fence contractor	Same time as R8	Separate fence permit (same permit as R8)	Phase 2	No rough-in needed (if R8 permit is pulled, add to same scope)	$1,200–$2,500 incremental	Same as R8	Minimal; included in R8 O&M
R10	EV charging (2 dual-port Level 2 EVSE, ChargePoint CT4000)	Near approved parking/arrival area at main entry (C005)	Electrical contractor	After primary electrical service is energized; before Phase 2 opening	Separate electrical permit	Phase 2 (or deferred)	During Phase 1 electrical rough-in: pull 2" conduit from main panel to EVSE pad location (25–50 LF); cap conduit ends. Install EVSE units when Phase 2 opens.	$6,000–$14,000 installed (conduit rough-in: $500–$1,500 in Phase 1; EVSE units + finish: $5,500–$12,500 in Phase 2)	8–12 weeks (ChargePoint hardware lead time)	ChargePoint network fee ($150–$300/yr); annual EVSE inspection
R11	Pickleball court striping (if appropriate concrete slab	Verify against C007/C008 that a 20'×44' concrete area exists in	Paving/striping contractor	After concrete is placed and cured	No permit (striping on existing slab)	Phase 2 (slab timing)	None	$800–$2,500	1–2 weeks	Net replacement every 2–3 seasons ($250–$500);

#	What Gets Specified/Installed	Where on Approved Plans	Trade	Construction Sequence	Permit	Phase	Rough-In Now for Deferral	Rough Cost	Lead Time	Post-Opening O&M
	exists within approved open space/amenity area)	approved plans; if yes, apply thermoplastic striping								striping refresh every 5 yrs ($400–$800)
R12	Shade trees (50 live oaks, 3" caliper) along approved open space strips between Phase 1 and Phase 2 site rows	Within approved open space areas on C006, C007, C008 — no impervious added	Landscape contractor	Final landscaping phase, after grading and pad construction	No permit	Phase 1 trees at Phase 1 final landscaping; Phase 2 trees at Phase 2	No rough-in	$9,000–$15,500	4–8 weeks (nursery stock lead time for 3" caliper)	Irrigation watering bag (first 2 yrs, $200/yr/tree avoided if Bahia base used); annual mulch refresh; pruning at years 3 and 7
R13	Entry monument sign and native landscape beds within approved sign/entry footprint	Approved entry area on C005	Masonry contractor (monument); landscape contractor	After site entry is graded; before Phase 1 opening	Separate sign permit (illuminated monument sign)	Phase 1	Pull 120V conduit to sign location during site electrical rough-in	$8,000–$20,000	6–10 weeks (fabrication lead time for monument sign)	Seasonal plant replacement ($300–$500/yr); bulb/LED replacement ($100/yr)
M1	LED site lighting specification	All approved light pole locations (electrical plan TBD)	Electrical contractor	During electrical rough-in and pole setting	Separate electrical permit (already required for site lighting)	Phase 1 and Phase 2	Specify LED in electrical permit documents	+$1,500–$4,000 incremental vs. HPS	Same as electrical schedule	Annual photocell testing; LED replacement at 10+ yrs
M2	Argentine Bahia + centipede grass spec confirmation	All sodded areas per C002	Landscape contractor	Final grading and stabilization phase	No permit	Both phases	No rough-in	$0–$500 incremental	Same as landscaping schedule	Mowing (monthly growing season); no supplemental irrigation needed after

#	What Gets Specified/Installed	Where on Approved Plans	Trade	Construction Sequence	Permit	Phase	Rough-In Now for Deferral	Rough Cost	Lead Time	Post-Opening O&M
										establishment
M3	Premium commercial pedestals (Milbank 50/30/20A with meter lugs)	All 215 site pedestal locations per electrical design (TBD)	Electrical contractor	During electrical installation	Separate electrical permit (already required)	Phase 1 and Phase 2	Specify in electrical design documents now	+$32,250–$43,000 incremental vs. basic pedestals	Same as electrical schedule	Annual pedestal inspection; breaker replacement as needed
M4	Erosion control blanket on pond bank slopes before sod	WRA-1 (C009), WRA-2, WRA-3 (C010) pond banks — all 2:1 and 4:1 slopes	Landscape/erosion contractor	During pond construction, immediately after bank grading and before sod	No permit	Phase 1 (WRA-1); Phase 2 (WRA-2, WRA-3)	No rough-in	$1,500–$2,400	Same as pond construction	Annual bank inspection; reseed bare spots after storms
M5	Fiberglass manholes at SSMH 6, 7, 8, 9	Deep sewer manholes per C016 (Phase 1 sewer)	Sewer contractor	During sewer main installation	No permit (material substitution within approved sewer design)	Phase 1	No rough-in	+$4,000–$8,000 incremental vs. precast concrete	Same as sewer schedule	Same inspection frequency as concrete MH; no corrosion lining required
M6	HDPE storm pipe (ADS N-12) for 15"–24" runs in lieu of RCP	All 15"–24" storm drain runs on C009, C010 (excluding main 36" trunk — confirm with EOR)	GC/storm contractor	During storm drain installation	No permit (material substitution within approved design; C002 allows PP/HDPE per AASHTO MP21-11)	Both phases	No rough-in	-$10,000–$15,000 (cost savings, not additional cost)	Same as storm drain schedule	Same O&M as RCP
M7	Wax myrtle and sparkleberry	Open space strips between designat	Landscape contractor	Final landscaping phase	No permit	Both phases	No rough-in	$0–$500 incremental vs.	Same as landscaping schedule	Zero irrigation after establishment (yr

#	What Gets Specified/Installed	Where on Approved Plans	Trade	Construction Sequence	Permit	Phase	Rough-In Now for Deferral	Rough Cost	Lead Time	Post-Opening O&M
	native hedgerow screening for premium sites	ed premium sites (within approved open space C006–C008)						ornamentals		3+); annual shaping
M8	Data conduit sleeve (1.25" ENT) joint-trenched with sewer/water throughout both phases	Along all utility trench runs (C013–C018)	Utility contractor	During utility trenching (add to sewer/water trench as part of joint trench)	No permit (conduit sleeve in private trench)	Both phases — must be done during utility installation	THIS IS THE ROUGH-IN: conduit sleeve must be installed now; Wi-Fi, low-voltage, and future fiber runs will use it.	$3,000–$6,000 (conduit material and extra trench labor for joint sleeve)	Same as utility trench schedule	Pull wire as needed; seal conduit ends after each pull

Part 4 — Summary

Top 25 Ideas Ranked by NOI Impact per Dollar of Cost

Rank	Idea	Category	Incremental Cost (mid)	Annual NOI Lift	NOI per $1 of Cost	Cumulative NOI	Value @ 7.5% cap
1	Reservation system + dynamic pricing	Revenue/Ops	$1,500	$57,500	$38.33	$57,500	$767,000
2	Snowbird monthly rate card	Revenue/Ops	$0	$60,000	N/A (no capital)	$117,500	$800,000
3	HDPE storm pipe (cost savings at install)	Cost reduction	-$12,500 (savings)	$0 O&M benefit	N/A (upfront savings)	$117,500	savings reduce cost basis
4	Bahia/native grass sod spec	Cost reduction	$250	$4,500 avg (irrigation + fert savings)	$18.00	$122,000	$60,000
5	Wi-Fi backbone	Revenue	$26,500	$78,000	$2.94	$200,000	$1,040,000

Rank	Idea	Category	Incremental Cost (mid)	Annual NOI Lift	NOI per $1 of Cost	Cumulative NOI	Value @ 7.5% cap
6	LED site lighting	Cost reduction	$2,750	$3,200 (electricity + avoided relamping avg)	$1.16	$203,200	$42,700
7	Propane/camp supply vending	Revenue	$1,750	$23,000	$13.14	$226,200	$307,000
8	Laundry facility	Revenue	$17,000	$73,000	$4.29	$299,200	$973,000
9	Golf greens fee/rental program	Revenue	$3,500	$28,000	$8.00	$327,200	$373,000
10	Dog park (basic, with agility)	Revenue	$6,625	$13,820	$2.09	$341,020	$184,000
11	Erosion control blanket on pond banks	Cost reduction	$1,950	$1,375 avg savings	$0.71	$342,395	$18,300
12	Native hedgerow screening (wax myrtle)	Cost reduction	$250	$5,000 avg (labor savings yr 3+)	$20.00 (once established)	$347,395	$66,700
13	Pond-front premium site package (60 sites)	Revenue	$82,500	$101,000	$1.22	$448,395	$1,347,000
14	Fishing access/stocking (3 ponds)	Revenue	$2,590	$960	$0.37	$449,355	$12,800
15	Pickleball striping (if slab available)	Revenue	$1,650	$13,750	$8.33	$463,105	$183,000
16	End-cap/corner premium sites (12 sites)	Revenue	$24,000	$18,476	$0.77	$481,581	$246,000
17	Fiberglass manholes at deep SSMHs	Cost reduction	$6,000	$1,200 avg (avoided relining)	$0.20	$482,781	$16,000
18	Premium pedestals (Milbank)	Cost reduction	$37,625	$4,300 avg (avoided replacement)	$0.11	$487,081	$57,300
19	EV charging (2 EVSE units)	Revenue	$10,000	$15,600	$1.56	$502,681	$208,000

Rank	Idea	Category	Incremental Cost (mid)	Annual NOI Lift	NOI per $1 of Cost	Cumulative NOI	Value @ 7.5% cap
20	Data conduit sleeve (joint-trench rough-in)	Cost reduction (future)	$4,500	$0 current (enables Wi-Fi without future trenching)	Enables $78K Wi-Fi NOI	$502,681	Enabling item
21	Dog park dual-zone divider	Revenue	$1,850	$1,360	$0.74	$504,041	$18,100
22	Entry monument sign + landscaping	Revenue	$14,000	$5,670	$0.40	$509,711	$75,600
23	Shade trees (50 live oaks)	Revenue	$12,250	$10,000 (10-yr avg)	$0.82	$519,711	$133,000
24	Self-priming maintenance pump	Cost reduction	$450	$225 avg	$0.50	$519,936	$3,000
25	Corrosion-resistant sign posts	Cost reduction	$450	$75 avg	$0.17	$520,011	$1,000

Note: Items 1 and 2 (reservation software and snowbird rate card) are ranked 1–2 not because they have the best ROI ratio (Item 2 has infinite ROI — $0 capital) but because they represent the largest single NOI decisions and must be made before opening. The "per dollar" ranking is most meaningful for capital allocation decisions.

Decide Before the Trade Mobilizes

The following decisions must be locked in before the relevant trade begins work, because changing them after mobilization causes rework:

1. **Pedestal specification (Milbank 50/30/20A vs. basic 30A)** — Must be in the electrical design documents before the electrical contractor bids. Cannot swap pedestals after conduit is set.
2. **Data conduit sleeve (1.25" ENT joint-trenched with water/sewer)** — Must be added to the utility trench scope before the first trench is pulled. Retrofitting conduit after site is graded costs 5–10× more.
3. **Laundry rough-in (drain stubs and 240V conduit in bathhouse slab)** — Must be specified before the bathhouse slab is poured. Cannot add drain lines or high-voltage circuits after slab is poured without saw-cutting.
4. **EV charging conduit stub (2" from panel to EVSE pad)** — Must be added to the electrical rough-in scope before electrical work begins. Installing conduit in the ground after paving: $2,000–$5,000 vs. $500 in the trench.
5. **Concrete patio finish specification for pond-front premium sites** — Must be specified on the concrete bid (broom finish vs. exposed aggregate vs. stamped surface). Cannot change after pour.
6. **Storm pipe material (HDPE vs. RCP)** — Must be in the storm drain bid documents. HDPE and RCP require different bedding and joining methods; cannot substitute in the field without EOR approval.

7. **Pond bank erosion control blanket** — Must be on the grading contractor's scope. After the banks are shaped and before sod is delivered — a 48-hour window that is easy to miss if not pre-specified.
8. **Sod species confirmation (Argentine Bahia + centipede)** — Must be on the landscape contractor's purchase order before nursery stock is ordered.
9. **Reservation system vendor selection** — Must be contracted before marketing begins. Campspot, RoverPass, and Camplife have different commission structures; switching mid-season is disruptive.
10. **Native hedgerow species (wax myrtle vs. ornamentals)** — Must be on landscape contractor PO. Substituting after planting is costly (plant removal + replanting).

Rough In Now, Build Later

Item	Rough-In During Construction	When to Build Out	Rough-In Cost	Build-Out Cost (Deferred)
Wi-Fi backbone	1.25" ENT data conduit sleeves joint-trenched with utility runs throughout both phases; conduit stub to future Phase 2 distribution cabinet location	Before Phase 1 opening (backbone and Phase 1 nodes); Phase 2 nodes at Phase 2 opening	$3,000–$6,000 (in utility trench)	$15,000–$29,000 (nodes, fiber, cabinet)
EV charging (Phase 2)	2" conduit from main panel to EVSE pad during Phase 1 electrical rough-in; cap conduit ends	Phase 2 opening (12–18 months after Phase 1)	$500–$1,500 (conduit only)	$5,500–$12,500 (EVSE units + finish wiring)
Laundry rough-in	Drain stubs (4× 2" lines), hot/cold supply pairs (4× 3/4" PEX), 240V conduit stubs (4× 30A) in bathhouse slab during pour	Phase 2 opening (equipment order after slab is poured)	$1,500–$3,000 (in slab during pour)	$10,500–$19,000 (equipment + installation)
Future gas fire pit connection at pond-front sites	4" PVC sleeve in concrete pad at fire pit location during slab pour	When gas distribution is designed (if propane system is added in future)	$50–$100/site × 60 sites = $3,000–$6,000	Gas line installation: $500–$1,000/site (future, not in current scope)
Pickleball net post anchors	4" diameter steel sleeves in concrete at net post locations if any amenity slab is poured during construction	When pickleball program is launched	$200–$400 (sleeves during slab pour)	$600–$1,800 (nets, posts, striping)

Total NOI Lift and Value Creation Summary

Category	Incremental Cost	Annual NOI Lift	Value @ 7.5% Cap
Revenue-generating amenities (all Part 1 items)	$170,000–$310,000	$450,000–$510,000	$6,000,000–$6,800,000
Cost reduction / maintenance savings (all Part 2 items, including HDPE savings)	-$12,500 (net savings at install) + $54,000–$90,000 additional spec cost	$20,000–$25,000/yr O&M savings	$267,000–$333,000

Category	Incremental Cost	Annual NOI Lift	Value @ 7.5% Cap
TOTAL	**$210,000–$400,000 net incremental**	**$470,000–$535,000/yr**	**$6,267,000–$7,133,000**

Important caveats:

- These figures assume full implementation of all recommended items and stabilized occupancy (52–58% blended). They are not a guarantee.
- The two largest contributors — reservation/dynamic pricing ($57,500) and snowbird rate card ($60,000) — require $0 and $1,500 capital respectively. Excluding them, the remaining capital-intensive ideas contribute ~$352,000–$418,000 NOI on ~$210,000–$400,000 of capital.
- The $470,000–$535,000/yr NOI lift is relative to a base of $0 (greenfield expansion). The existing portfolio NOI of $24,425/quarter ($97,700/yr from 91 existing sites) suggests a blended operating efficiency that these amenities are designed to replicate and exceed on the new 215 sites.
- At a 7.5% cap rate, the $470,000–$535,000 NOI lift implies **$6.3M–$7.1M of incremental property value** against **$210,000–$400,000 of amenity and upgrade capital** — a compelling case for full implementation.

Excluded — Would Require Plan Amendment

The following ideas were considered and excluded because they would alter site count, road alignment, grading, utility routing, impervious area, building footprints, setbacks, stormwater facilities, or a condition of approval:

1. Adding a swimming pool or splash pad (changes building footprint and impervious area; not in approved plans).
2. Adding a pavilion or covered picnic structure (changes building footprint; requires site plan amendment).
3. Widening any road from 16 ft to 20 ft to resolve the NFPA 1 fire lane issue (changes road footprint and surface area — flagged in errors register as a code issue but excluded from VE).
4. Adding a dump station (not shown on approved plans; new structure and utility connection).
5. Adding a playground structure (not shown; new footprint).
6. Adding a mail kiosk or package locker structure (not shown; new footprint).
7. Adding a maintenance building (not shown; new footprint).
8. Installing a formal irrigation system throughout the site (would add impervious area if piped above grade, and changes grading drainage patterns if subsurface).
9. Relocating sites 1-68/1-69 away from railroad ROW (changes site boundaries and road layout).
10. Adding a boat launch or lake access trail to an off-site water body (off the approved plans entirely).
11. Converting any RV sites to MH lots (changes site type, a condition of approval).
12. Adding a cell tower or communications monopole (new structure, height change, separate FCC/zoning process).
13. Expanding the golf course or adding a driving range (outside this expansion's approved area).
14. Adding a second bathhouse building at a new location (new footprint, not in approved plans).

Archived: Jan 2026 concept-exhibit review (superseded) ARCHIVED — SUPERSEDED

Archived: Concept-Exhibit Review (Jan 2026 single-page exhibit)

This is the OLD review, kept for reference only. It has been superseded by the full 21-sheet engineered plan set review above -- the concept exhibit was a single-page, AACE Class 5 order-of-magnitude sketch; the real plan set (Menadier Engineering, Job 2024-30-001, sealed 09/16/2026) supersedes it with actual grading, water, and sewer design.

Old basis: Menadier Engineering concept exhibit (Sheet EX001), Job No. 2024-30-001 -- single-page master site plan, AACE Class 5 / order-of-magnitude only

Old cost estimate: [{"key": "cost", "label": "Cost estimate", "sites": 215, "total": 13385000, "per_site": 62256, "range_low": null, "range_high": null, "regional_factor": 0.82, "components": [], "phase1": null}]

Old register: 29 items (10 critical, 9 moderate, 8 potential, 2 low):

- 1 (Critical): EX001, entire new Phase 1 and Phase 2 area -- Wastewater system entirely absent. No wastewater collection, treatment, or disposal system of any kind is shown for any of the 215 new sites. Existing septic tank lids (labeled for the existing 95-site area) are shown in thin/gray lineweight and serve only the existing resort. No force main, gravity sewer, grinder pumps, drainfield, lift station, package WWTP, or municipal sewer connection is shown or referenced for the new work.
- 2 (Critical): EX001, entire new Phase 1 and Phase 2 area -- Water supply system entirely absent. No potable water source, main sizing, service laterals, meter vault, fire hydrants, or connection point is shown for any of the 215 new sites. The only water source visible on the exhibit is an existing private well (thin/gray lineweight) near the existing one-story residence — almost certainly insufficient for 215 additional full-hookup RV sites.
- 3 (Critical): EX001, new sites 1-68 through 1-78 (Phase 1, north loop, adjacent to CSX ROW) -- RV sites potentially inside CSX railroad setback. Sites 1-68 through 1-78 appear, at 1"=80' scale, to be sited in close proximity to the 200'-wide CSX right-of-way crossing the north-central portion of the site. The exact setback distance from the ROW line to the nearest pad edge is not dimensioned on this sheet. CONFIRMED (2026-09-30 regulatory verification pass, direct review of Madison County's own LDC Chapters 4, 6, and 7): the county has NO railroad-specific setback provision. The applicable fallback is the general district dimensional standard (e.g. 40 ft in Ag-1/Ag-2, per Schedule 1.0) -- RV parks have no dedicated dimensional-standards section, unlike mobile home parks, which do get an explicit 35/100-ft buffer requirement.
- 4 (Critical): EX001, entire site — at least 6 wetland systems labeled -- Wetland impact analysis and SRWMD permits entirely absent. At least six distinct delineated wetland systems (Wetlands A, B, C, D, E, H, and possibly R) are shown on or adjacent to the site. Referenced wetland detail sheets ("SEE DETAIL ON SHEET...") were not included in this PDF. No wetland buffer dimensions, impact acreages, mitigation calculations, or SRWMD permit references appear on this exhibit. New roads, pads, and detention ponds appear to be routed in areas that may impact or be immediately adjacent to multiple wetland systems.
- 5 (Critical): EX001, entire new Phase 1 and Phase 2 area -- Grading plan entirely absent; earthwork volumes unknown. No grading plan, contour data, spot elevations, finish pad elevations, road profile, or cut/fill balance is shown for any of the 215 new sites. Existing building FFE elevations (101.21, 111.08, 117.54) confirm

approximately 16 feet of topographic relief across the site. Without a grading plan, it is impossible to confirm that pads drain away from structures, that roads do not pond, that wetland buffers are not graded into, or that detention ponds are sized for the actual drainage shed.

- 6 (Critical): EX001, entire new Phase 1 and Phase 2 area -- Electrical distribution system entirely absent. No utility company service point, primary feeder, transformer pads, secondary distribution, conduit routing, pedestal locations, or site lighting is shown for any of the 215 new sites. The existing resort area shows "ELEC. TRANS./ELEC. CABINET" in thin/gray lineweight — it is unknown whether this existing transformer has any spare capacity for new sites, and at this scale it almost certainly does not for 215 additional full-hookup RV sites.
- 7 (Critical): EX001, north-central area, new Phase 1 sites 1-68 through 1-78 -- Noise and vibration impact from active CSX Class I freight railroad not addressed. No noise study, vibration analysis, setback note, or site design mitigation (berm, wall, or planting buffer) is shown for sites adjacent to the 200'-wide CSX ROW. CSX is an active Class I freight railroad; freight trains generate significant noise (95–105 dB at track centerline) and ground vibration.
- 8 (Critical): EX001, entire site -- NPDES Construction General Permit (SWPPP) not referenced. The expansion footprint, while not precisely calculated, almost certainly exceeds 1 acre of land disturbance (the federal NPDES threshold). Florida's NPDES Construction General Permit requires a Stormwater Pollution Prevention Plan (SWPPP) and a Notice of Intent (NOI) filed with FDEP before any earth-disturbing work begins. No SWPPP, erosion control plan, silt fence layout, construction entrance location, or NPDES reference appears on this exhibit.
- 9 (Critical): EX001, ADA — entire new site area -- No ADA-accessible sites designated, and no accessible route shown. None of the 215 new sites is labeled as ADA-accessible. No accessible route from any site to the Bath House or any existing amenity is shown. CORRECTED (2026-09-30 regulatory verification pass): the widely-repeated "1 accessible site per 25 total sites" ratio (and a "2010 ADA Standards Section 1102" citation for it) does NOT actually apply here -- that numeric table is from the 2013 ABA Accessibility Standards, which the U.S. Access Board states apply only to FEDERAL outdoor recreation sites, not privately-owned RV resorts. The 2010 ADA Standards (the ones that actually bind this private Florida resort under Title III) have no RV/campsite-specific accessible-unit table at all -- RV parking areas fall under the GENERAL Section 208 accessible-parking scoping instead, with no dedicated site count for RV pads themselves. An accessible route to shared amenities (Bath House, pool, pro shop) is still a real Title III obligation regardless of site count. Confirm the actual applicable design approach with a Florida-licensed accessibility specialist before assuming any specific ratio.
- 10 (Moderate): EX001, new Bath House location (shown in bold near Phase 1/Phase 2 boundary) -- Bath House is located but entirely undesigned. The Bath House is shown as a bold symbol with a label but no floor plan, fixture count, square footage, construction type, foundation type, utility connections, or accessibility compliance detail is provided. For 215 new RV sites, a resort-tier Bath House must meet Florida Building Code plumbing fixture counts for recreational vehicle parks (FBC Plumbing, Chapter 29, or as amended), which are based on the number of sites served.
- 11 (Moderate): EX001, new Phase 2 western edge — "30' BUFFER" callout near WD-flagged wetland area -- 30' Buffer purpose and jurisdictional basis are unlabeled. The buffer callout appears between Phase 2 sites and what appears to be the WD-flagged wetland area, but the exhibit does not label whether this is a property-line buffer, a SRWMD-required wetland buffer, a county LDC buffer, or a voluntary setback. CORRECTED (2026-09-30 regulatory verification pass): SRWMD's own published guidance states a fixed 25-foot upland buffer from jurisdictional wetland edges for projects requiring an ERP -- the 25'-50' range in an earlier version of this finding appears to have mixed in St. Johns River Water Management District's buffer standard, a different district entirely. Use 25 feet as the reference figure, not a range, until SRWMD confirms

otherwise for this specific project. If the 30' buffer is a wetland buffer, it must be measured from the delineated wetland edge (shown on the missing detail sheets), not from a property line.

- 12 (Moderate): EX001, Pro Shop — northeast corner, "PROPOSED ADDITION" callout -- Pro Shop addition is shown but not dimensioned, designed, or utility-connected. The bold "PROPOSED ADDITION" callout points to a small bump-out at the northeast corner of the existing pro shop, but no floor plan, square footage, structural type, or utility connection detail is shown. The existing pro shop has a labeled FFE of 117.54, and the addition's finished floor elevation is not called out.
- 13 (Moderate): EX001, entire new Phase 1 and Phase 2 area -- Stormwater detention ponds are located but not sized, graded, or outlet-designed. Two new detention areas are shown in bold (stacked-ring symbols) but carry no dimensions, pond bottom elevations, normal pool elevations, seasonal high water table reference, outlet structure design, or volume calculation. SRWMD ERP requires engineered stormwater calculations demonstrating pre/post-development runoff attenuation (typically 25-year, 72-hour storm in SRWMD jurisdiction) and water quality treatment (typically 1" of runoff over the impervious area).
- 14 (Moderate): EX001, new road network — entire Phase 1 and Phase 2 area -- Road width, surface type, and cross-section are unspecified. No road cross-section detail, pavement thickness, surface type (asphalt, chip seal, or gravel), subbase specification, or curb and gutter callout appears on this sheet for any of the new internal roads. For a resort-tier RV park, roads must accommodate Class A motorhomes (up to 45' long, 13'6" tall, 40,000 lbs GVW) and fire apparatus (IFC Section 503.2.4 itself does not set a specific turning-radius number -- it delegates that to the local fire code official/AHJ; 25 ft inside / 45-50 ft outside are commonly-cited reference figures from IFC Appendix D or local amendments, not a fixed number in the base code -- confirm Madison County's actual requirement with the fire marshal directly). No turning radius dimensions appear on this exhibit.
- 15 (Moderate): EX001, entire site -- FEMA floodplain designation not shown or referenced. No FEMA FIRM panel number, flood zone designation, Base Flood Elevation (BFE), or floodplain boundary appears on this exhibit. Given the six-plus delineated wetland systems and rural north-central Florida location (Suwannee River watershed), portions of the site may fall within FEMA Zone AE or AH. CONFIRMED (2026-09-30 regulatory verification pass, direct review of Madison County's LDC): the county has its own Floodplain Management Article (Chapter 6, Article II, Sections 6.6-6.12) with a real permit process administered by the Building Official as Floodplain Administrator, and it explicitly lists "placement of recreational vehicles" within its scope -- this is not a conditional "only if in a mapped flood zone" question, the county's own process applies to RV placement generally. Separately: the company's own insurance-schedule review (Ricky's insurance audit, Sep 2026) independently attempted a FEMA flood-zone lookup for this exact address and could not get a reliable result either (geocoder mismatch) -- two independent efforts have now hit the same wall, which argues for a real parcel-level survey/GIS lookup rather than another address-based attempt.
- 16 (Moderate): EX001, entire site -- Geotechnical investigation entirely absent. No soils report, boring log, SPT N-value, bearing capacity, USDA soil series reference, or percolation test result appears on this exhibit or is referenced in any note. The site's wetland density strongly suggests the presence of hydric soils (poorly drained, high organic content) across significant portions of the expansion footprint. Hydric soils have low bearing capacity, poor subgrade strength for roads, and very limited percolation for drainfield use.
- 17 (Moderate): EX001, new Phase 1 and Phase 2 site pad details (detail insets, lower portion of sheet) -- Pad surface type and utility connection details are underspecified in the detail insets. The pad cross-section details show a gravel base and a concrete runner strip, but do not specify: concrete mix design (compressive strength, fiber reinforcement, or rebar schedule for the runner), whether the runner is one continuous strip or two wheel-track strips, aggregate base gradation specification, compaction standard (95% modified Proctor or equivalent), or how the water/sewer/electrical service lateral enters the pad from the road-side utility

trench. Without service lateral entry points shown, the GC cannot sequence utility rough-in relative to pad construction.

- 18 (Moderate): EX001, entire new area -- Dump station not shown. No RV dump station is shown anywhere in the new Phase 1 or Phase 2 area, or in the existing resort area. For a 310-site resort (95 existing + 215 new), a dump station is a standard operational requirement. Many guests use self-contained RVs and require a dump station for holding tank discharge. Without a dump station, a resort at this site count cannot achieve resort-tier marketing positioning. It is also a potential regulatory requirement depending on the Madison County LDC definition of "full hookup."
- 19 (Potential): EX001, new sites — all, but specifically sites adjacent to Wetlands A, E, H, and the WD-flagged area -- Setback distances from delineated wetland edges to new RV pads are not dimensioned and cannot be confirmed from this exhibit. SRWMD ERP typically requires a minimum 25'–50' upland buffer from the jurisdictional wetland edge (specific distance depends on wetland type, function, and any agency-negotiated mitigation). The wetland polygons are shown on EX001, but the wetland detail sheets (referenced but not included) are required to confirm the exact delineated edge locations. The 30' buffer callout in the Phase 2 western area suggests the designer is aware of at least one buffer, but the other five-plus wetland systems have no visible buffer callout.
- 20 (Potential): EX001, existing gravel RV area (north-central, thin/gray lineweight) — existing septic system -- Existing septic system capacity for 95 existing sites is unknown, and its relationship to any new wastewater system for the 215 new sites is undefined. The existing septic tank lids ("SEPTIC TANK LIDS (TYP. FOR 3)") are shown but not sized, permitted-capacity-labeled, or connected to any expansion pathway. If the existing system is already near its permitted capacity, the addition of 215 new sites may require completely independent wastewater infrastructure for the new sites and possibly an upgrade to the existing system as well.
- 21 (Potential): EX001, new Phase 1 sites 1-1 through 1-88 — road loop geometry -- Fire apparatus turning radii at loop road terminations and cul-de-sacs cannot be confirmed from this exhibit. The new road network shows curved loop roads serving both phases, but no turning radius dimensions, cul-de-sac bulb radii, or fire apparatus access geometry is called out. IFC Section 503.2.4 requires a minimum 25' inside and 48' outside turning radius for fire apparatus. At 1"=80' scale, it is not possible to confirm compliance or identify specific locations where the geometry may be insufficient.
- 22 (Potential): EX001, new sites — site width callouts (16' standard, 22' apparent at select locations) -- ADA site width and pad configuration are unconfirmed. The 22'-wide sites visible at certain locations may be intended as ADA-accessible sites, but they are not labeled as such. ADA-accessible RV sites require specific pad configuration, clear floor space, accessible hookup heights, and connection to an accessible route — a 22'-wide pad alone does not satisfy ADA without the full accessible site design package.
- 23 (Potential): EX001, title block — plan date -- Plan date conflict between title block, PDF metadata, and filename. The title block shows "1/21/2025" but the PDF creation date is 2026-01-21, the modification date is 2026-02-19, and the source filename reads "CONCEPT 1-21-2026." The most likely explanation is a title block year typo (should be 2026).
- 24 (Potential): EX001, western boundary — adjacent "NOT PLATTED" parcels (Brian William Basilius; Dale L. Schulls Sr. and Priscilla L. Schulls; Samuel M. Hicks and Brenda S. Hicks; Benjamin and Cynthia Mabry; John Thomas Burnette) -- Western boundary survey may be unresolved due to unplatted adjacent parcels. Several adjacent parcels along the western boundary are explicitly marked "NOT PLATTED," meaning their boundaries are not recorded in the official plat system and must be established by metes-and-bounds survey. Any Phase 2 sites or the "30' BUFFER" zone near the western edge depend on knowing exactly where the property line is.

- 25 (Potential): EX001, entire new site area -- Electrical service provider, voltage, and primary feeder route are unconfirmed. The electric utility serving this portion of Madison County (likely Duke Energy Florida or Suwannee Valley Electric Cooperative, but unconfirmed) has its own service extension standards, easement requirements, and lead times for primary feeder extensions. Neither the utility company name nor any existing primary line is labeled on this exhibit.
- 26 (Potential): EX001, entire site -- Municipal water and sewer availability along Country Club Rd is unconfirmed. The site is addressed at 445 Country Club Rd, Madison FL. It is unknown whether the City of Madison or Madison County utilities has a water or sewer main within practical extension distance of the site. If municipal water or sewer is available at or near the site boundary, this significantly changes the utility design strategy and cost. If neither is available, the developer must design and permit a private system (well, package WWTP or septic) — a more expensive and time-consuming pathway.
- 27 (Low): EX001, sheet-wide -- Single-sheet concept exhibit references multiple detail sheets that do not exist in this PDF. Multiple "SEE DETAIL ON SHEET..." callouts appear throughout EX001 (for Wetlands A, H, E, and others). These referenced sheets are not included in this PDF package. Any reader, reviewer, or contractor working from this PDF alone cannot access the referenced detail information.
- 28 (Low): EX001, existing gravel RV area (north-central, thin/gray lineweight) -- Existing resort infrastructure (septic tanks, electrical pedestals, propane tank, weir) appears on the exhibit without disposition notes. It is unclear from this exhibit whether any existing infrastructure conflicts with proposed new roads or grading. While the existing area appears to be north of the main new development zones, the absence of disposition labels ("REMAIN," "REMOVE," "RELOCATE") for existing elements near any new work is a documentation gap.
- 29 (Critical): Madison County approval process -- not a plan defect, a real timeline driver -- CONFIRMED (2026-09-30 regulatory verification pass, direct review of Madison County's LDC): a 215-site RV resort expansion is NOT an administrative approval in Madison County. RV parks are listed as a Special Exception use, and every Special Exception requires full Site Plan Review. The real process is multi-step: Development Review Committee (DRC) review, likely escalation to "Major Development" review given this project's scale, then Planning & Zoning Board recommendation, then final Madison County Commission approval. Each step carries its own built-in minimum notice/scheduling period -- this realistically runs several months on its own, before any of the wastewater/water/wetland permitting timelines even start.

---

1. What Will This Cost?

The single estimate in the grounding material puts this project at **$13.4 million total, or about $62,300 per new site**, with a realistic accuracy range of roughly –20% to +35% — meaning the real number could land anywhere from $11 million to $18 million or higher. That spread isn't padding; it's because five major cost items have zero engineering behind them yet: wastewater treatment ($800K–$3.5M swing depending on system type), water source ($150K–$1.2M), earthwork ($400K–$2.5M), primary electrical service ($200K–$800K), and wetland mitigation ($100K–$600K). There is only one estimate in the grounding material, so there's nothing to cross-check it against — treat it as directional, not final.

---

2. What Can We Do to Reduce the Cost?

The single best-supported cost reduction is **running a joint utility trench** — putting water, electrical conduit, and communications in one shared trench rather than three separate ones, saving an estimated **$114,000–$297,000** in excavation and backfill with no quality or revenue impact. Right behind it and nearly as well-supported: specifying **chip seal over compacted aggregate base** instead of full-depth asphalt for all internal

roads, saving another **$130,000–$185,000** — chip seal is standard practice at resort-tier RV parks in Florida, the fire marshal accepts it, and guests don't notice the difference. Combined, those two changes save roughly $250,000–$480,000 before you touch anything else, and both are decided at contract time, not in the field.

---

3. What Else Would You Recommend?

The most important thing Jay needs to know is that **this plan cannot be permitted or built as drawn** — the concept exhibit is missing the three systems that every regulatory agency will require before issuing a single permit: wastewater (no collection, treatment, or disposal shown for any of the 215 new sites), potable water (no source, no mains, no fire hydrants), and a grading plan (no contours, no cut/fill balance, on a site with confirmed 16-foot elevation change). These aren't documentation gaps — they're missing scope worth $1–5 million each, and the wastewater decision alone controls 6–18 months of your permitting timeline. The fix that is cheapest right now and most expensive to skip: commission a wastewater feasibility study and schedule a SRWMD pre-application meeting immediately, before spending another dollar on design. Separately, Phase 1 sites 1-68 through 1-78 sit close to the CSX railroad — Madison County has no specific railroad setback rule (the fallback is the general 40 ft district setback), but CSX itself requires coordination within 25-50 ft of its right-of-way regardless of local zoning, so this still needs a direct check before those pads are treated as final. One more thing worth knowing up front: this isn't a simple permit — Madison County treats a resort expansion this size as a Special Exception requiring full Site Plan Review (DRC, then Planning & Zoning Board, then County Commission), which realistically runs several months on its own, in addition to the wastewater/water/wetland permitting above.

PLAN FILES

- Full plan set (21 sheets, PDF) (plans)
- Archived: concept exhibit (Sheet EX001, superseded) (plans)
- Plan cover sheet (sheet)

CROSS-RUN NOTES LOG

1. Replaced 2026-09-30: the single-page concept exhibit review has been superseded by a full review of the real, signed-and-sealed 21-sheet construction plan set (Menadier Engineering, Job 2024-30-001). The old review is archived and still viewable via the 'Archived: Jan 2026 concept-exhibit review' option in the dropdown above.
2. This review's 81-item register was built by 5 parallel specialist visual reviews (using Opus, per Louis's request to use a stronger model for this real, higher-stakes plan set), each reading every sheet image directly at full resolution -- not a text-only pass. Every finding cites the specific sheet(s) it came from.

3. THE SINGLE MOST IMPORTANT FINDING: the plan's own general notes describe the sewer system as "on-site septic," but the actual design (confirmed independently by 2 of the 5 review groups, reading different sheets) is a centralized gravity collection system feeding a duplex lift station whose force main leaves the property with NO destination named anywhere in the 21 sheets -- no treatment plant, no disposal facility. At ~306 total sites (215 new + 91 existing, since the existing park's septic tanks are being abandoned and rerouted into this same system) this is also almost certainly too much flow for a septic permit (Florida's DOH septic ceiling is ~10,000 gpd; this system is estimated at 30,600-46,000 gpd) and needs a full FDEP wastewater permit instead.

4. Two other Critical, independently-confirmed findings: (1) the existing drainfield serving the 91 currently-occupied RV sites is demolished with no replacement named -- a live-operations and health-code risk, not just a design gap; (2) all three stormwater retention ponds' design-storm stages are higher than several connected storm-inlet rims and the bathhouse finished floor, meaning pond water backs up onto roads and into the building in a design storm -- confirmed by direct elevation comparison, not an assumption.

5. A separate, unrelated clubhouse addition/rebuild project (Hammond Design Group, real county permits 26-0243/26-0261) exists at the same property. This RV plan set runs new water/sewer service stubs to "the existing clubhouse building" and shows a "PROPOSED ADDITION" footprint there, but neither project's drawings reference the other -- this needs direct coordination between the two design teams before either is built.

6. Cross-model check (Gemini), 2026-10-05: independent review, zero exposure to this page's findings. It independently confirmed the septic/sewer contradiction (this review's #1 finding) AND caught something new: a sinkhole/karst repair detail on Sheet D001, not in the 81-item register. Full text under each question's "cross-model second opinion" toggle.

7. FEMA check, 2026-10-05 (tested, real API call): both known addresses on this parcel = Zone X (minimal flood hazard). Refines, doesn't close, the earlier "mixed X/A" note -- the 25+ acres of wetlands on this 140-acre site weren't individually checked.

Prepared from Jennings Realty's internal development-review data, generated 2026-10-06. This page is a static snapshot of the plan-review data as of 2026-09-30 and does not update automatically.